ENME408 END OF YEAR REPORT

Underwater Drone

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Abstract

Context: Mussel sampling is vital for commercial mussel farms to determine optimal harvesting opportunities, but is costly and labour-intensive. UC Vision is developing a remotely operated vehicle (ROV) to collect samples autonomously. **Objectives:** Improve the ROV's mussel position prediction, image processing in turbid water, tether management, and collection mechanism. **Method:** The interim unscented Kalman filter (UKF) was re-benchmarked and rebuilt with a damped-velocity model. Image enhancement and segmentation methods were compared on recorded footage. The tether reel was motorised, given a current-based controller and trialled at sea. Claw teeth were field-tested by hand before a multi-mussel claw and rope pusher were built. **Findings:** Ten frames ahead, the interim UKF had 70.5% more error than holding the last known position; the rebuilt UKF was better on average at every horizon, but only by 0.9–4.0% at the nominal 5–10 s. No enhancement method led on every quality measure. Depth segmentation was fast but noisy, SAM3 worked only on selected footage, and the combined pipeline reached 8.3 frames per second (FPS), below the camera's 15 FPS. Powered assistance eased tether payout and retrieval; the automatic controller worked on the bench, but the speed controller failed before in-water testing. Triangular teeth detached mussels best, informing a bent stainless-steel multi-mussel claw, and the rope pusher produced about 100 N in the laboratory. **Conclusions:** Each component provides a basis for development, but the components are unproven together. Field testing and integrated farm trials are the next steps.

Graphical abstract

UNDERWATER DRONE



Background

Mussel monitoring is a vital part of the mussel farming industry, which is costly and labour-intensive. This underwater drone autonomously identifies and collects green mussels, saving time and money for these companies.

Objective

Optimization of key aspects of the drone, including refining the computer vision system, redesigning the collection system, developing motion prediction for the drone, and designing an automated tether management system.

Mussel Grabber

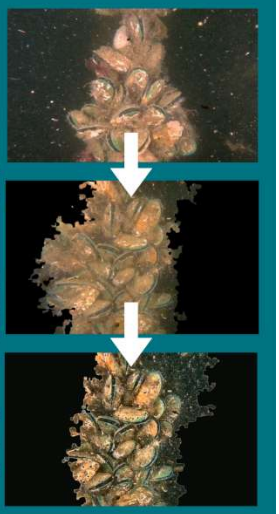


Problem: The ROV needs to be able to consistently grab mussels for sampling. The original solution – one that grabbed a single mussel at a time – was unreliable and demanded over-precise ROV navigation.

Solution: A claw was developed that grabbed an entire clump of mussels at once and is more forgiving of imprecise ROV navigation. A separate "rope pusher" was developed to overcome the mussels' grip on the rope.

Results: Initial testing showed that the system was able to detach mussels and was more consistent than the original single-mussel grabber.

Image Denoising



Problem: Background clutter and image noise (turbidity) can obscure mussel ropes and make them harder to analyse.

Solution: Mussel ropes were separated from the background before applying denoising methods, which were compared using image quality metrics.

Results: SAM3 provided more accurate segmentation but was slower. Segmentation using depth was less accurate but could operate in real time.

Motion Prediction

Problem: Mussels on a dropper line constantly sway with waves and currents, so by the time the ROV's grabber arm reaches a mussel, it may have already moved.

Solution: A real-time algorithm to forecast each mussel's 3D position was developed.

Results: An Unscented Kalman Filter (UKF) was selected. It predicted positions 5% more accurately than assuming the mussels stayed still. This was over a small range of test cases and does not fully reflect the efficacy of the method.



Automated Tether Management



Problem: Manual tether handling increases operator workload, while tangles and excessive tension can restrict ROV movement.

Solution: We developed a powered tether-management prototype to support automated deployment and retrieval.

Results: Field trials demonstrated assisted tether handling and reduced demand on the ROV's thrusters. Automatic operation requires further refinement.



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UWAI Robotics

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1 Introduction

1.1 Context

Mussel sampling provides information about the health, size, and maturity of mussels growing on commercial farms, and helps determine the optimal time for harvesting. The collection process is expensive, labour intensive, and time consuming. UC Vision is developing a remotely operated vehicle (ROV) with autonomous functionality to collect mussel samples from growing lines to reduce the expenses associated with collection.

The existing ROV, Figure 1, provides the platform for this project. Further development was undertaken this year to optimise several key aspects of the existing drone to improve its ability to identify and collect mussels under changing underwater conditions. This project addresses four areas that affect sampling: mussel position prediction, image clarity in turbid water, tether management, and the grabber and collection system.

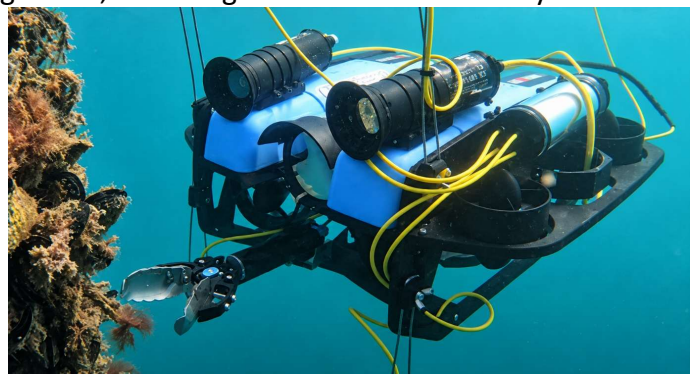


Figure 1: UC Vision's ROV for mussel sampling

1.2 Area under examination

The project focused on improving the existing ROV through four workstreams. Each addressed a separate operational problem, with the shared aim of supporting more reliable mussel collection and reducing operator involvement.

1.2.1 Prediction of mussel position

Waves, currents, and the ROV's own motion move mussels relative to the camera by several centimetres within seconds, so a target can move while the vehicle approaches. The client estimated that the grabber needs to aim 5–10 s ahead of the moment it commits to a grab. This workstream developed a filter that predicts the future position of each tracked mussel from the ROV's stereo tracking data, tested whether it beats assuming the mussel stays where it is, and evaluated targeting the mussel cluster and timing the grab to a calm moment. The aim was to allow the ROV to account for this movement when positioning the grabber.

1.2.2 Minimising turbidity noise

The ROV relies on computer vision to identify mussels and guide collection. Turbidity reduces image clarity and can interfere with these functions, restricting the conditions in which the vehicle can operate effectively. This workstream investigated image enhancement methods to reduce the effects of turbidity on the camera feed and improve its usefulness for the ROV's vision system.

1.2.3 Automated tether management

The tether connects the ROV to the surface and must be deployed and retrieved as the vehicle moves. Excess tether can become tangled, while insufficient length can restrict movement or apply unwanted forces to the vehicle. Manual management also requires continued attention from an operator. This workstream focused on developing an automated tether management system to control the deployed length while limiting its effect on vehicle movement.

1.2.4 Grabber and collection system

The existing grabber collected mussels individually, requiring repeated collection actions and over-precise alignment of the ROV. This workstream focused on developing a grabber capable of collecting multiple mussels from a section of growing line to increase the commercial viability of the drone without relying on precise alignment.

2 Literature

2.1 Underwater robotic systems

Underwater robotic systems are increasingly used in marine industries to perform inspection, monitoring, and intervention tasks in environments that are difficult or hazardous for human divers. Remotely operated vehicles (ROVs) are commonly deployed for subsea inspection, scientific research, and infrastructure monitoring due to their ability to collect data and perform tasks while being remotely controlled or operating autonomously [1].

Within the aquaculture industry, underwater robots are becoming an increasingly common tool for data collection. These systems can be used to monitor fish cages, inspect mooring systems, and collect data about the environment and condition of the farm [2]. The use of robots reduces operational overheads, equipment investment and labour requirements while allowing for an increased frequency of inspections compared to traditional methods [3].

2.2 Prediction of mussel position

Existing target-tracking methods range from constant-velocity (CV) models to detailed descriptions of manoeuvring behaviour [4]. A CV model extrapolates the current velocity over the whole prediction interval. Applying Singer's correlated-acceleration model [4] to velocity gives a damped-velocity model, whose forecast follows the current motion for a characteristic time and then levels off. Oscillatory models capture repeated movement more directly but need a consistent frequency.

The unscented Kalman filter (UKF) handles nonlinear functions using sigma points [5], but with these linear models and direct 3D position measurements, it matches a conventional Kalman filter. The process and measurement noise covariances, (Q) and (R), should be identified from data and can be adapted online [6], and a forecaster should be scored against the naive forecast that the next value equals the last [7].

For ship motion, the more useful output is often a quiescent period of low motion [8], and for stereo-tracked tree branches, every predictor was degraded by tracking noise and missed frames [9]. No published evaluation exists for mussel dropper lines viewed from an ROV, where tracks are short, gappy and measured from a moving camera. The achievable horizon

is therefore unknown, as is whether any predictor improves on holding the last known position.

2.3 Minimising turbidity noise

Underwater environments are challenging locations for autonomous vehicles. A contributing factor to this reason is the turbidity of the water and the weak illumination due to the optical properties of water. An important metric for image denoising is being able to assess how effective this process is in terms of restoring the original image. This is particularly challenging underwater as it is often impossible to compare a filtered image with what it should look like. Therefore, four main metrics have been developed in this field for evaluating underwater image filtering performance. These are peak signal to noise ratio (PSNR), structural similarity index measure (SSIM), underwater image quality measure (UIQM), and underwater colour image quality evaluation (UCIQE) [10], [11].

There are several methods for tracking objects of interest across a video. Some popular ones are fast iterative shrinkage-threshold algorithm (FISTA) and segment anything model two and three (SAM2, SAM3) [12], [13]. Both algorithms achieving tracking in a similar way; by segmenting objects of interest from the background to focus on. This can be made challenging with changing camera position, changing backgrounds and changing illumination levels which are all increased further in underwater applications [12]-[15].

Therefore, this project assesses the application of these aforementioned methods to the ROV with a focus on the SAM methods for segmentation as suggested by the client.

2.4 Automated tether management

Current research into tether management systems focuses on optimising deployed tether length, maintaining tether tension, and reducing slack. Examples include buoy–ballast systems [16] and autonomous surface vehicles that maintain a position directly above the underwater vehicle [17]. However, limited research has focused on compact, low-cost tether winch designs that mechanically control the length of tether deployed based on input from the previously mentioned optimisation system.

The project therefore examines whether a compact retrofit can manage deployment reliably while limiting forces on the ROV. Motor feedback must be assessed against the behaviour of the assembled winch and deployed tether.

2.5 Grabber and collection system

One of the key functionalities of the underwater drone is the mechanism to pull the mussels off their growing lines. Mussels attach to growing lines via byssal threads, which end in adhesive plaques [18], the tighter the mussel's cluster the harder it is to detach. An effective underwater grabber must grip the shell, or cluster of mussels, and overcome the attachment force. Different solutions have been tested in the aquaculture industry for removal of different species. Solutions range from rigid claws to soft material mechanisms with results showing that rigid claw grippers provide stronger holding force than soft material solutions [19]. Because mussels have a naturally high attachment force to the mussel line, a solution that can produce a high controlled detachment force will be prioritised.

The mechanism must therefore consider accessibility and claw material, as well as the physical constraints of the ROV. This requires evaluating the grabber together with its interaction with the mussel line.

3 Methods and Materials

3.1 Project objectives

The project aimed to optimise UC Vision's ROV for mussel sampling through four workstreams: predicting mussel position to support grabber alignment, minimising turbidity noise, developing an automated tether management system, and developing a grabber capable of detaching multiple mussels. These workstreams had individual approaches which are detailed in Section 3.2.

3.2 Approach

3.2.1 Prediction of mussel position

A small sample of recorded mussel tracking data was accessed as CSV files containing x, y and z coordinates and mussel identifiers (four 2023 scans without timestamps and one 4 s scan at 15 fps from 2026). Fourier decomposition excessively attenuated small movements, and linear and quadratic extrapolation overshot when tracking was lost, so a UKF with a CV model and fixed Q and R was adopted for the interim report.

After errors were found in the interim benchmark (Section 4.1.2), the UKF was rebuilt as a drop-in replacement with one filter per tracked ID. It uses a damped-velocity model (time constant 1.5 s), R set from the measured noise, Q adapted online from the normalised innovation squared (NIS) [6], and an outlier gate that rejects measurements with an NIS above 25 lying more than 30 mm from the prediction [20]. Its settings were chosen on the same scans, so a leave-one-scan-out check was also run.

Accuracy was assessed using root mean square error (RMSE) against subsequent recorded positions, relative to a zero-order hold (ZOH) that retains the last valid measurement. Results are reported as skill, $1 - \mathrm{RMSE}_{\mathrm{method}}/\mathrm{RMSE}_{\mathrm{ZOH}}$, where positive values indicate an improvement [7]. All methods were scored frame by frame on the same valid frames, excluding dropouts and depth outliers. As the 2023 frame interval is unknown, prediction intervals were set in frames (1–50), covering 5–10 s for frame intervals of 0.2 s or more (Appendix A).

3.2.2 Minimising turbidity noise

Initial work examined image enhancement using CLAHE and UDCP. Later investigation focused on segmentation using SAM2, SAM3, and depth thresholds. SAM2 outputs were reviewed with the client, and further development was discontinued because the segmentation was unsuitable for the intended task.

SAM3 and depth thresholds were evaluated using the same recorded footage. Several SAM3 prompts were trialled, with "water" selected because the alternatives did not produce useful segmentation. The detected water region was removed, retaining the remaining image as the

candidate mussel-covered line. Cases where water was not recognised were treated as segmentation failures.

Depth thresholds were selected through trial and error informed by client-suggested values. These retained image regions within the selected distance range. Performance was compared visually, considering whether the line and mussels were retained and unwanted regions excluded. Processing time was measured to assess suitability for real-time operation (Appendix B).

3.2.3 Automated tether management

A compliant buoy–ballast arrangement was initially preferred because it provided feedback near the ROV while limiting transmitted forces [16]. However, its submerged sensing and communication requirements exceeded the project timeframe. Motor current, available through the electronic speed controller (ESC), was therefore selected as an indirect loading input without requiring additional submerged sensors. As current also reflected drivetrain resistance and acceleration, it was not treated as a calibrated tension measurement.

The existing reel was retrofitted to retain compatibility with the tether and electrical passthrough while avoiding the cost and integration constraints of commercial alternatives. A client-supplied motor drove the reel through a timing belt with a 6.67:1 reduction and adjustable idler. Mechanical revisions targeted durability and friction through an aluminium mounting plate, stainless fasteners, a revised torque-transfer adapter and a bearing-supported idler.

Control logic was transferred from the ESC to an ESP32 communicating through UART. Filtered current determined retraction, hold, or extension, with hysteresis, minimum time between state changes, and speed ramping to reduce abrupt movement. Final automatic speed and motor current were limited to 100 rpm and 4 A. Manual override disabled automatic operation, with manual speed limited to 500 rpm. Communication-loss responses commanded zero speed without disconnecting power.

Exploratory bench tests monitored current during restrained and unrestricted retraction, including tests with the reel brake tensioners removed to investigate drivetrain resistance. Manually controlled field trials examined tether behaviour, payout, retrieval, and cable distribution. No calibrated tension measurements were obtained, and autonomous control was not tested in water. Post-trial inspections informed mechanical revisions. Detailed drawings, controller code, and test procedures are included in the Appendix C.

3.2.4 Grabber and collection system

Several weeks of review were conducted, including experimentation with the existing design and meetings with various people who have previously worked on the project. During these meetings and conversations with the client, the need to redesign the system for the collection of multiple mussels developed – it optimised the arm design, reducing degrees of freedom, and simplified the projected collection procedure in a finalised version of the robot. Several additional weeks were then spent developing designs and concepts for this redesigned grabber – see Figure 2.

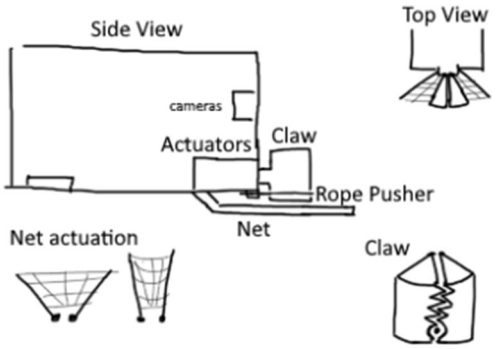


Figure 2: The grabber concept selected for further development

The original set of jaws developed for this concept were tested on a mussel-sampling boat using a growing line lifted from the water. Difficulty obtaining a secure grip prompted a redesign with backward-sloping teeth and outward bends, intended to imitate the way human hands engage mussels.

A rope pusher was developed to provide detachment force once the jaws had established a grip. An underwater stepper motor drove a pair of gears and a leadscrew, producing linear movement of a steel rod and flat plate against the growing line. The original concept sketch is shown in Figure 3.

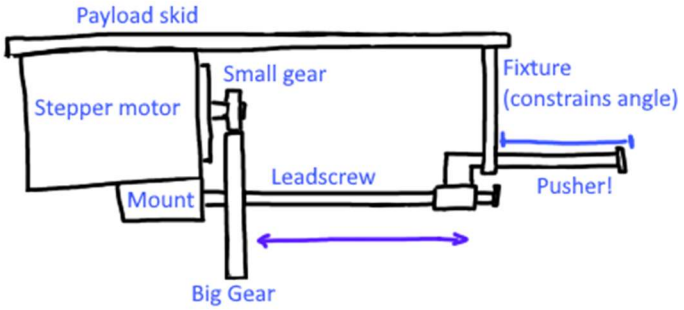


Figure 3: The rope pusher concept

Once the build was completed, testing used loads, including weights and a basket of rocks, to assess pusher force output. Field testing of the redesigned jaws and rope pusher is planned for an upcoming boat trip that has not occurred yet at the time of writing.

4 Results

4.1 Prediction of mussel position

4.1.1 Data

Typical frame-to-frame movement on the 2023 scans was 2.5–4.4 mm, against 0.3–1.9 mm of measurement noise. Tracking was lost in 0.2–33% of frames, with gaps of up to 26 frames on RD_004, and eight frames had depth errors. The scans differ in character: RD_003 is dominated by a steady approach of the ROV towards the rope, while RD_001 wanders within about ±1 cm with no preferred direction. On the 15-fps scan, the mussels moved about 14 mm per frame (about 210 mm/s) relative to the camera, as the ROV moved around the rope.

4.1.2 Verifying initial UKF

The interim benchmark was reproduced exactly from the project code, giving the same 4.6% average improvement for the UKF. Checking it found problems in both the evaluation and the filter:

- The ZOH baseline held the raw current sample, so whenever tracking was lost it 'held' (0, 0, 0), making the baseline appear far less accurate than it is.
- The UKF was scored on different frames from the other methods, and the 4.6% was an average of four percentages (−6%, +9%, +68% and −53%).
- The UKF's measurement noise setting corresponded to about 316 mm of noise, against about 1 mm measured, so it lagged the mussel. Its process noise was also applied to the wrong states.
- Its CV model overshot whenever the sway reversed, and depth errors and ID changes were accepted as measurements, creating large false velocities.

Once a true hold was used and all methods were scored on the same valid frames, the UKF was 70.5% worse than the ZOH at its operating point of 10 frames ahead.

4.1.3 Finalised filter

Averaged over the four 2023 scans, the finalised filter was better than the ZOH at every horizon from 1 to 50 frames, but by a small margin: at most 6.4%, and 0.9–4.0% at 10–20 frames, a nominal 5–10 s (Table 1). Pooled over the scans, the RMSE 10 frames ahead fell from 14.3 to 13.4 mm. RD_003 and RD_004, whose motion is the most persistent, gained up to about 16%, as Figure 4 shows, and RD_002 gained 0.2–3.1%. RD_001, which wanders with no preferred direction, was made worse at almost every horizon, by up to 19.9% (1.2 mm) at 5 frames. The bottom row of Table 1 shows how the error of the ZOH itself grows with horizon.

Table 1: Skill relative to the ZOH: finalised UKF by scan, and the mean of the four 2023 scans for the finalised and interim filters

Scan	1 frame	5 frames	10 frames	20 frames	40 frames
RD_001	−3.9%	−19.9%	−8.6%	−7.8%	−1.5%
RD_002	+1.7%	+0.4%	+2.1%	+2.2%	+1.4%
RD_003	+12.7%	+12.7%	+8.5%	+4.8%	+1.9%
RD_004	+15.2%	+14.9%	+10.1%	+4.5%	+0.2%
Mean, finalised UKF	+6.4%	+2.0%	+3.0%	+0.9%	+0.5%
Mean, interim UKF	−128%	−22%	−70%	−29%	−56%
ZOH RMSE, all scans (mm)	**3.0**	**8.7**	**14.3**	**24.7**	**45.4**

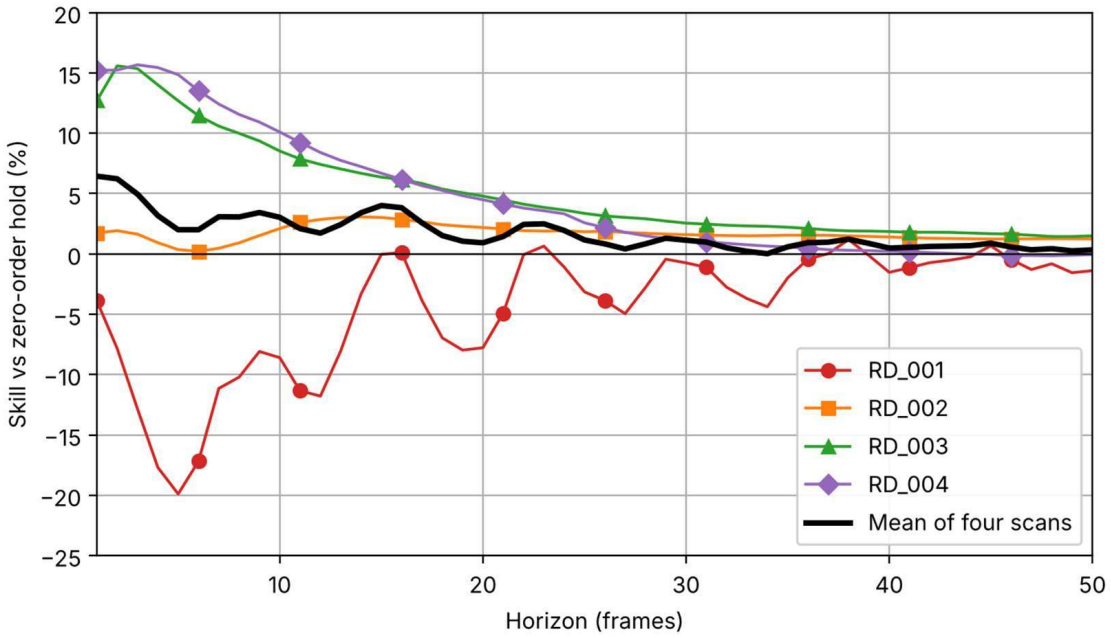


Figure 4: Skill of the finalised UKF relative to the ZOH by scan (10–20 frames are a nominal 5–10 s)

Removing each change in turn showed what made the difference. Restoring the interim measurement noise made the filter 138% worse than the ZOH at 10 frames and removing the damping (the CV model) gave -43% at 10 frames and -117% at 40. The outlier gate mattered operationally: when forecasts were also made from glitch frames, it let the filter beat a hold that carried the glitch by 78–86% on RD_001. Of seven motion models compared, only damped velocity with a position random walk was clearly positive. The settings were chosen on the same scans and did not reliably transfer between them in a leave-one-scan-out check, so the true gain may be smaller.

4.1.4 Verifying the timing

The 15-fps scan is the only data with a known frame rate. With one filter per ID, as at runtime, the finalised filter beat the ZOH by 11–19% from 2 to 20 frames (0.13–1.33 s ahead), against at most 5.2% for the interim filter (Appendix A, Figure 6). One mussel was reported under five tracker IDs in 4 s, and the ID changes moved its reported position by 20, 26 and 62 mm, compared with a typical step of 6.8 mm. A single filter run across these changes was 7% worse than the ZOH at 10 frames, so a new filter should be started for each ID, as the runtime pipeline does.

4.1.5 Targeting mussel clusters

ID changes were common, with 69 IDs appearing in 60 frames, and the median ID lasting only 8 frames (0.53 s). The cluster centroid used in Runtime.py never removes a lost ID, so by the end of the scan it averaged 69 IDs while only 19 mussels were in view, sitting a median of 22 mm (up to 49 mm) away from the mussels in view. Three other ways of defining the cluster target were compared. The mean and median of the current detections jumped whenever mussels appeared or disappeared (a median of 11.1 and 9.6 mm per frame). A target moved by the average displacement of the IDs seen in both the previous and current frame

(common-ID displacement) moved a median of 5.5 mm per frame, about half as much, and is not affected by IDs changing.

4.1.6 Lull detection

Starting the grab when the target is moving slowly gives a modest benefit (Figure 5). Over the four 2023 scans, frames in the slowest quarter of recent speed were followed by 5–18% less movement than an average frame over the next 1–40 frames, while the fastest quarter was followed by 11–27% more. The result is dominated by RD_002, which supplies most of the frames, and was negligible on RD_003, where the steady approach makes speed uninformative. The LullDetector in the runtime pipeline applies the same idea online.

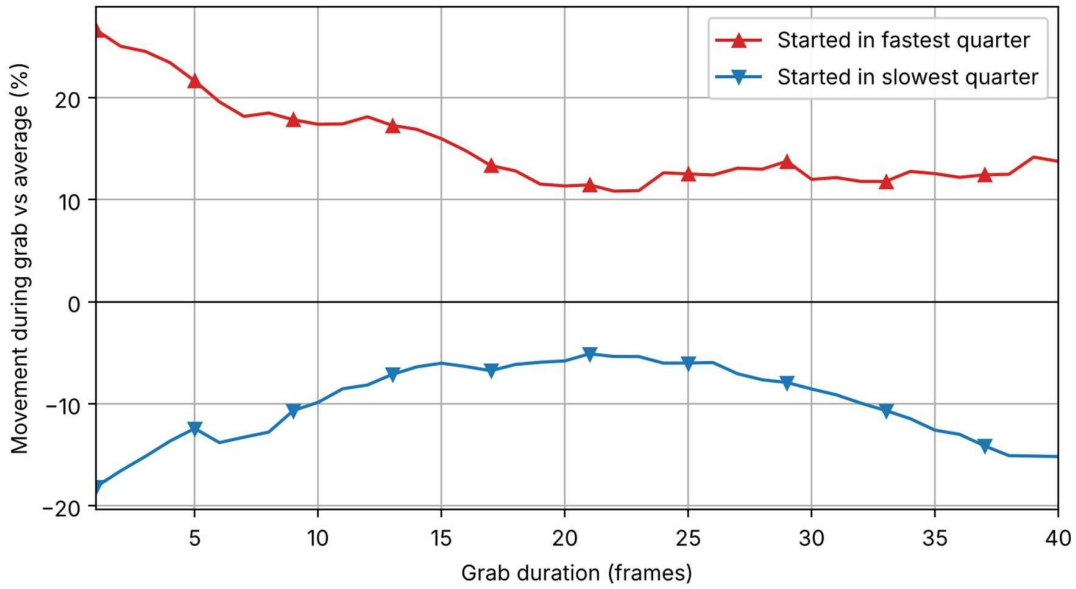


Figure 5: Movement during a grab of a given duration relative to the average, for grabs started in the slowest and fastest quarter of recent speed

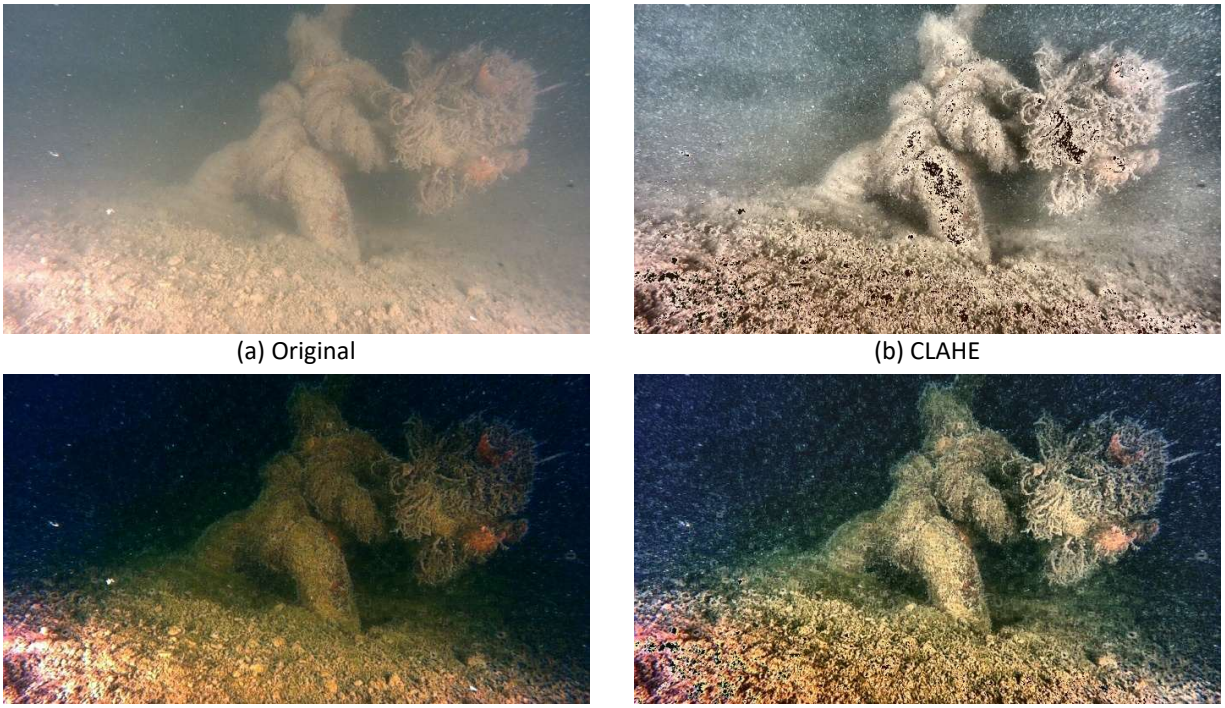
4.1.7 Processing time

Processing time is not a constraint. The finalised filter took about 0.34 ms per mussel per frame for a 10-frame forecast and 0.63 ms for a 5 s forecast at 15 fps, against 1.2 and 7.3 ms for the interim filter, which steps its forecast forward one frame at a time. For the 21 mussels per frame seen at 15 fps, a 5 s forecast needs about 0.2 s of processing per second of video, against 2.3 s for the interim filter. These timings were measured on a desktop processor, not the ROV's Jetson.

4.2 Minimising turbidity noise

4.2.1 Image enhancement

CLAHE and UDCP were implemented as separate Python modules and evaluated on recorded underwater imagery, both individually and in combination. The outputs demonstrated different changes in image appearance. CLAHE increased local contrast and made surface detail more prominent, while UDCP changed colour and contrast. Figure 6 compares the same frame before and after processing, allowing these effects to be assessed without differences in scene content.



(a) Original (b) CLAHE

(c) UDCP (d) CLAHE and UDCP

Figure 6: Image-enhancement comparison using the same underwater frame: (a) original; (b) CLAHE; (c) UDCP; (d) CLAHE and UDCP.

Mean image-quality scores are reported in Table 2. CLAHE achieved the highest underwater image quality measure (UIQM), at 1.488, while UDCP achieved the highest underwater colour image quality evaluation (UCIQE), at 0.513. The combined output scored between the individual methods on both measures. Applying both algorithms therefore did not produce the highest score, despite changing both contrast and colour characteristics.

Table 2: Mean image-quality scores for the enhancement algorithms.

Algorithm	**PSNR (dB)**	**SSIM**	**UIQM**	**UCIQE**
CLAHE	15.26	0.506	1.488	0.323
UDCP	9.02	0.448	1.097	0.513
CLAHE and UDCP	11.31	0.347	1.390	0.422

PSNR and SSIM were calculated against the original underwater images because clean reference images were unavailable. CLAHE retained the greatest similarity to the input, with a PSNR of 15.26 dB and SSIM of 0.506. UDCP produced larger pixel-level differences, with a PSNR of 9.02 dB. These values describe changes relative to the degraded input; they do not measure restoration accuracy. Consequently, the scores did not establish which output most accurately represented the underwater scene.

Visual inspection and the underwater quality metrics provided complementary comparisons, but no single configuration performed best across all measures. The results demonstrated image enhancement rather than a measured improvement in mussel detection, localisation, or collection. Those downstream tasks were not evaluated through controlled comparisons in this workstream.

4.2.2 Mussel-rope segmentation

Depth-based segmentation used distance measurements recorded alongside the video to retain the mussel-covered rope and exclude surrounding regions. Initial masks contained substantial noise and holes. Median filtering followed by erosion and dilation improved their continuity. Applying CLAHE and UDCP to the retained region then changed its contrast and colour, producing the example shown in Figure 7.

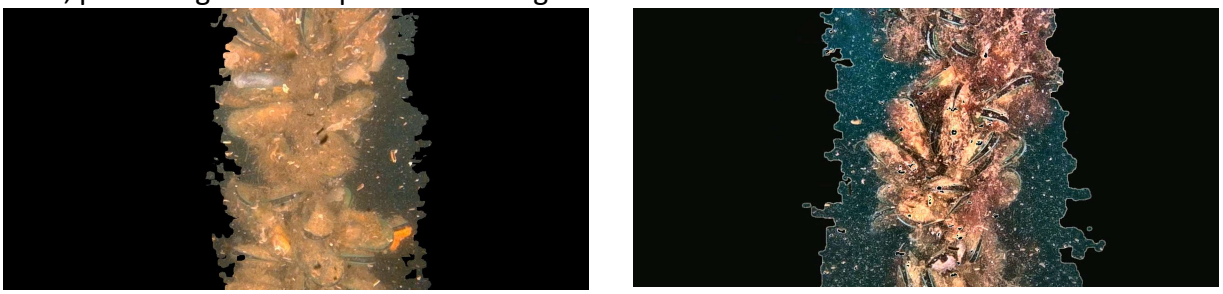


Figure 7: Depth-based rope segmentation: (a) segmented region, left; (b) segmented region after CLAHE and UDCP enhancement, right.

Although selected outputs retained a useful section of rope, noisy depth measurements prevented segmentation from meeting the client's requirements. The filtering operations improved mask continuity but did not consistently recover the rope boundary. Segmentation was assessed visually without manually labelled reference masks, so boundary accuracy and the proportion of rope retained were not quantified. The favourable processing speed therefore did not establish that the masks were sufficiently reliable for autonomous use.

SAM2 also produced inconsistent results. Some frames retained a recognisable rope region, while others excluded parts of the rope or retained unwanted background. Figure 8 shows examples of the difference between successful and unsuccessful outputs. The slow processing and inconsistent segmentation led to further development being discontinued following review with the client. Enhancement was not subsequently applied to the SAM2 outputs.

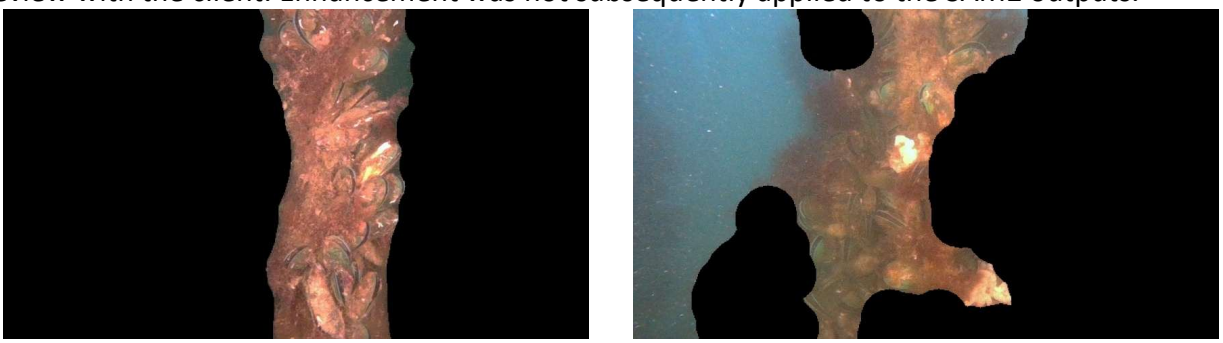


Figure 8: SAM2 segmentation examples: (a) a more successful output, left; (b) an unsuccessful output, right.

SAM3 produced effective segmentation in selected footage using the prompt "water". Removing the detected water region retained the rest of the image as the candidate mussel-covered rope. Figure 9 shows a successful example. However, the model failed to recognise water in most other tested videos and did not produce useful segmentation. Its performance therefore varied substantially between recordings, rather than providing a reliable improvement over the other approaches.

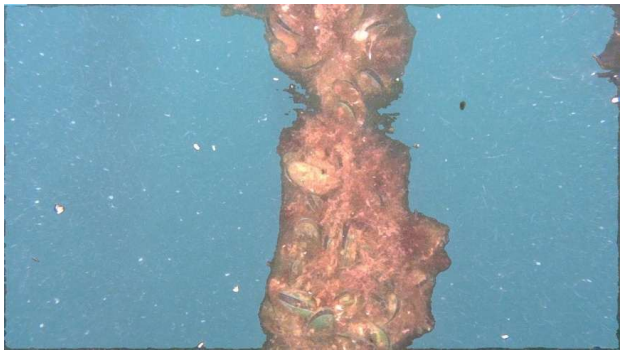


Figure 9: Successful SAM3 segmentation example using the prompt "water".

4.2.3 Processing performance

Processing performance was measured on a computer using recorded footage, with 621 frames processed for each configuration in Table 3. The camera operated at 15-fps, providing approximately 66.7 ms between frames. Depth segmentation was fastest, averaging 26.4 ms per frame and 37.89 FPS. CLAHE, UDCP and their combination also exceeded 15 FPS on average when tested separately from segmentation.

Table 3: Processing performance measured over 621 frames per configuration.

Configuration	**Mean time per frame (ms)**	**Average FPS**
Depth segmentation	26.4	37.89
CLAHE	39.0	25.65
UDCP	52.1	19.20
CLAHE and UDCP	55.9	17.89
Depth segmentation, CLAHE and UDCP	120.7	8.29

Combining depth segmentation with both enhancement algorithms increased mean processing time to 120.7 ms and reduced performance to 8.29 FPS, approximately 55% of the camera's frame rate. The complete pipeline therefore could not process every incoming frame at the recording rate. This differed from the satisfactory average speeds of the individual components and demonstrated the additional computational cost of combining them.

Frame times also varied considerably. Depth segmentation reached a slowest recorded time of 88.7 ms, exceeding the camera's frame interval despite its favourable average. The combined enhancement methods reached 163.6 ms. The complete segmentation and enhancement pipeline ranged from 70.5 to 160.4 ms per frame, exceeding the 66.7 ms interval throughout the recorded test. Average FPS alone therefore did not demonstrate consistent real-time responsiveness.

CLAHE and UDCP also exceeded the camera interval on their slowest frames, taking 111.1 and 109.4 ms respectively. Thus, none of the configurations maintained the required processing time on every frame. The combined enhancement stage averaged 55.9 ms, leaving only around 10.8 ms of the camera interval for other processing. Adding depth segmentation removed that remaining margin, even though segmentation alone was faster than either

enhancement algorithm. The measurements therefore showed why component-level timing did not establish the performance of the integrated pipeline.

Observed SAM runs exceeded 10 s. Neither tested implementation approached the camera rate, so frame-by-frame benchmarking was not performed. Measurements used recorded footage on a computer and excluded the ROV's acquisition and control chain. Detailed results are provided in Appendix B.

4.3 Automated tether management

4.3.1 Powered reel development

The existing hand-operated reel was retrofitted with a motor, belt transmission, adjustable tensioner and electronics enclosure (Figure 10). The assembly retained the tether, electrical passthrough and manual handle. Powered operation was demonstrated in both directions, providing the platform for testing the automated controller.

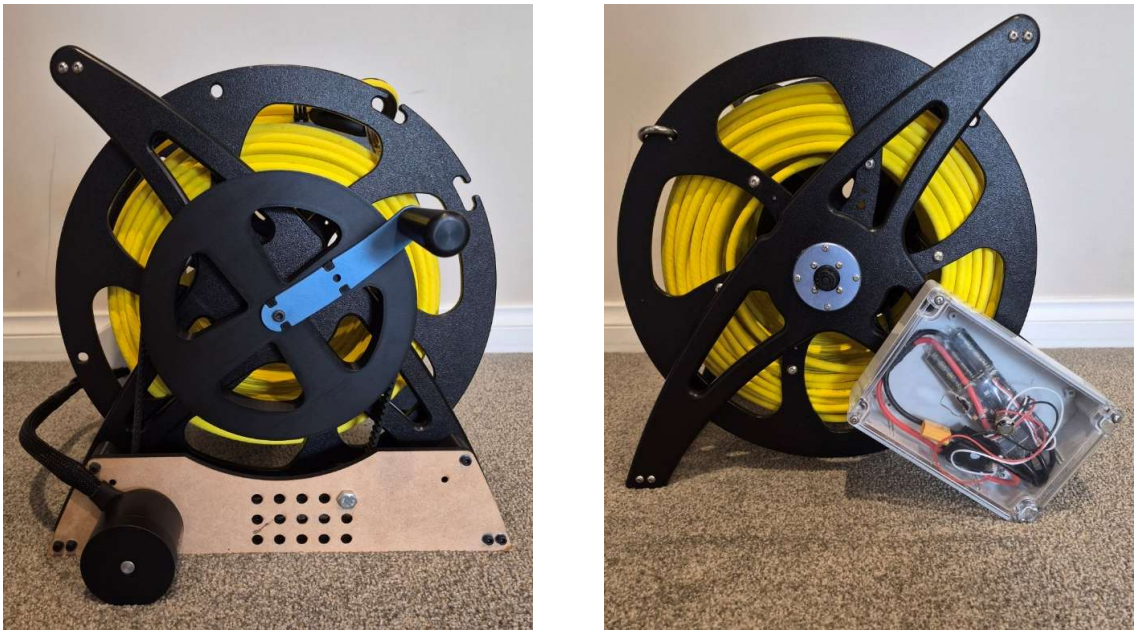


Figure 10: Initial powered tether reel prototype: (a) motor and belt-drive arrangement, left; (b) reel and electronics enclosure, right.

4.3.2 Initial controller behaviour

Initial testing demonstrated extension, retraction and hold commands using motor current as the control input. Changing from current to velocity commands produced smoother movement through the ESC's acceleration ramping. However, intermittent scripting failures remained, so the control logic was transferred to an ESP32 communicating through UART.

With the tether restrained and the lower threshold set to 5 A, current repeatedly crossed the threshold, fluctuating by approximately 6 A (Figure 11).

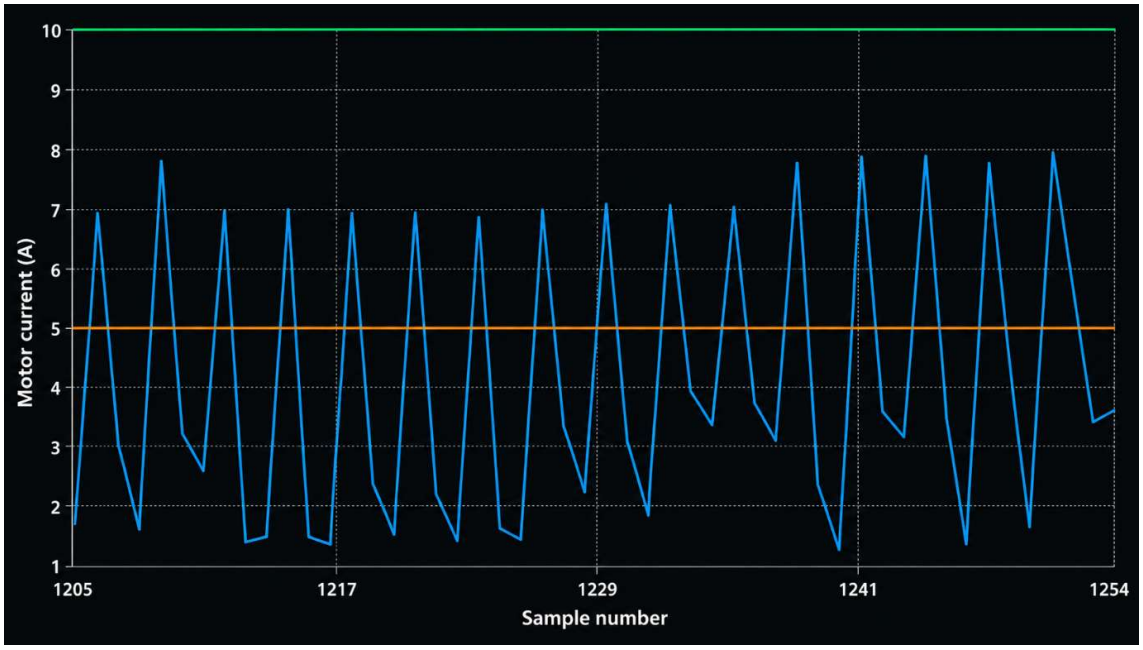


Figure 11: Motor current during restrained-tether testing with a lower control threshold of 5 A.

During unrestricted retraction, the threshold was deliberately increased to 9 A to prevent transitions into hold. The large oscillations disappeared, with current varying more gradually between approximately 4 and 7 A for a complete revolution (Figure 12).

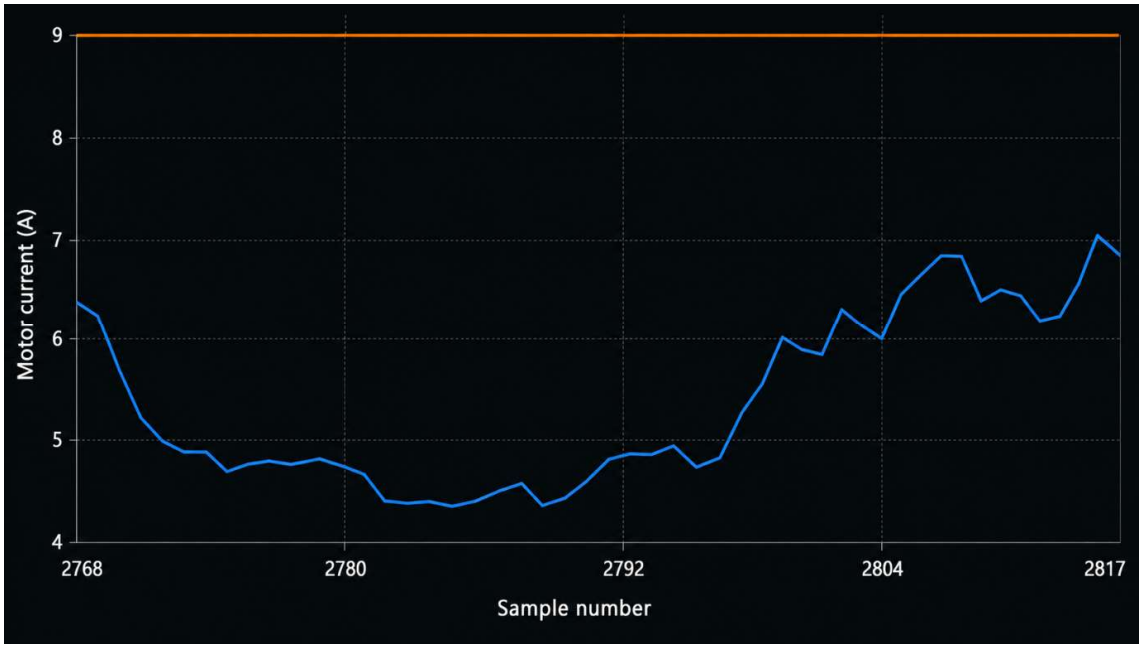


Figure 12: Motor current during unrestricted retraction with a lower control threshold of 9 A.

4.3.3 In-water tether handling

During mussel-rope scanning, the mechanical resistance in the reel affected the vehicle's stability and increased the thruster power draw observed on the live display when the ROV was allowed to pull the tether out itself. Applying approximately 0.2-0.5 A assisted payout by

overcoming this resistance. Suitable settings were established experimentally from tether and vehicle behaviour. Faster payout produced loose cable and tangles (Figure 13). Powered surface retrieval reduced thruster use, and fewer battery swaps were required. The 4 A limit provided sufficient torque for mussel-farm handling while keeping the reel stationary on the deck.

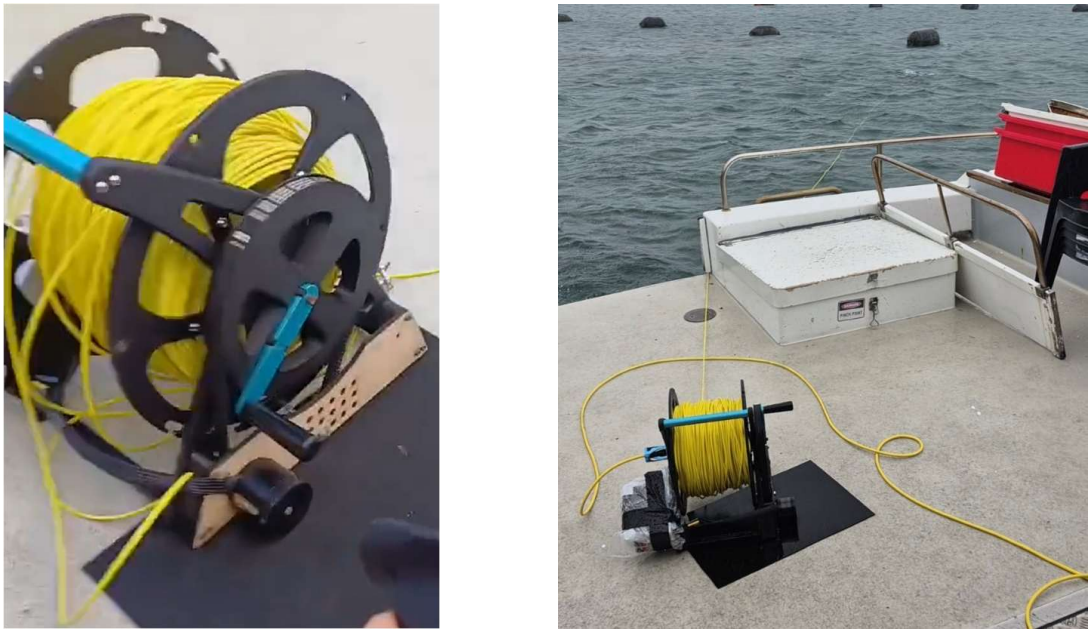


Figure 13: Field trials: (a) tether tangling on the reel, left; (b) powered reel during mussel-farm scanning, right.

Seabed scanning involved longer deployments, with approximately 100–150 m recovered during retrieval. Uneven winding and trapped cable caused more frequent tangles. Water currents pulled excess tether from the reel when resistance was too low, while retrieval at 4 A sometimes stalled. Tether resistance could also impede or stall the ROV. Powered retrieval remained useful, but manual tether management was necessary to not affect the operation of the ROV during the commercial operation.

4.3.4 Final controller behaviour

After field testing, the ESP32 was powered from the ESC's 5 V supply. Investigation traced the retract-to-hold oscillation to the ESC treating zero rpm as an inactive output. Drive ceased during hold, causing current to fall and retraction to resume.

Changing the speed-control dead zone from 900 electrical rpm to zero enabled active holding. During qualitative bench testing, the reel remained stationary in hold, the oscillation stopped, and all three states responded as intended to manually vary loading.

Manual override disabled automatic operation until restart. Scaling commands to 500 motor rpm and ramping their response produced smoother movement without the abrupt behaviour that had caused tangles. Filtering, hysteresis and minimum state timing were also incorporated, although their individual effects were not measured separately.

The ESC subsequently failed before quantitative recordings were collected. A replacement was procured but lead time resulted in it not arriving in time for writing this report. Final controller performance was therefore demonstrated qualitatively on the bench, without automatic in-water validation.

4.3.5 Mechanical redesign

Post-trial inspection found swelling of the MDF mounting plate, corrosion of steel components and binding of the printed idler. The printed handle-engagement tabs also broke during manual retrieval.

Aluminium mounting plates and a bearing-supported idler were installed (Figure 14). The idler rotated freely when secured and allowed belt tension adjustment. Its resistance was higher than the original idler before binding, but more consistent. The bearings removed sliding contact between the printed pulley and its sleeve.



Figure 14: Revised reel assembly with aluminium mounting plates and bearing-supported idler.

A metal torque-transfer adapter was designed but had not been manufactured or installed due to workshop availability. Inspection also identified exposed motor windings without a visible protective coating. The revised assembly had not undergone further powered or marine testing. Full results, drawings and controller code are provided in Appendix C.

4.4 Grabber and collection system

4.4.1 Jaws

Originally the jaws were designed using 2 mm aluminium sheet metal and had a single 90-degree bend for the teeth on either side to create the closing claw shape. Two teeth designs were initially tested including triangular teeth, Figure 15, and square teeth, Figure 16. The

sharp-teeth design includes the “duck head” piece designed to connect the claw to the existing actuation system. This was 3D-printed out of PETG and will also be present (but is not shown) in the square teeth design.

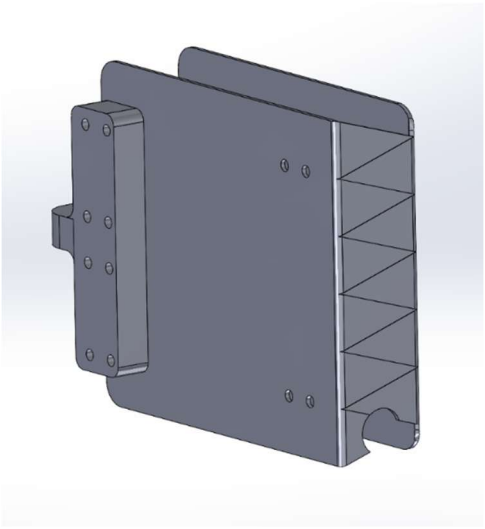


Figure 15: Triangular teeth – front view

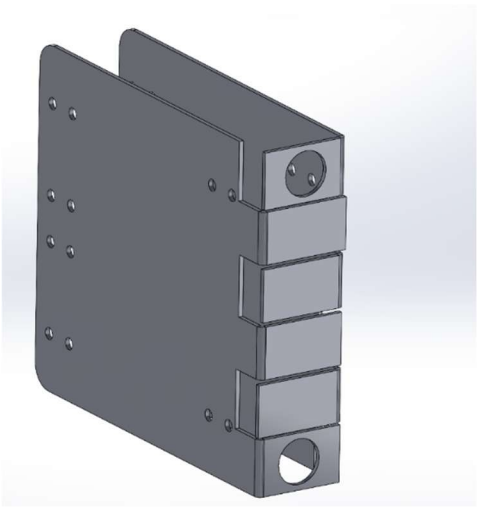


Figure 16: Square teeth – front view

Both of these designs left a circular hole for the “rope pusher” actuator, designed to push the ROV back from the rope, prying whatever mussels the claw has gripped with it. In later designs it was decided to mount the pusher below the claw and have complete clearance, so the circular hole was excluded from the design.

These designs were with a manual grabber – Figure 17 – in a marina environment where mussels were located on pylons.



Figure 17: Manual mussel grabber

Results from this testing demonstrated that the triangular claw was much more effective in detaching mussels from their substrate. These were mostly singular mussels attached directly to the pylon. Further testing in Pigeon Bay, on a commercial mussel farm, confirmed that the triangular claws were more effective in detaching clumps of mussels compared to the square teeth. One major issue found was that the 90-degree bend was unable to reach around large clumps and thus the teeth could not secure themselves to the base of the mussels. Beyond this, the 90-degree bend had the potential to unintentionally grip the rope which would result in the ROV getting stuck.

The next iteration built on these results from testing, including two additional 50-degree bends in the flat plate to make the claw more 'rounded', resembling the way a human hand negotiates mussels. The triangular teeth were chosen over the square teeth for the additional grip, and were angled 20 degrees inwards, such that the rope would not get caught during use. This iteration can be seen in Figure 18, with a drawing of the jaws in Appendix F.

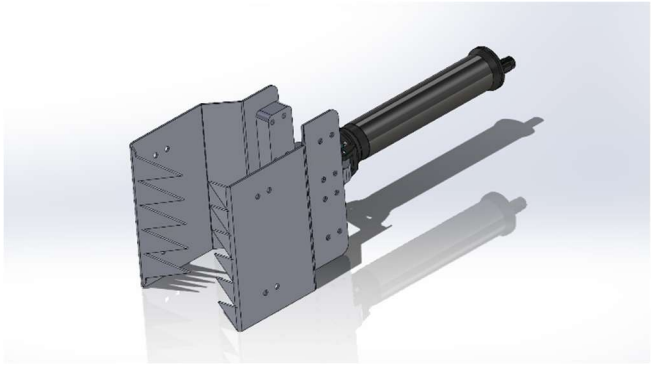


Figure 18: The redesigned set of jaws

This design was manufactured out of 1.6 mm stainless steel sheet metal for additional strength, as the previous aluminium design had begun to deform during field testing. This design is intended to be tested in a near-future boat trip.

4.4.2 Rope pusher

Working from the initial concept, several weeks of iterative CAD work eventually led to the completed rope pusher design as presented in Figure 19.

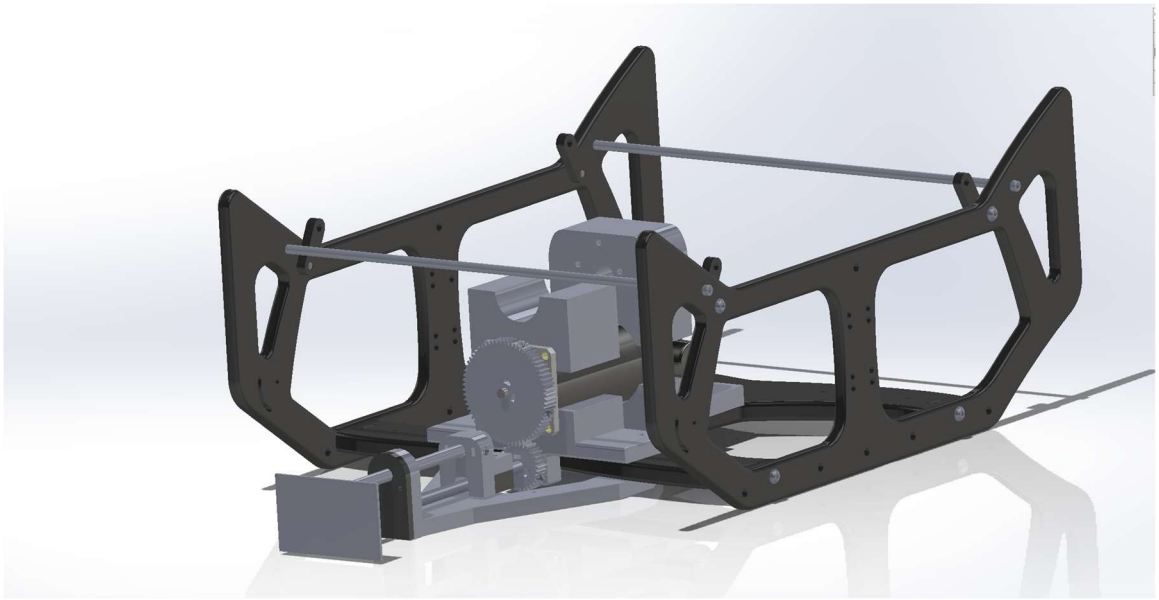


Figure 19: A CAD model of the finished rope pusher design

The design, in principle, is based upon a waterproofed stepper motor, linked to a leadscrew by a pair of gears, which actuates a steel rod and flat plate (of the same material) to push against the mussel rope.

To begin, a stepper motor was selected for its precise and repeatable control of rotational position – and hence horizontal leadscrew position. A major concern was the ability of the rope pusher to return to a neutral home position after actuating against the rope; a stepper motor allows the steps taken forward to simply be taken in reverse, returning the rope pusher to its original position. A pre-made linear actuator was considered as an alternative, though underwater linear actuators are a niche product, and none could be identified within a reasonable price range. The stepper motor needed to be IP68 rated, in order to survive immersion below a metre; capable of sufficient torque to drive the rope pusher; and compatible with the ROV's 16 V power supply – which made identifying a suitable model challenging. A drylin® E NEMA23 stepper motor with water protection was eventually selected and purchased through the client. One-quarter microstepping was used to smooth the motion and reduce noise upon client suggestion.

Considering now the gears, the principal concern was offsetting the axial force developed by the rope pusher from the sensitive axle of the stepper motor. Gears accomplished this conveniently, while also affording additional control over the torque developed by the motor – initially, they were set to increase the torque at the leadscrew; following testing, the client requested faster actuation, and the gear ratio was inverted to accomplish this. The gears were printed out of PETG for additional strength over PLA, and integrated set screws were used to constrain them against the leadscrew and motor axle. The leadscrew itself was made of stainless steel and had a pitch of 2 mm.

The rope pusher was constructed from a simple 8 mm steel rod welded to a flat plate. The flat plate was incorporated to provide a wider base with which to push against the rope, reducing the possibility of slipping. The force output of the system was trialled in lab, using a set of weights alongside a simple basket of heavy rocks – weighed after the fact. An upper push-force limit of around 100 N was established, which is on the lower end of suitable: future work could look to improve this by adjusting micro-stepping, gear ratio, or leadscrew pitch. A field test of the rope pusher design is planned alongside the jaws at a future date.

4.4.3 Deliverables

The main deliverable in this project is the functional prototype of the collection mechanism. While the final prototype is currently untested, the design is an iteration of previous designs, accounting for the observations seen in previous testing. This payload includes the grabber arm, with stainless steel claw, stainless steel pusher and pusher mounting system, printed gears, and motor mount. The entire design can be seen in Figures 20 and 21.

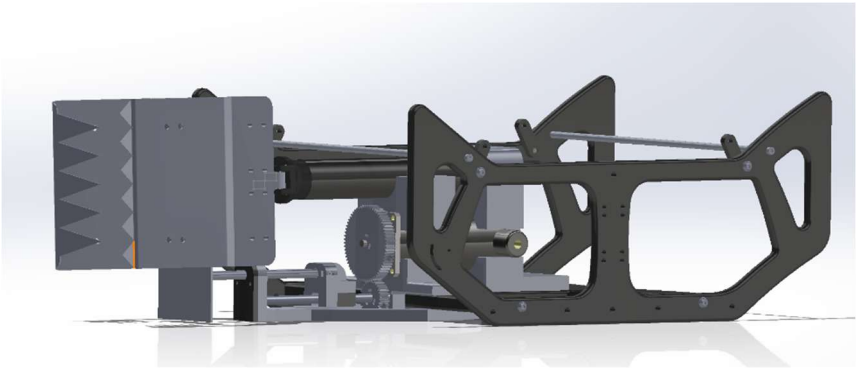


Figure 20: CAD assembly of grabber system

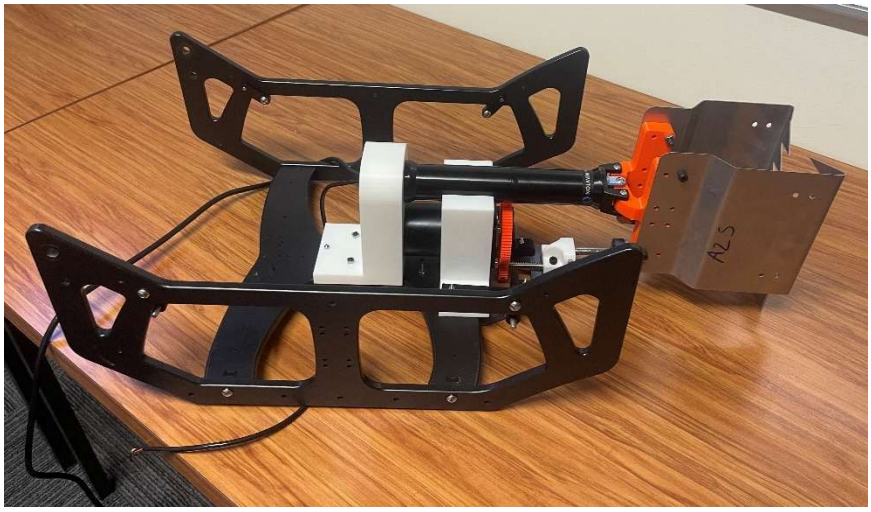


Figure 21: Manufactured assembly of grabber system

As well as the prototype, CAD files of every individual part will be supplied to the client, alongside a document outlining the electronic wiring and code to control the motor and arm.

The final payload attaches to the bottom of the ROV, as seen in Figure 22:



Figure 22: Grabber payload attached to UC Vision's underwater drone

5 Discussion

5.1 Findings

The project developed and evaluated components intended to improve the ROV's ability to locate and collect mussels autonomously. The results demonstrate improvements in prediction, image processing, tether management, and collection-mechanism design. However, the components were assessed separately, so their contribution to a complete sampling operation has not yet been established.

5.1.1 Mussel position prediction

The prediction work showed that reliable evaluation and tracking data were more important than increasing model complexity. Correcting the interim benchmark removed the previously reported advantage of the original UKF. The revised filter produced modest average improvements over holding the last known position, with greater benefits where motion persisted in one direction. Performance deteriorated on less predictable movement, and settings did not transfer reliably between scans.

These findings suggest that prediction should be used selectively rather than assumed to improve every forecast. The cluster-targeting and lull-detection results also indicated that selecting a stable target and a suitable moment to grab could complement position forecasting.

5.1.2 Minimising turbidity noise

The underwater image enhancement appeared to improve the visibility of mussel ropes both visually and quantitatively with the analyses of the UCIQE and UIQM algorithms. Depth-based segmentation was a much more computationally efficient method of segmentation than the SAM methods, however, SAM3 provided a much more accurate segmentation. This was largely due to noisy depth values in the sensor.

Despite the image enhancement and depth segmentation processing fast enough to work in real-time separately (below the camera's 15 FPS), when both pipelines were combined, they were too slow.

5.1.3 Automated tether management

The powered reel helped the ROV pull tether out during scanning and return to the boat after surfacing. Testing established suitable settings for mussel-farm operation. Moving the control logic to the ESP32 allowed further improvements, and correcting the ESC's response to a zero-speed command stopped the repeated switching between retract and hold. The final controller then behaved as intended during qualitative bench testing.

ESC failure prevented quantitative assessment and automatic testing in water. Current also varied with reel resistance, while longer seabed deployments caused more tangling and made retrieval harder. These findings showed that improving the controller alone would not remove the need for an operator to guide the tether. Further testing of the final controller and revised assembly, together with better cable guidance, is needed before the system can manage the tether unattended.

5.1.4 Grabber and collection system

Initial testing showed that tooth geometry and access around the mussel cluster affected grip. Triangular teeth performed better than square teeth, while difficulty reaching around larger clusters prompted the development of a 'rounded' redesign. This demonstrated the value of testing against actual growing conditions, where access and attachment differed from simplified design assumptions.

The rope pusher produced approximately 100 N during laboratory testing. The detachment force study in Appendix D recorded individual forces up to 105.32 N, but a mean detachment force of 67.58 N in the farming environment. 100 N appears broadly suitable, though this study also outlines the high discrepancies between samples due to environmental factors. Testing the assembled jaws and pusher is therefore necessary to establish whether sufficient detachment force can be achieved from this design.

5.2 Implications

For the client, the main contribution is a clearer understanding of which developments offer practical benefits and which remain barriers to reliable sampling. The powered reel reduced tether resistance during payout and helped return the ROV to the boat under manual control; the prediction and image-processing software provide components for further integration; while the revised claw and rope pusher provide a physical collection prototype. These outcomes support continued development, but do not yet demonstrate reduced sampling time, staffing requirements, or operating cost.

The findings also show that the workstreams are closely connected. Prediction depends on reliable positions and track identities; segmentation depends on usable imagery or depth data; processing delay affects how current the position information remains; tether forces can disturb vehicle motion, changing the target's position relative to the camera and grabber. Improvements to one component therefore need to be assessed within the complete system.

Collecting a mussel cluster could reduce reliance on maintaining precise tracking for one mussel. The smoother common-ID displacement target identified in Section 4.1 may be useful for this approach. However, collecting a larger group also changes gripping and detachment requirements. These changes align with the modifications and testing conducted by the grabber arm sub-project.

5.3 Limitations

The prediction assessment used few recordings, all replayed offline. The 2023 scans have no timestamps, so horizons in seconds are estimated, and the only timed recording lasted about 4 s. The filter was tuned and tested on the same scans, and its settings did not transfer reliably between them. The results therefore do not establish reliable prediction 5–10 s ahead.

While image filtering metrics indicated an improvement in visibility, this could not be confirmed without a completely denoised image, which would be extremely difficult to obtain. Processing measurements were obtained on a computer using recorded footage rather than on the ROV. Therefore, the performance on the ROV is unclear. Both the SAM2 and SAM3 algorithms were hugely limited by processing time and lack of training. This resulted in them taking a large amount of time to run and sometimes not working entirely.

Tether field trials used manual control because the external supply prevented ESP32 operation. Later ESC failure prevented quantitative final-controller assessment, and a replacement could not be procured before reporting. The revised mechanical assembly consequently remained untested under power and at sea. A separate boat trip for mechanical testing alone was not considered justified without also assessing the final controller. Workshop availability also prevented manufacture of the metal torque-transfer adapter.

The final jaws and rope pusher have not undergone field testing at the time of reporting. Laboratory pusher tests provided initial evidence of viability but will require in field testing to confirm reliable and consistent results.

5.4 Future work

Further work is needed across the project to improve the reliability of the complete system. For tether management, the replacement ESC needs to be installed, the torque-transfer adapter manufactured and fitted, and the final controller tested with the revised assembly. Testing should record motor current, control state and speed alongside observations of tether and ROV movement.

The planned collection testing should assess whether the revised claws successfully detach mussel clusters. Also, the speed of detachment and effectiveness of the pusher system will need to be observed to establish whether changes to the gear ratio are necessary.

Further perception testing should use longer recordings with known timestamps, ROV navigation data, stable tracking identities and representative underwater conditions. Prediction settings should be evaluated on recordings excluded from tuning. SAM3 should attempt to be improved through training locally to increase speed and accuracy. Alternatively, the depth segmentation method must be improved to meet the 15 FPS threshold. The complete processing pipeline then needs to be timed on the ROV's hardware.

Once the components are sufficiently reliable, an integrated sampling trial should measure collection success, cycle time, operator intervention and energy use. This would establish whether the combined developments meet the client's operational objectives. Improved cable distribution, potentially through a level-wind mechanism, and protection of exposed motor components would also support longer and less supervised deployments.

6 Conclusions

This project designed and evaluated improvements to UC Vision's ROV, further developing mussel position prediction, underwater image processing, tether management, and the collection mechanism. The work produced software and physical prototypes, as well as identifying the remaining barriers for future work.

The revised position predictor was slightly more accurate than holding the last known position, with gains depending on the motion and tracking data. Targeting the cluster and timing the grab to calm periods also reduced target movement. Image enhancement improved contrast and colour characteristics, but segmentation remained inconsistent, and the complete segmentation and enhancement pipeline did not meet the camera's 15 FPS processing rate.

The tether retrofit demonstrated powered payout and retrieval during field operation. Suitable mussel-farm settings were established experimentally, and the final automatic controller demonstrated the intended behaviour during qualitative bench testing. Automatic in-water operation remained unverified following ESC failure. Field trials also identified cable distribution and mechanical durability as requirements for reducing operator involvement.

The claws of the collection mechanism were redesigned with the intent to grip larger mussel clusters for improved sampling, while the rope pusher demonstrated approximately 100 N of force in laboratory tests. Field validation is still required to establish whether the assembled mechanism can consistently grip and detach samples, and is planned for a near-future boat trip.

The project therefore provides a strong basis for further development. The next priority is to complete component validation and then conduct integrated trials under mussel-farm conditions to measure collection success.

Declarations

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Conflict of interest statement: The authors declare that there are no financial conflicts of interest regarding this work.

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AI usage: AI tools, namely ChatGPT6.1 (OpenAI) and Copilot (Microsoft) were used to assist with research, broadening knowledge, and clarifying or improving human written text. Grammarly was used for spell-checking, grammar and improving the readability of the original text. All AI outputs were verified, and all ideas, opinions, and analysis in this report are the authors' own.

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Appendix A - Mussel Position

ENME408-2026 Individual Final Report

Predicting Mussel Motion for Autonomous ROV Mussel Sampling

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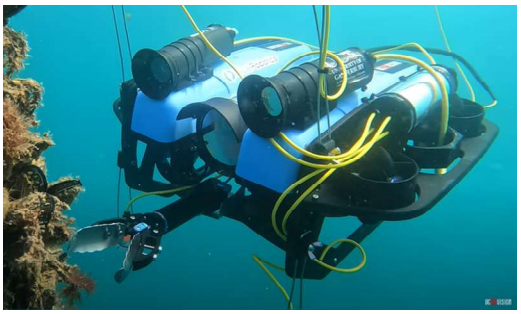
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Abstract

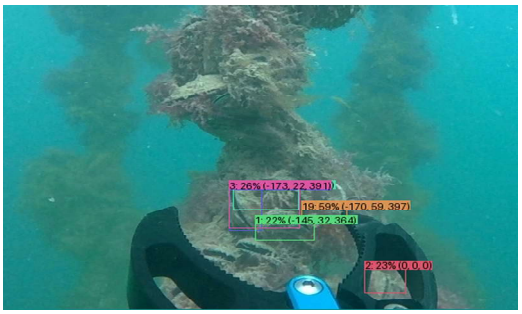
Context: UC Vision and UWAI Robotics are developing a stereo-camera remotely operated vehicle (ROV) to sample mussels from farm dropper lines. Automating the grab requires knowing where the target mussels will be when the grabber arrives, a nominal 5–10 s after it commits.
Objective: Develop a lightweight, real-time predictor using only the ROV's stereo tracking that is consistently more accurate than a zero-order hold (ZOH), which assumes the target does not move.
Method: The interim unscented Kalman filter (UKF) was diagnosed and rebuilt with a damped-velocity motion model, measured noise levels, an outlier gate and adaptive process noise. It was scored against the ZOH on four 2023 scans and one 15 fps scan, with dropouts and depth outliers excluded, at horizons of 1–50 frames.
Findings: Ten frames ahead, the interim UKF was 70 % worse than the ZOH; the finalised filter was 3.0 % better. Averaged over the four scans, it was better than the ZOH at every horizon, though by at most 6.4 %, with gains of up to 16 % on scans with persistent motion but none on random-walk motion. At 15 fps it was 11–19 % better between 0.13 and 1.3 s ahead. Measured noise levels and the damped velocity made the difference. Committing to a grab when the target was slow reduced later movement by 5–18 %.
Conclusions: The finalised UKF is recommended for alignment, with a cluster target built from mussels in view and a calm-window gate. Gains at 5–10 s are small; timestamps and ROV motion should be recorded before relying on longer horizons.

Graphical abstract

Predicting mussel motion for autonomous ROV sampling



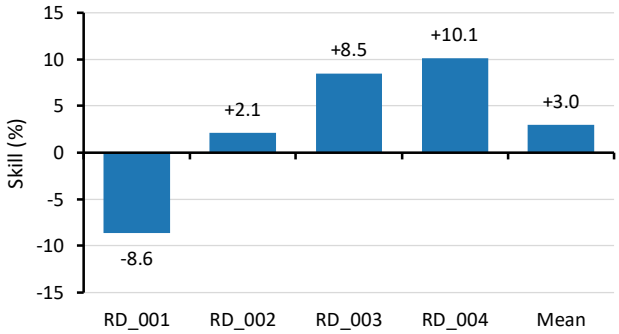
The ROV has to aim its grab at mussels that sway with waves and current



A stereo tracker gives the 3D position of each mussel in every frame

Unscented Kalman filter for each mussel
Damped-velocity motion model, measured noise and an outlier gate; forecasts h frames ahead

Finalised UKF vs zero-order hold, 10 frames ahead



Mean of the four scans, 10 frames ahead:

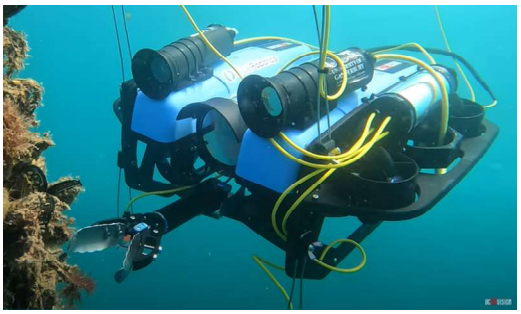
Interim UKF: 70 % worse than the hold

Finalised UKF: 3.0 % better

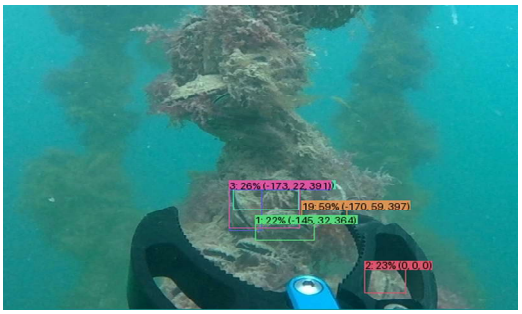
Largest gains where the motion is persistent; none on random-walk motion (RD_001).

Recommendation: align on the predicted cluster position and commit to the grab in a calm window.

Predicting mussel motion for autonomous ROV sampling



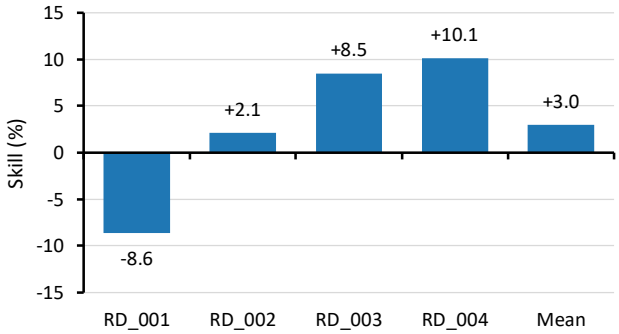
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Largest gains where the motion is persistent; none on random-walk motion (RD_001).

Recommendation: align on the predicted cluster position and commit to the grab in a calm window.

1 Introduction

1.1 Context

The University of Canterbury Computer Vision group (UC Vision) and its industry partner UWAI Robotics are developing a remotely operated vehicle (ROV) that collects mussel samples from the dropper lines of longline mussel farms (Figure 1) [1]. Farmers sample lines through the season to monitor growth and condition before harvest. Sampling currently relies on a crane on a large vessel to lift dropper lines out of the water, which is costly and labour-intensive; an ROV operated from a small boat would reduce both. Aquaculture is a growing share of world seafood production [2], so cheaper sampling has wide value.

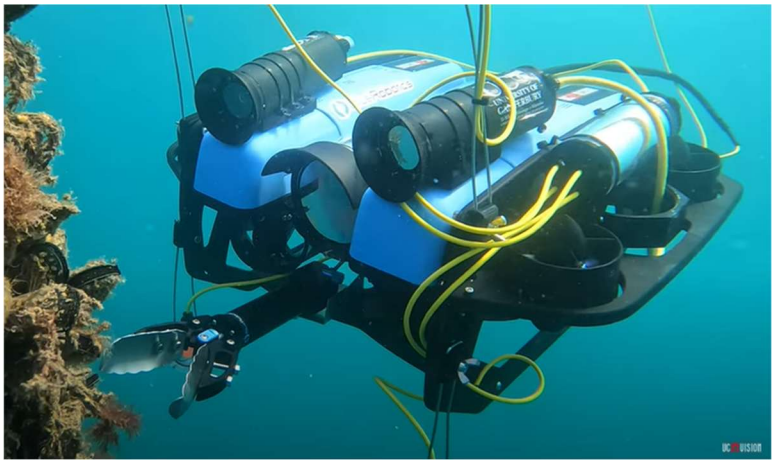


Figure 1: UC Vision mussel-rope scanning ROV. From: https://ucvision.org.nz/aquaculture/

The ROV carries a stereo camera. UC Vision's detector and tracker report, for every detected mussel in every frame, its three-dimensional position relative to the camera and a mussel instance identifier (ID) (Figure 2). The ROV can already follow a line and grab mussels under manual control. Within the wider M26 project, team members are redesigning the grabber, automating the tether and optimising the vision software; this sub-project addresses predicting where the target mussels will be.

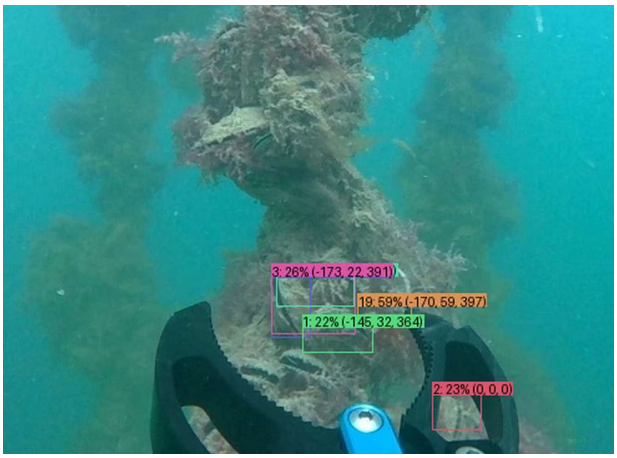


Figure 2: Tracker output on a 2023 scan (supplied by P. Nguyen). Each box shows ID, detection confidence and camera-frame position (mm).

1.2 Problem statement

Dropper lines move. Waves, currents and the ROV's own motion make mussels sway relative to the camera by several centimetres within seconds (Section 3.3). A grab takes time: the client estimated that the grabber needs to aim 5–10 s ahead of the moment it commits. If it aims at the target's current position, all motion during that interval becomes positioning error. The problem addressed is therefore: given only the stream of stereo-tracked positions available on the ROV, how accurately can the position of a target mussel or cluster be predicted over the grab horizon, and is the prediction better than assuming the target stays where it is?

1.3 Motivation

Manual alignment is slow and tedious, and it is the step that prevents the ROV from replacing crane sampling. A controller that only knows the current position of a mussel must chase it; one that knows where the mussel is going can lead it, and can wait for the moment when it is moving least. Even a modest improvement over holding the last position reduces the error the grabber has to absorb, and a predictor that is never worse than the hold can be adopted without risk to the rest of the system.

1.4 Scope and structure

The project began as hydrodynamic modelling of dropper-line motion from swell and tide [3]. Without access to the ROV simulator or site measurements, and with tracked positions available from the ROV itself, the scope moved in April 2026 to data-driven prediction. The interim (work-in-progress, WIP) report selected an unscented Kalman filter (UKF), reporting a 4.6 % improvement over a zero-order hold (ZOH). A verification study in October reproduced that result, showed that it came from how it was measured, and set out a corrected evaluation protocol under which no method, including the interim UKF, beat the ZOH. The interim filter was then diagnosed and rebuilt.

This report presents the finalised filter, evaluates it under the corrected protocol, identifies which changes produce its accuracy, and evaluates two supporting strategies: cluster targeting and timing the grab to a calm moment. Section 2 reviews the literature, Section 3 describes the data, the filter and the evaluation, Section 4 gives the results and Sections 5 and 6 discuss them.

2 Literature

2.1 Why dropper lines move

In a longline farm, vertical dropper lines hang from a buoyed backbone line anchored at both ends [2]. As mussels grow, the droppers present a large drag area. Plew [3] showed that farms measurably slow and divert currents, and field measurements at an offshore farm confirmed reduced currents inside the farm and accelerated flow around it [4]. These studies describe mean flow over minutes to days; none gives the second-by-second motion of one dropper relative to a moving vehicle, which is what a grab needs. Typically, the period of the waves sets the time scale. At the Canterbury wave buoy off Banks Peninsula, the dominant wave period between 26 February and 26 March 2026 had a median of 7.3 s [5]. Sheltered farm sites will differ, but a 5–10 s horizon is of the order of one full wave period.

2.2 ROVs and grasping moving targets

ROVs routinely hold station and manipulate objects, but wave and current disturbances make precise station keeping difficult, and manipulation from a floating base couples vehicle and arm dynamics [6]. Grasping moving objects in industrial settings succeeds by predicting the object's pose a short time ahead and replanning continuously [7]; the objects there move smoothly and are observed from a fixed camera. The ROV case is harder on both counts; the target moves irregularly, and the camera moves with the vehicle.

2.3 Predicting wave-driven motion

The closest precedents are wave and ship-motion prediction. Autoregressive (AR) models predict swell elevation accurately up to about two wave periods ahead, but less so for broader-banded wind seas [8]. A review of ship-motion prediction found that methods using only past motion predict about 10–15 s, and that for operations such as landing the useful output is often the quiescent period, a window in which motion stays below operational limits [9]. Closest to this project, Rycroft Sinclair et al. [10] compared predictors for swaying tree branches tracked by stereo vision and found AR models most reliable, spectral models poorly suited when motion periods exceed the analysis window, and all methods degraded by tracking noise and missed frames. Conventional DFT-based extrapolation treats the record as one period of a repeating signal, a limitation that super-resolution methods try to overcome [11]. Table 1 summarises the horizons reported.

Table 1: Prediction horizons reported for related problems, and the conditions behind them.

Study	Target and data	Best method	Useful horizon
Fusco and Ringwood [8]	Sea-surface elevation at a fixed point; long, uniformly sampled records	Autoregressive model	Up to about two wave periods (swell); shorter for wind seas
Cademartori et al. [9] (review)	Ship motions from past motion only, or from wave measurements	Various; prediction then quiescent-period detection	About 10–15 s; about 1 min with wave data
Rycroft Sinclair et al. [10]	Tree branches tracked by stereo camera	Autoregressive model	Short-term (about 1 s evaluated)
This work	Mussels on dropper lines, camera on moving ROV; short, gappy tracks, nominal timing	Damped-velocity Kalman filter	Gains of 0.9–4.0 % at 10–20 frames (nominal 5–10 s); 11–19 % up to 1.3 s at 15 fps

These successes share three conditions: long, uninterrupted and uniformly sampled records, narrow-band motion of a large, inertial body, and measurements in an earth- or vessel-fixed frame. The ROV data violate all three: tracks are short and gappy, frame timing is irregular, and positions are relative to a camera that itself moves. The literature therefore suggests that useful horizons will be a fraction of a dominant period, and it points to an alternative: predicting when motion will be small.

2.4 State estimation for moving targets

The Kalman filter [12] is the standard recursive estimator for moving targets, and its forecasts depend mainly on the motion model. Constant-position (random-walk) and constant-velocity (CV) models are the simplest. Singer [13] modelled manoeuvres as an exponentially correlated acceleration; the same idea applied to velocity gives a damped-velocity model, in which a forecast follows the current motion for a characteristic time and then levels off. Li and Jilkov [14] survey these and related models. The unscented Kalman filter [15, 16] propagates sigma points through non-linear models. With a linear model, it reproduces the Kalman filter. FilterPy [17] implements both.

Two practices are well established. Noise covariances should be identified from data rather than assumed, and can be adapted online from the innovations, the differences between each measurement and its prediction [18]. Filter consistency should be checked with the normalised innovation squared (NIS), which also provides a statistical gate for rejecting outliers [19]. The interim UKF used a CV model with assumed noise levels and neither practice.

2.5 Evaluating forecasts

A forecaster is effective only if it beats the naive forecast that the next value equals the last one, and accuracy should be scaled against that forecast [20]. For this application, the naive forecast is the ZOH, which is what an aligning controller does by default when it holds the last measured position. Comparisons are fair only when every method is scored on the same forecast origins and targets, and when its parameters are not tuned on the data used to score it. The interim benchmark met neither condition.

2.6 Research gaps

- No published evaluation exists of short-horizon motion prediction for targets on aquaculture dropper lines viewed from an ROV-mounted camera.
- It is not known which motion model suits the camera-frame motion of swaying, densely packed targets with fragmented IDs, irregular sampling and depth errors, or how far ahead such motion can usefully be predicted.
- Timing manipulation to calm periods, the quiescent-period idea from ship operations, has not been applied to underwater sampling.

3 Method

3.1 Research objectives

The interim objective required a model that is “provably accurate and computationally lightweight”. It is refined here into four measurable objectives; O3 and O4 were added as the scope evolved.

- O1 – Accuracy: Develop a predictor, applied causally to the ROV's stereo-tracking data, that is consistently more accurate than the ZOH, measured by root-mean-square error (RMSE), effective at horizons up to 5–10 s. In this instance, consistently means positive skill on average and on most scans.
- O2 – Cost: Show that its computation per mussel per frame allows real-time use for the number of mussels detected at the camera frame rate.
- O3 – Understanding: Explain which design choices produce the accuracy observed, and where prediction does not help, so the client knows what would have to change.
- O4 – Alternatives: Evaluate cluster targeting and lull timing as ways to reduce the effect of target motion on the grab.

3.2 Research design

The work proceeded in four stages. Spectral and polynomial predictors were built in May and June and replaced by the UKF in July. In October, the interim benchmark was audited and a

corrected evaluation protocol established. The interim UKF was then diagnosed against that protocol and rebuilt (Section 3.4). Finally, the finalised filter and the two supporting strategies were evaluated. Every analysis streams the recorded data one frame at a time, so each method sees only past and present frames, as it would on the ROV. Figure 3 shows the runtime pipeline the finalised filter sits in. Appendix C compares the plan with what happened, and Appendix D gives the hazard assessment for the planned field tests.

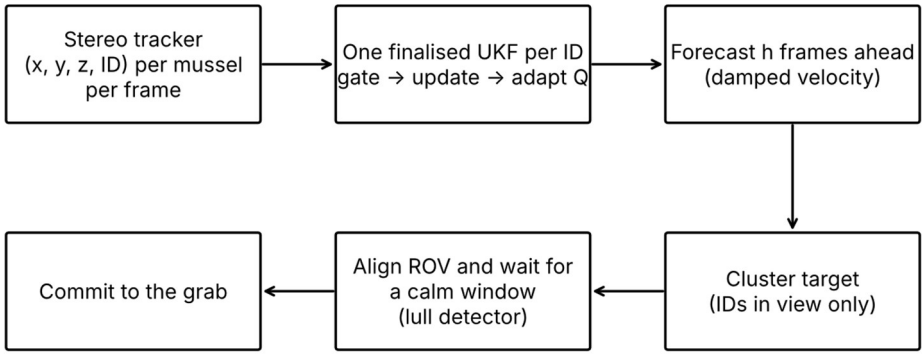


Figure 3: Runtime pipeline. One finalised UKF runs per tracked ID; its forecasts feed a cluster target, and the grab is committed only when the ROV is aligned and the cluster is calm.

The interim report compared three predictors ten frames ahead, a nominal 5 s at an assumed 0.5 s per frame. A streaming fast Fourier transform (FFT) predictor, which removed small spectral components from a 10-sample buffer and extended the result periodically, had a mean RMSE of 0.305 m and was 626 % worse than the hold used at the time. First- and second-order polynomial extrapolation through the same buffer had mean RMSEs of 0.147 m and 0.566 m (97 % and 756 % worse). Both families assume that a short buffer holds a representative piece of the motion, which short, gappy tracks rarely provide [10], and both amplify jumps and glitches. They were retired after the interim report and are not evaluated further.

3.3 Data

Five tracks were analysed (Table 2). RD_001–RD_004 come from 2023 grab scans supplied by P. Nguyen as camera-frame positions in metres. Following her guidance, only the first 80 rows of RD_001 were used (later rows are not consecutive frames), IDs 1 and 15 of RD_003 were treated as the same mussel, and the ID-0 rows of RD_004 were treated as frames in which mussel 3 was not detected. TF_260721 is the only scan recorded at a known frame rate (15 fps, July 2026). It contains 60 frames with up to 29 detections each; rows were mapped back to their frames from the tracker printout. One mussel, identified by P. Nguyen under IDs 1, 8, 36, 86 and 149, was stitched into a single track.

The 2023 scans carry no timestamps. Frames were saved when the strobe-lit image was bright, not at a fixed rate, and file numbers advance in steps of three. The tracking video supplied with the scans plays at 2 frames per second (353 frames over 176.5 s), which matches the 0.5 s interval assumed in the interim work, but it shows the saved frames, not when they were captured. Results are therefore reported in frames, and seconds are nominal values at 0.5 s per frame. The finalised filter needs a time step because its damping is set in seconds; 0.5 s was used for the 2023 scans, and 1/15 s for the 15-fps scan.

A row of exactly (0, 0, 0) means no measurement. A second failure mode is a depth error, which moves the stereo point along the camera ray so that x, y and z all scale by the same factor

(RD_002 frame 538 collapsed to 4 mm from the camera). A sample was flagged as a depth outlier when its range differed by more than 12 % from the median range of measured samples within ±10 frames, while its nearest unflagged neighbours on both sides did not. Measurement noise was estimated robustly from second differences of consecutive valid samples: 0.3–1.9 mm per axis.

Table 2: Tracked scans used. Valid frames exclude dropouts and depth outliers. RMS step is the root-mean-square movement between consecutive frames.

Scan	Frames	Valid	Dropouts	Longest gap	Depth outliers	Range x / y / z (mm)	RMS step (mm)
RD_001	80	69	9 (11 %)	7	2	21 / 23 / 22	4.4
RD_002	543	538	1 (0.2 %)	1	4	140 / 54 / 226	2.5
RD_003	115	111	4 (3.5 %)	2	0	106 / 36 / 165	3.0
RD_004	248	164	82 (33 %)	26	2	118 / 171 / 193	4.4
TF_260721	60	55	5 (8.3 %)	3	0	199 / 150 / 49	13.9

3.4 Finalised prediction filter

The finalised filter is a drop-in replacement for the interim class: one UKF instance per tracked ID, the same constructor and the same predictPositionUKF() call. It keeps FilterPy's UKF and the state $\mathbf{x} = [x, y, z, v_x, v_y, v_z]^T$, but makes six changes, each a response to a fault found in the interim filter (Section 4.1).
The system now uses a damped-velocity motion model. Each axis follows

$$p_{k+1} = p_k + \tau\left(1 - e^{-\Delta t/\tau}\right)v_k, \qquad v_{k+1} = e^{-\Delta t/\tau}v_k + w_k \tag{1}$$

where p is position, v velocity, Δt the frame interval, τ the velocity time constant (1.5 s) and w_k the random change in velocity over one frame. The velocity estimate decays, so a forecast follows the current motion for about τ and then levels off instead of extrapolating a straight line. Applying (1) h times without noise gives the forecast in closed form:

$$\hat{p}(t + h\Delta t) = p(t) + \tau\left(1 - e^{-h\Delta t/\tau}\right)v(t) \tag{2}$$

So, a forecast never moves the target more than τ·v(t) from its current position. As $\tau \to \infty$, (1) becomes the CV model of the interim filter.

Process noise, Q, is the exact discretisation of white-noise acceleration with spectral density q = 10^{-3} m^2 s^{-3} driving (1), plus a random walk of density q_p = 10^{-3} m^2 s^{-1} on position to allow for slow drift of the line. It is assembled for the actual state order [x, y, z, v_x, v_y, v_z].

Measurement noise, R, has standard deviations of 1.4, 0.8 and 2.0 mm in x, y and z, at the upper end of the noise in Table 2, with depth the noisiest. The interim value, 0.1 m^2, is a standard deviation of 316 mm.

The filter starts at the first real measurement rather than at a dropout, with zero velocity and standard deviations of 10 mm and 0.1 m/s. The interim filter used 10 m^2 and 10 m^2/s^2.
For each measurement, the normalised innovation squared is

$$\varepsilon_k = v_k^T S_k^{-1} v_k \tag{3}$$

where v_k is the innovation and S_k its predicted covariance [19]. A measurement is rejected when $\varepsilon_k > 25$ and it lies more than 30 mm from the prediction; both conditions are needed, so a statistically surprising but physically small step is still accepted. If two consecutive rejected

measurements lie within 20 mm of each other, either the mussel has genuinely moved or been re-acquired, and the filter restarts there.

For a consistent three-dimensional filter, the mean of ε_k is 3. After each accepted update, a scale s on Q is nudged towards that value, following the innovation-based approach of Mehra [18]:

$$s_k = s_{k-1} \exp[\alpha(\min(\varepsilon_k/3, 10) - 1)], \qquad Q_k = s_k Q_0 \tag{4}$$

with α = 0.05, the ratio capped at 10 so one bad frame cannot inflate Q, s bounded between 0.01 and 10^4, and no adaptation for the first three updates after a restart. On the 2023 scans, s settled at 0.01–0.05 (the lower bound), and on the 15-fps scan at about 0.3–0.8, so a single fixed Q would not suit both.

FilterPy's update reuses the sigma points drawn before Q is added, so Q is missing from the innovation covariance [17]; the finalised filter redraws them from the predicted mean and covariance, as in the additive-noise UKF [16]. The forecast pushes the state mean through (1) h times, which equals (2) and the mean of repeated predict() calls, without saving and restoring the filter state.

Table 3 lists the parameters. τ and q were chosen by a grid search over seven motion models. Their sensitivity under the protocol used here is examined in Section 4.3. Because every element of the model is linear, the UKF is numerically identical to a linear Kalman filter with the same settings (forecasts agree to within 10^{-10} mm), which allowed the model comparison to use plain matrix arithmetic.

Table 3: Parameters of the finalised filter.

Parameter	Value	Basis
Velocity time constant τ	1.5 s	Grid search over 0.75, 1.5 and 3 s
Acceleration noise density q	10^{-3} m^2 s^{-3} (initial)	Grid search over 10^{-5} to 10^{-2}
Position random-walk density q_p	10^{-3} m^2 s^{-1} (initial)	Set equal to q
Measurement noise (x, y, z)	1.4, 0.8, 2.0 mm	Measured noise (Table 2)
Initial uncertainty	10 mm, 0.1 m/s	Scale of typical motion
Outlier gate	ε > 25 and distance > 30 mm	Far beyond the 99 % χ^2 bound (11.3)
Re-acquisition	2 rejected frames within 20 mm	Tracker re-acquisition behaviour
Adaptation	α = 0.05; ratio cap 10; s in [0.01, 10^4]; 3-update delay	Developer's choice
Sigma points (α, β, κ)	0.1, 2, 0	As interim filter

3.5 Evaluation protocol

The evaluation follows the corrected protocol of the verification study unchanged, so that results are comparable with it. Forecasts are issued at every valid frame from frame 10 onwards for horizons h = 1–50 frames. A forecast is scored only if both its origin t and its target t + h are valid frames, and only pairs scored for every method are kept. The error is the three-dimensional distance between forecast and measured position, and accuracy is summarised per scan and horizon by the skill relative to the ZOH [20].

$$\mathrm{SS}(h) = 1 - \frac{\mathrm{RMSE}_{\mathrm{method}}(h)}{\mathrm{RMSE}_{\mathrm{ZOH}}(h)} \tag{5}$$

So positive skill means better than holding position. Across scans, the mean skill (equal weight per scan) and the number of scans improved by more than 0.5 % are reported. The ZOH holds the last valid measurement. Because the 15 fps tracker assigns a new ID whenever it re-

acquires a mussel, that scan is scored as the runtime pipeline sees it: every ID as its own track with its own filter, pooled over all 69 IDs, with no warm-up because the median ID is detected in only 8 frames. The stitched single-mussel track is also reported.

Two further views test robustness. In the operational view, forecasts are issued at every frame, including dropout and depth-outlier frames, where the ZOH carries the last raw measurement and the filter coasts or gates. This shows the value of rejecting glitches, which the main protocol deliberately excludes. Second, the benchmark was re-run and reconciled with the main protocol (Appendix A.2). Table 4 sets out the main evaluation decisions.

Table 4: Evaluation decisions and their justification.

Decision	Alternatives considered	Reason for the choice
Baseline: hold the last valid measurement	Hold the raw current sample.	The naive persistence forecast is the standard reference for skill [20] and is what an aligning controller does by default. Holding a (0, 0, 0) dropout is not a forecast anyone would make.
Metric: 3-D RMSE, reported as skill per scan	Per-axis RMSE; mean of % improvements	The grab misses by a 3-D distance, and RMSE penalises the large misses that cause failed grabs. Skill normalises across scans with different motion.
Score only valid origins and targets, identical for every method	Each method on its own frames (interim); every frame	Differences must come from the forecasts, not from which frames were scored. A depth outlier is a sensor error, not motion; its effect on the hold is reported separately (operational view).
Data rules from P. Nguyen	All rows of every file	Later RD_001 rows are not consecutive frames; IDs 1 and 15 of RD_003 are one mussel.
Horizons in frames, 1–50	A single 5 s horizon	Frame timing is nominal for the 2023 scans. A range shows where skill appears and covers 5–10 s for intervals of 0.2 s or more.
Developer's default parameters, with sensitivity and leave-one-scan-out checks	Leave-one-scan-out tuning only	The defaults were chosen on the same scans under the developer's benchmark; the checks show how much the result depends on them.

3.6 Supporting analyses

Each change was removed from the finalised filter in turn: the damping ($\tau \to \infty$), the gate, the adaptation, and the measurement noise (restored to the interim 0.1 m^2).

Seven motion models (random walk, CV, constant acceleration, Singer, damped velocity, damped velocity with a position random walk, and a damped oscillator) were compared over the parameter grids of UKF_Model_Exploration.py, each with the same measurement noise, gate, initialisation and adaptation, and scored at 1, 5, 10 and 20 frames. The finalised filter was run for τ = 0.75, 1.5 and 3 s and q = 10^{-5} to 10^{-2}. In a leave-one-scan-out check, each 2023 scan was scored with the setting that had the highest mean skill on the other three.

Mussel ID lifetimes were measured on the 15-fps scan. The cluster target used in Runtime.py (the mean of the latest value of every ID ever registered) was compared with the mussels in view, and three alternative targets were compared by their frame-to-frame jitter: the mean of the detections, their per-axis median, and a common-ID target moved by the mean displacement of the IDs seen in both the previous and the current frame.

For each valid frame, the causal speed (path length over the last 10 frames, with at least 6 valid) was compared with the subsequent movement $D_h(t) = |p(t + h) - p(t)|$. D_h is exactly the error of committing to the current position for a grab lasting h frames. Frames in the slowest and fastest quarter of speed in each scan were compared with that scan's average frame.

4 Results

4.1 Faults with initial approach

When run through the interim evaluation code, the interim UKF reproduced the published result exactly (RD_001 −6.1 %, RD_002 +9.2 %, RD_003 +67.9 %, RD_004 −52.7 %; mean +4.6 %). That code holds dropout zeros in the baseline and scores methods on different frames. Under the corrected protocol, the same filter was 70.5 % worse than the ZOH ten frames ahead, and worse on average at every horizon (by 18 % to 128 %). Four faults explain this:

- R = 0.1 m^2 implies 316 mm of measurement noise, against about 1 mm measured. The filter gave new measurements little weight and lagged behind the mussel; the verification study found its mean NIS to be 0.0002–0.007 instead of 3.
- Q was built for the state order [x, v_x, y, v_y, z, v_z] but applied to [x, y, z, v_x, v_y, v_z], placing process noise on the wrong, cross-coupled states.
- The CV model extrapolates the current velocity over the whole horizon. Mussel sway reverses within seconds, so long forecasts overshoot (Figure 4).
- Depth errors and ID changes were accepted as measurements and turned into large false velocities.

Table 5 removes each change from the finalised filter in turn. The measurement noise matters most: restoring the interim R makes the filter 138 % worse than the ZOH at 10 frames even with every other change in place. Removing the damping (the CV model) costs most at long horizons: the mean skill falls from +3.0 % to −43 % at 10 frames and −117 % at 40, with RD_001 and RD_002 134 % and 54 % worse than the ZOH at 10 frames. Only RD_003, the steady approach, gained more without damping (+18 % against +8.5 %).

On valid origins, the gate is neutral or slightly harmful: without it, the mean skill at 10 frames is +5.4 %, and on the 15-fps scan one frame ahead, +11.6 % instead of −0.3 %, because it rejects some large frame-to-frame jumps and restarts the filter with zero velocity. Its value is operational (Section 4.3). Without adaptation, the results are similar, except at 10 frames on RD_001: after two dropouts, the wider innovation covariance let the depth glitch at frame 67 through the gate, and the false velocity produced a single forecast 256 mm wrong and a mean skill of −91 %. Adapting Q keeps the filter's uncertainty, and therefore the gate.

Table 5: Ablation of the finalised filter: mean skill relative to the ZOH over the four 2023 scans at each horizon (frames), and pooled skill on the 15-fps scan at 10 frames.

Variant	1	5	10	20	40	15 fps at 10
Finalised UKF	+6.4 %	+2.0 %	+3.0 %	+0.9 %	+0.5 %	+16 %
No damping (CV model)	+3.9 %	−27 %	−43 %	−78 %	−117 %	+15 %
No outlier gate	+6.1 %	+1.7 %	+5.4 %	+1.1 %	+0.6 %	+14 %
No adaptive Q	+8.2 %	+2.4 %	−91 %	+1.1 %	+0.5 %	+15 %
Interim R (0.1 m^2)	−913 %	−247 %	−138 %	−63 %	−25 %	−39 %
Interim UKF (as built)	−128 %	−22 %	−70 %	−29 %	−56 %	+4.5 %

4.2 Finalised filter against the hold

Table 6 and Figure 4 show the main result. Averaged over the four 2023 scans, the finalised filter was better than the ZOH at every horizon from 1 to 50 frames. The margin was small: 1.7–6.4 % up to 17 frames and 0.0–2.5 % beyond. Pooled over the scans, the RMSE ten frames

ahead fell from 14.3 mm to 13.4 mm. Three of the four scans were improved at most horizons. RD_003 and RD_004, whose motion is the most persistent (Table 2), gained the most: up to +15.6 % and +15.7 %, falling to +1.9 % and +0.2 % by 40 frames. RD_002, a slow irregular drift, gained 0.2–3.1 % at every horizon. RD_001, which wanders within about ±1 cm with no preferred direction (persistence 0.01), was made worse at almost every horizon, by up to 19.9 % at 5 frames (7.1 mm against 5.9 mm). The interim UKF, by contrast, improved on the ZOH only on RD_003 and briefly on RD_001.

Table 6: Skill relative to the ZOH at selected horizons (frames). Scan rows are the finalised UKF; per-horizon values for every scan are in Appendix A.1.

Scan	ZOH RMSE at 10 frames (mm)	1	5	10	20	40
RD_001	7.9	−3.9 %	−19.9 %	−8.6 %	−7.8 %	−1.5 %
RD_002	10.3	+1.7 %	+0.4 %	+2.1 %	+2.2 %	+1.4 %
RD_003	21.8	+12.7 %	+12.7 %	+8.5 %	+4.8 %	+1.9 %
RD_004	21.8	+15.2 %	+14.9 %	+10.1 %	+4.5 %	+0.2 %
Mean, finalised UKF	-	+6.4 %	+2.0 %	+3.0 %	+0.9 %	+0.5 %
Mean, interim UKF	-	−128 %	−22 %	−70 %	−29 %	−56 %

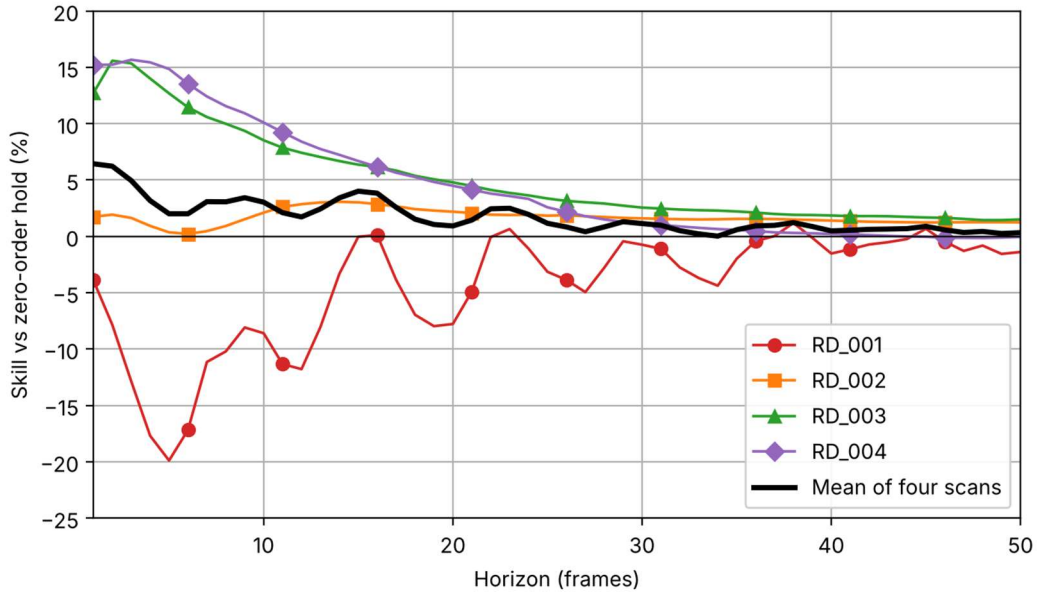


Figure 4: Skill of the finalised UKF relative to the ZOH by scan and horizon (positive is better). At a nominal 0.5 s per frame, 10–20 frames is 5–10 s.

Figure 5 shows the difference on RD_004. The interim filter's ten-frame forecasts overshoot the reversal near frame 115 and then trail far below the mussel as it rises, while the finalised filter's forecasts stay close to the measured sway. Both forecasts hold still through dropouts, when there is nothing to update them.

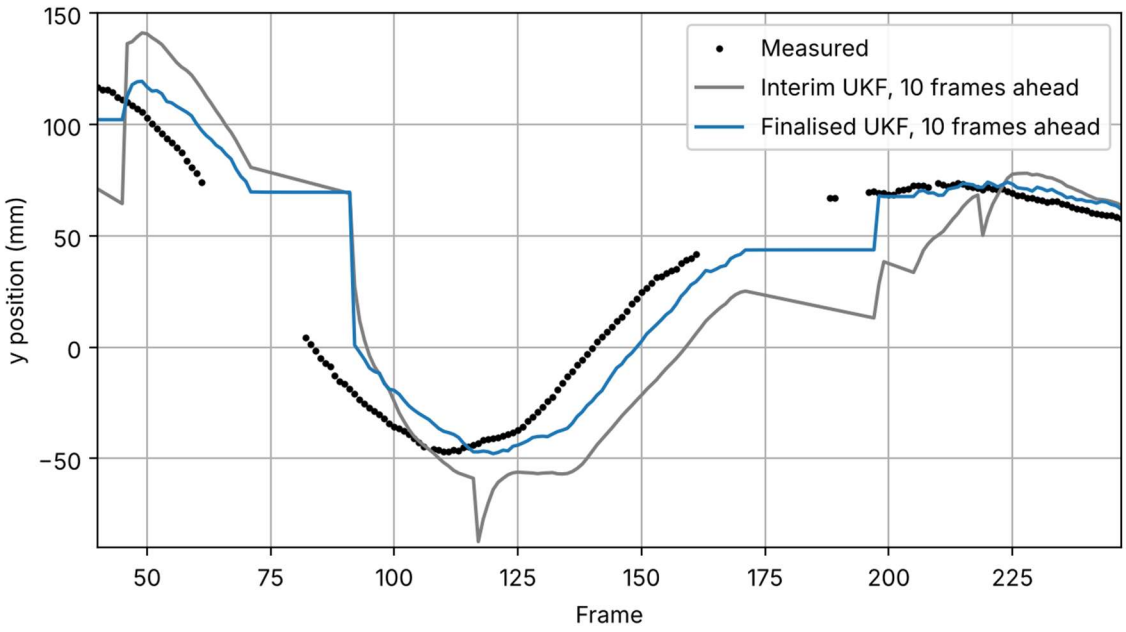


Figure 5: RD_004 y-position (dots) and the forecasts made ten frames earlier by the interim and finalised filters, plotted at the frame they forecast. Gaps are dropouts.

4.3 Robustness

Table 7 compares the seven motion models, each at its best setting on the grid and with every other part of the filter identical. Only damped velocity with a position random walk, the model used in the finalised filter, reached clearly positive mean skill. Models that extrapolate a trend over the whole horizon (CV, constant acceleration, Singer) helped on RD_003, the steady approach, but overshot badly elsewhere (mean skill −38 to −329 %). The random walk, a gated and smoothed hold, stayed within 2 % of the ZOH: it can smooth but not predict. The damped oscillator, which assumes the sway has one dominant frequency, was below the ZOH on average; with about one sway cycle per track, its frequency cannot be identified, as the September oscillator study also found.

Table 7: Motion models compared with identical noise, gate, initialisation and adaptation: best setting by mean skill over the four 2023 scans and horizons of 1, 5, 10 and 20 frames, the worst single scan-horizon value, and scans improved at 10 frames.

Motion model	Best setting	Mean skill	Worst case	Scans improved at 10
Random walk (hold)	$q = 10^{-2}$	−0.2 %	−1.7 %	0 of 4
Constant velocity	$q = 10^{-5}$	−38 %	−206 %	1 of 4
Constant acceleration	$q = 10^{-5}$	−329 %	−1843 %	0 of 4
Singer (damped acceleration)	$\tau = 0.75$ s, $q = 10^{-4}$	−75 %	−435 %	1 of 4
Damped velocity	$\tau = 3$ s, $q = 10^{-5}$	−0.1 %	−36 %	2 of 4
Damped velocity + position random walk	$\tau = 3$ s, $q = 10^{-3}$	+3.9 %	−44 %	3 of 4
Damped oscillator	$f = 0.2$ Hz, $\zeta = 1$, $q = 10^{-3}$	−2.0 %	−59 %	2 of 4
Damped velocity + position random walk, chosen	$\tau = 1.5$ s, $q = 10^{-3}$	+3.1 %	−19.9 %	3 of 4

Skill depends strongly on τ and q (Table 8). A longer time constant extrapolates further, which helps RD_003 and RD_004 (up to +23 % at 10 frames with τ = 3 s) but hurts RD_001 (down to −30 %). The largest q lets depth glitches through the gate after dropouts and produced single forecasts hundreds of millimetres wrong on RD_001. The chosen setting had a mean of +3.1 %

and was among the best on average, but not the most consistent: $\tau = 0.75$ s with $q = 10^{-3}$ kept the worst case to −5.2 % at the cost of a near-zero mean (+0.1 %).

When each scan was scored with the setting that did best on the other three (leave-one-scan-out), RD_003 kept its gains (the chosen setting was the default), and RD_002 and RD_004 kept small gains at 10–20 frames (+1.1 % to +2.8 %). RD_002, however, lost 24 % one frame ahead, and RD_001 lost 571 % at 10 frames from one glitch. The default is therefore a reasonable compromise rather than a setting shown to transfer, and its in-sample selection is a limitation (Section 5.3).

Table 8: Sensitivity of the finalised filter: mean skill over the four 2023 scans and horizons of 1, 5, 10 and 20 frames, with the worst single scan-horizon value in brackets.

	$q = 10^{-5}$	$q = 10^{-4}$	$q = 10^{-3}$	$q = 10^{-2}$
$\tau = 0.75$ s	−16 % (−125)	−7.8 % (−58)	+0.1 % (−5)	+1.3 % (−6)
$\tau = 1.5$ s	−6.5 % (−57)	−1.5 % (−16)	+3.1 % (−20)	−20 % (−386)
$\tau = 3$ s	+2.6 % (−24)	+3.2 % (−33)	+3.9 % (−44)	−31 % (−571)

When forecasts are issued at every frame, including frames whose measurement is a depth glitch, the ZOH carries the glitch, and the finalised filter rejects it. The filter's advantage then grows to 78–86 % at 1–10 frames on RD_001, +61 % one frame ahead on RD_002 and +54 % on RD_004. A gated hold (the same filter with the velocity damped within a fraction of a frame, so that it does not extrapolate) matched these gains on RD_001 and RD_004 (+80 % and +9.5 % at 10 frames, against +78 % and +7.2 %) but none on RD_003 (−1.3 %, against +8.3 % for the finalised filter). The two contributions are therefore separate: the gate removes glitches, and the damped velocity predicts.

Benchmarking the finalised filter shows that it is ahead of the ZOH on every test case at 1, 5, 10 and 20 frames, including +45.9 % on RD_001 at 10 frames (Appendix A.2). Applying the rules of the main protocol one at a time shows where the difference comes from: forecasts made at frames whose measurement is a depth glitch, which the ZOH carries. With forecasts scored only from valid frames, RD_001 returns to −14 % at 10 frames.

4.4 Timing verification

On the 15-fps scan, with one filter per ID as at runtime, the finalised filter was better than the ZOH from 2 to 20 frames (0.13–1.33 s ahead), by 11.0–18.6 % (Figure 6), with no gain one frame ahead (−0.3 %). The interim filter was worse than the ZOH except at 9–14 frames (at most +5.2 %). The ZOH error grew quickly on this scan, from 8.9 mm one frame ahead to 72.9 mm at 1 s, because the camera moved around the rope at about 210 mm/s.

Within constant-ID segments of the stitched single-mussel track, the finalised filter gained 11–29 % at 2–7 frames and nothing by 10 frames (0.67 s; 14 pairs). Across ID changes, the stitched track was harder. The three ID changes between consecutive frames shifted the reported position by 20, 26 and 62 mm (the fourth followed a dropout), against a within-ID median step of 6.8 mm. The two smaller jumps fall inside the 30 mm gate and are treated as motion, so the filter extrapolated a false velocity: under the main protocol, the stitched track was 7 % worse than the ZOH at 10 frames and 52 % worse at 20 (24 pairs). Starting a new filter for each new ID, as the runtime pipeline does, avoids this.

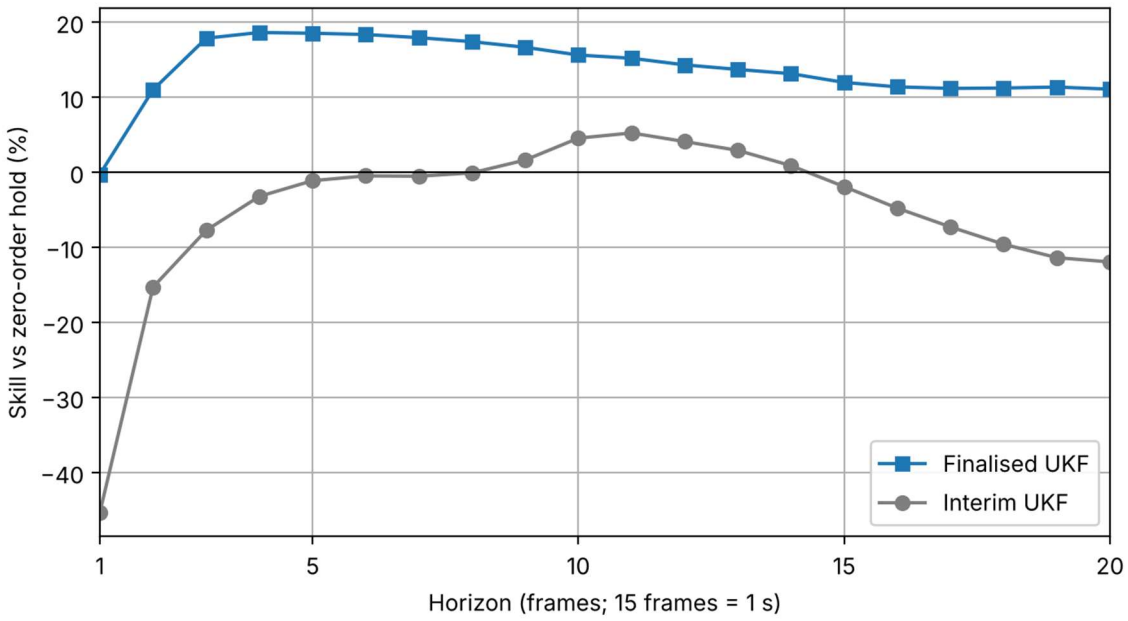


Figure 6: 15-fps scan, one filter per ID: pooled skill relative to the ZOH. Scored pairs fall from 609 at 1 frame to 73 at 20 frames.

4.5 Multiple mussels and the cluster target

ID fragmentation on the 15 fps scan was severe: 69 IDs in 60 frames. In the densely detected part of the scan (frames 31–59), there were 21 detections per frame and, after the 14 IDs that appeared together at its start, 1.6 new IDs per frame. The median ID was detected in 8 frames, and only 17 of 69 in 15 frames (1 s) or more.

Runtime.py never removes a lost ID, so by the last frame its centroid averaged 69 IDs while 19 mussels were in view, and it sat a median 22 mm (up to 49 mm) from the mean of the mussels in view. On the synthetic SD_001 stream, the gap was a median 70 mm (up to 120 mm), with departed mussels counted in 150 of 248 frames. Among the alternatives, the mean and the per-axis median of the detections jumped whenever detections appeared or disappeared (median 11.1 and 9.6 mm per frame, up to 29 and 25 mm). The common-ID target moved a median 5.5 mm per frame, close to the within-ID motion of a single mussel, with one step of 23 mm.

4.6 Lull detection

Committing when the target is slow helps (Figure 7). Over the four 2023 scans, moments in the slowest quarter of causal speed were followed by 5–18 % less movement than average over grabs of 1–40 frames (18 % over 1 frame, 10 % over 10), while moments in the fastest quarter were followed by 11–27 % more. The pooled result is dominated by RD_002, which supplies most of the frames and shows the effect consistently (10–21 % less). On RD_001 and RD_004, the benefit lasted only a few frames (27 % less over 1 frame on RD_004), and on RD_003 the steady approach makes speed uninformative. The LullDetector in the runtime pipeline applies the same idea online, flagging a calm window when the windowed speed stays below a threshold for a minimum time; the threshold depends on sea state and frame rate and has to be set for each deployment.

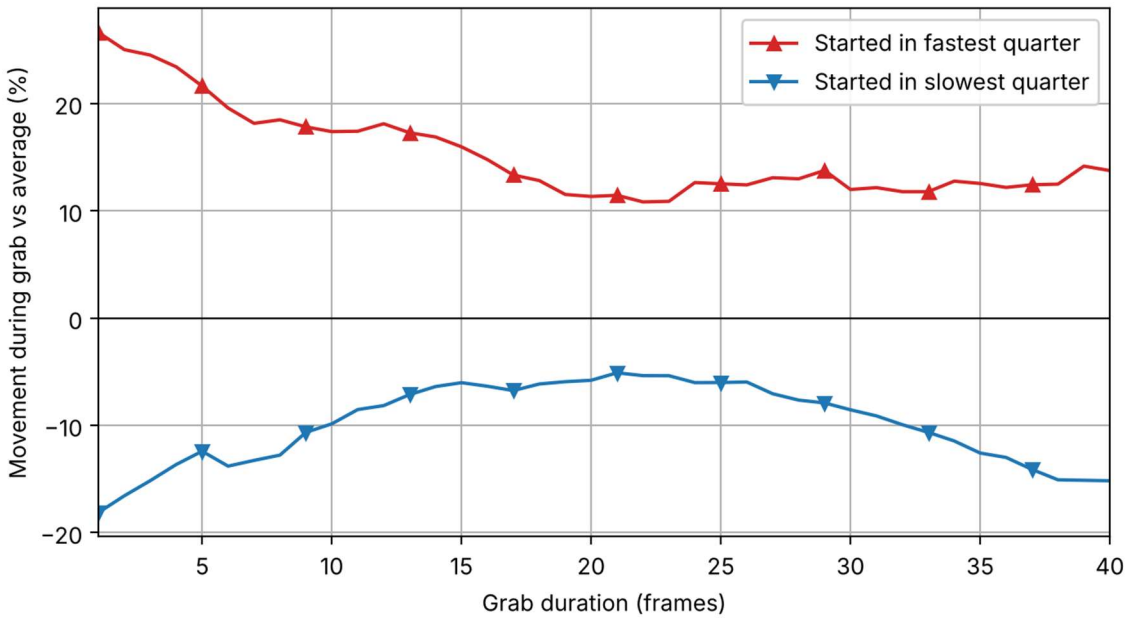


Figure 7: Movement during a grab of a given duration, relative to the average, for grabs started in the slowest and fastest quarter of causal speed (pooled over the four 2023 scans).

4.7 Computational cost

The finalised filter took about 0.34 ms per mussel per frame with a 10-frame horizon, 0.63 ms with 75 frames (5 s at 15 fps) and 0.98 ms with 150 frames (10 s), against 1.2 ms, 7.3 ms and 14.1 ms for the interim filter, whose forecast rolls the sigma points forward one step at a time. For the 21 mussels per frame seen at 15 fps, a 5 s forecast needs about 0.2 s of processor time per second with the finalised filter, against 2.3 s with the interim one. Using the closed form (2) would make the forecast cost independent of the horizon.

5 Discussion

5.1 Findings

The central finding is that the ROV's own stereo tracking can support a predictor that is, on average, never worse than holding the last position, and clearly better when the motion is persistent. The finalised UKF met the accuracy objective as defined, but narrowly: averaged over four scans its skill was positive at every horizon but never above 6.4 %, and 0.9–4.0 % in the nominal 5–10 s band (10–20 frames). This reverses the verification study's conclusion that no predictor beat the hold, and it was achieved by fixing the filter, not by adding data.

Three design choices explain the improvement, and the filter type is not one of them. First, the noise levels: with R matched to the measured millimetre-level noise, the filter follows the mussel instead of lagging behind it. Second, the damped velocity: forecasts follow the current motion for about $\tau = 1.5$ s and then level off, which suits motion that changes direction within seconds. A CV forecast assumes the motion continues for the whole horizon and is penalised every time the mussel sways back; the damped forecast never moves the target more than $\tau \cdot v$. Third, glitch handling: the gate keeps depth errors and some ID jumps out of the velocity

estimate, and adaptation keeps the gate tight on quiet scans. The unscented transform adds nothing for this linear model, which is why a linear Kalman filter reproduces it exactly.

Where prediction helps follows from the motion. Skill tracks persistence (Table 2): the scans whose successive displacements tend to point the same way (RD_003, RD_004) gained 9–10 % at 10 frames, the slow irregular drift of RD_002 about 2 %, and the random walk of RD_001 nothing. Part of the persistent motion in RD_003 is the ROV's own approach, which the vehicle knows from its commands and could remove directly. As the literature suggested, the useful horizon is short compared with the wave period: the gains are largest within a few frames and close to zero by 40 frames, a nominal 20 s or about three wave periods. On the 15-fps scan, where timing is known, the gains (11–19 % between 0.13 and 1.3 s) are larger than on the 2023 scans, but the scan lasts only 4 s, and its IDs live for about half a second, so it says nothing about 5–10 s.

Robustness is the main caveat. The defaults were chosen on the same scans; the leave-one-scan-out check shows that settings chosen on three scans do not reliably transfer to the fourth, and a single glitch that passes the gate can produce a forecast hundreds of millimetres wrong. A runtime safeguard, such as limiting how far a forecast may move the target, would cap that risk at little cost.

The supporting strategies remain useful alongside prediction. Waiting for a slow phase reduced subsequent movement by 5–18 %, comparable to the prediction gain itself, and fits the ship community's shift from trajectory prediction to quiescent-period prediction [9]. Targeting the cluster by the IDs in view, or by common-ID displacement, roughly halves target jitter and avoids the stale-ID bias of Runtime.py.

What accuracy is good enough depends on the grabber. The ZOH's error grows from about 3 mm one frame ahead to 14 mm at 10 frames and 45 mm at 40 frames (pooled), and the finalised filter trims less than 1 mm from each (14.3 to 13.4 mm at 10 frames). Combined with the capture tolerance of the redesigned grabber, this curve sets the longest acceptable time between committing and contact, so faster final approaches and more tolerant grabbers remain as valuable as better prediction.

In hindsight, the year's most useful lesson concerned evaluation. The interim result looked encouraging because of how it was measured, and the finalised filter's results also move visibly with scoring choices (Appendix A.2). Fixing the scoring rules, and checking the evaluation on cases with a known answer, should come before comparing or tuning methods.

5.2 Implications for UC Vision and UWAI Robotics

- Cap the forecast: Limit how far a forecast may move the target (for example, to τ times the recent speed) and fall back to the filtered position after a re-acquisition, so that a glitch that passes the gate cannot cause a large miss.
- Time the grab: Gate the final approach on a calm phase as well as on alignment, and prefer short, fast final approaches, which shorten the horizon that matters.
- Target the cluster in view: Average only IDs seen in the last few frames, or move the target by the mean displacement of common IDs. This also suits P. Nguyen's suggestion

of grabbing a section of rope rather than a single mussel.

- Improve the data before relying on longer horizons: Record per-frame timestamps at a fixed rate of at least 15 fps, flag failed depth explicitly, keep IDs persistent where possible, and log ROV navigation (IMU, depth, thruster commands) so that the ROV's own motion can be removed.

5.3 Limitations

The evidence base is small: four 2023 scans, of which only RD_002 is long, and a 4 s scan at a known frame rate. The filter's defaults were selected on the same scans, and the leave-one-scan-out check shows that tuning does not transfer reliably between scans, so the true out-of-sample gain may be smaller than reported. Because the 2023 scans have no timestamps, horizons in seconds are nominal, and the damping time constant is only meaningful if 0.5 s is close to the real frame interval. All positions are relative to the moving camera, so mussel motion and ROV motion cannot be separated. The reference positions are the tracker's own bounding-box centres, which shift by tens of millimetres at ID changes. The depth-outlier rule is heuristic, although it affects only eight frames, and excluding glitch frames as origins understates the operational benefit of the gate. The lull results rest on few frames outside RD_002. Nothing was tested on the ROV or in the water, and grab success, the quantity that ultimately matters, was not measured. Timings were measured on a desktop processor, not the Jetson Orin NX.

5.4 Future work

- Record longer scans at 15 fps or faster with timestamps and ROV telemetry, and re-run the benchmark (Appendix B) with leave-one-scan-out tuning.
- Transform tracks into a world frame using ROV navigation and reassess how predictable the mussels' own sway is.
- Test filters that switch between hold, damped-velocity and oscillatory behaviour, and the autoregressive models favoured by related work [8, 10].
- Develop the lull detector into quiescent-period prediction and evaluate it, with the finalised filter, by grab success in simulation and then in the field.
- Implement the finalised filter, the forecast cap and the in-view cluster target in the runtime pipeline.

6 Conclusions

This project set out to predict the position of mussels on a swaying dropper line from the ROV's stereo tracking, accurately enough and cheaply enough to time an automated grab. The accuracy target was met, narrowly: Under the corrected protocol, the finalised UKF was better than the ZOH on average at every horizon from 1 to 50 frames (by up to 6.4 %) and on three of four scans at most horizons, with up to 16 % on scans with persistent motion and 11–19 % up to 1.3 s at 15 fps. The interim UKF had been 70 % worse at 10 frames. Gains in the nominal 5–10 s band are 0.9–4.0 %. The cost target was met, at about 0.63 ms per mussel per frame for a 5 s forecast at 15 fps, or 0.2 s of processor time per second for 21 mussels, against 2.3 s for the interim filter. For the understanding target, measured noise levels and a damped-

velocity model produce the gains; glitch gating and adaptive noise make them safer; the unscented transform adds nothing. Prediction helps in proportion to the persistence of the motion, and not for random-walk motion. Finally, investigation into alternatives found committing during a slow phase reduced movement during the grab by 5–18 %, and targeting the cluster by the IDs in view roughly halved target jitter. The centroid in Runtime.py must drop lost IDs.

The project delivered a finalised prediction filter, a verified evaluation framework, and tested strategies for cluster targeting and grab timing. The recommendation to the client is to align on the finalised filter's forecast of the cluster in view, commit in a calm window, cap the forecast, and record timestamps and ROV motion before relying on prediction several seconds ahead.

Declarations

Author contributions: Alexander Pirie: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Visualisation, Writing – original draft, Writing – review & editing. **Phuong Nguyen:** Data curation (tracked-mussel datasets and ID associations). **Tim Rensen:** Resources (15 fps scan, camera and frame-timing information). **Prof. Richard Green:** Project administration, Supervision (client). **Yilei Zhang:** Supervision (academic).

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Conflict of interest: None.

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Data and code availability: Tracking data belong to UC Vision and UWAI Robotics and were used under the project's data arrangements. The analysis scripts are listed in Appendix B.

AI usage: Claude (interim report) was used to draft literature-review talking points and to debug test software. For this final report, Claude (Anthropic) was used to help audit the evaluation code, help generate figure scripts, and check references. The author checked the analysis and is responsible for the content.

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Appendix A: Supplementary results

A.1 Skill by scan and horizon

Table A1 gives the values behind Figure 5. Scored pairs at 1, 10 and 40 frames were: RD_001 56, 47, 19; RD_002 523, 516, 488; RD_003 98, 87, 60; RD_004 153, 126, 83.

Table A1: Skill of the finalised UKF relative to the ZOH by scan and horizon, with the pooled ZOH RMSE.

Horizon (frames)	RD_001	RD_002	RD_003	RD_004	Mean	Pooled ZOH RMSE (mm)
1	−3.9 %	+1.7 %	+12.7 %	+15.2 %	+6.4 %	3.0
2	−7.8 %	+1.9 %	+15.6 %	+15.2 %	+6.2 %	4.6
3	−12.9 %	+1.6 %	+15.4 %	+15.7 %	+5.0 %	6.0
4	−17.7 %	+0.9 %	+14.0 %	+15.4 %	+3.2 %	7.4
5	−19.9 %	+0.4 %	+12.7 %	+14.9 %	+2.0 %	8.7
6	−17.2 %	+0.2 %	+11.5 %	+13.5 %	+2.0 %	9.9
8	−10.2 %	+0.9 %	+10.0 %	+11.6 %	+3.1 %	12.2
10	−8.6 %	+2.1 %	+8.5 %	+10.1 %	+3.0 %	14.3
12	−11.8 %	+2.9 %	+7.4 %	+8.4 %	+1.7 %	16.5
15	0.0 %	+3.0 %	+6.4 %	+6.7 %	+4.0 %	19.6
17	−3.9 %	+2.7 %	+5.8 %	+5.7 %	+2.6 %	21.7
20	−7.8 %	+2.2 %	+4.8 %	+4.5 %	+0.9 %	24.7
25	−3.1 %	+1.8 %	+3.4 %	+2.6 %	+1.2 %	29.7
30	−0.7 %	+1.6 %	+2.6 %	+1.1 %	+1.1 %	34.9
35	−2.0 %	+1.6 %	+2.2 %	+0.6 %	+0.6 %	40.0
40	−1.5 %	+1.4 %	+1.9 %	+0.2 %	+0.5 %	45.4
45	+0.7 %	+1.2 %	+1.7 %	0.0 %	+0.9 %	50.5
50	−1.4 %	+1.2 %	+1.5 %	0.0 %	+0.3 %	54.1

A.2 Benchmark reproduced and reconciled

Benchmark_UKF_Improved.py was run unchanged. Its strict protocol treats each ID as a track, so RD_003 splits into IDs 1 and 15; it uses all 98 rows of RD_001; it issues a forecast at every frame from the first detection, including dropout and glitch frames; and it scores every non-dropout target, including depth glitches. Table A2 reproduces its output. Under the interim evaluation code, the finalised filter would score +50.6 % on average (mean RMSE 32 mm against 86 mm for the interim filter), which shows how flattering that protocol is.

Table A2: Developer's strict benchmark as run: skill relative to the ZOH, interim / finalised UKF.

Horizon	RD_001	RD_002	RD_003/1	RD_003/15	RD_004	15 fps
1	+2 / +26	+19 / +22	−81 / +16	+3 / +22	−80 / +26	−20 / +5
5	−34 / +42	−2 / +0	−6 / +14	+46 / +16	−214 / +16	+3 / +19
10	−63 / +46	−4 / +1	−7 / +8	+57 / +10	−259 / +7	+5 / +16
20	−169 / +28	−19 / +1	−35 / +3	+62 / +6	−352 / +1	−11 / +12

Table A3 applies the main protocol's rules one at a time to RD_001. Restricting RD_001 to its first 80 rows changes little; excluding glitch targets raises the skill further, because the ZOH's errors at glitch origins then dominate; restricting forecasts to valid origins removes those and brings the result below zero.

Table A3: RD_001 reconciliation: finalised UKF skill relative to the ZOH as the main protocol's rules are applied in turn (ZOH RMSE at 10 frames in brackets).

Step	1 frame	5 frames	10 frames	20 frames
R0 Developer's strict protocol (98 rows, every origin, non-dropout targets)	+26 %	+42 %	+46 % (86 mm)	+28 %
R1 + first 80 rows only	+29 %	+40 %	+31 % (51 mm)	−0.5 %
R2 + depth-outlier targets not scored	+85 %	+82 %	+76 % (38 mm)	−7.7 %
R3 + forecasts only from valid origins	−5.9 %	−26 %	−14 % (8 mm)	−7.7 %
R4 + 10-frame warm-up (main protocol)	−3.9 %	−20 %	−8.6 % (8 mm)	−7.8 %

Appendix B: Software and reproducibility

All analysis was done in Python 3.13 with NumPy 2.5, SciPy 1.18, pandas 3.0, Matplotlib 3.11 and FilterPy 1.4.5. The analysis scripts are supplied as a Final_Analysis folder that sits in the root of the project repository; they import Prediction_Support.py and Prediction_Support_Improved.py unchanged. run_all.py regenerates every number, table and figure in about a minute; numbers are written to results/summary.txt and results/results.json, and timings depend on the machine. Table B1 lists the code used.

Table B1: Software used in this report.

Script	Origin	Purpose
Prediction_Support.py	Project	Interim UKF (as built), LullDetector, streamed file input
Prediction_Support_Improved.py	Project	Finalised UKF (Section 3.4)
Test_Verification/Benchmark_UKF_Improved.py	Project	Developer's benchmark (Appendix A.2)
Test_Verification/UKF_Model_Exploration.py	Project	Motion-model grid and linear Kalman equivalents (Table 7)
Data/Real/15fps/Exctract_15fps.py; Data/Synthetic/Synthetic_Data_Tool.py	Project	15 fps printout parser; synthetic multi-mussel stream SD_001
fa_core.py	Final analysis	Data rules, depth outliers, 15 fps frame mapping, forecasters, scoring
run_analysis.py	Final analysis	Every result in Sections 3.3 and 4 and Appendix A
make_figures.py	Final analysis	Figures 3–7 and the graphical abstract
report_content.py, build_report.py	Final analysis	Report text with numbers from the results; Word file assembly
run_all.py, README.md	Final analysis	Runs every step in order; setup and a map from scripts to sections

Appendix B - Minimising Turbidity

ENME408-2026 Individual student report

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

Image filtering and denoising using computer vision with an underwater robot and mussel farming application

Stevens T.E [1]

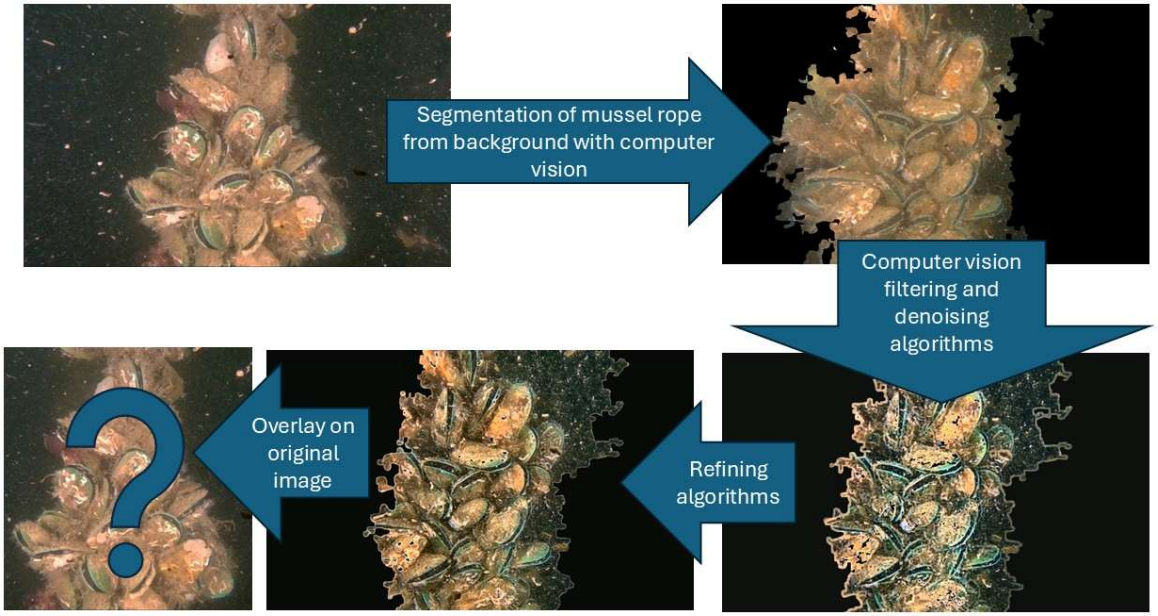
[1] Email: tst125@uclive.ac.nz

Document version: 18/10/2026

Abstract

Context: Autonomous underwater robots rely on stereo vision systems for navigation and object localisation. However, refraction, light attenuation, and scattering can reduce image quality and affect underwater computer vision. **Objectives:** This project investigated underwater stereo camera calibration methods and developed image enhancement, mussel rope segmentation, and performance evaluation techniques using footage from the University of Canterbury underwater robot. **Method:** A literature review examined underwater stereo calibration, focusing on refraction, intrinsic parameters, and checkerboard-based methods. CLAHE and UDCP were implemented in Python and tested on previously recorded underwater imagery. Depth-based segmentation using morphological filtering was developed and compared with SAM2 and SAM3 approaches. Image quality was assessed using PSNR, SSIM, UIQM, and UCIQE, while processing time and frame rate were measured for individual algorithms and the combined pipeline. **Findings:** CLAHE achieved the highest UIQM score (1.49), while UDCP achieved the highest UCIQE score (0.51). Depth-based segmentation was computationally efficient, averaging 37.89 FPS, but noise in the depth data limited segmentation accuracy. SAM2 and SAM3 produced inconsistent results, with SAM2 also requiring several seconds per image. Although individual enhancement and segmentation methods exceeded the camera's 15 FPS frame rate on average, the combined pipeline achieved only 8.29 FPS. **Conclusions**: The project identified practical image enhancement and segmentation approaches for future underwater robotic applications. However, further work is required to improve segmentation accuracy and optimise combined processing for real-time operation.

Graphical abstract



1 Introduction

1.1 Context

The University of Canterbury owns three underwater robots that have been developed over several years by postgraduate students. One of these robots is pictured in Figure 1. The purpose of these robots is to travel from the surface, down to an underwater mussel farm in which mussels grow on ropes. These robots would then collect several mussels to be sampled and tested by workers on a boat. Currently to achieve this, a muti-million-dollar ship must be hired to lift the mussel rope out of the water where workers can then take samples manually. During this process approximately a quarter to a third of the mussels will fall off. The objective of this project overall is to have a robot that can pick up several mussels for sampling so a much smaller and therefore cheaper boat can be used instead.

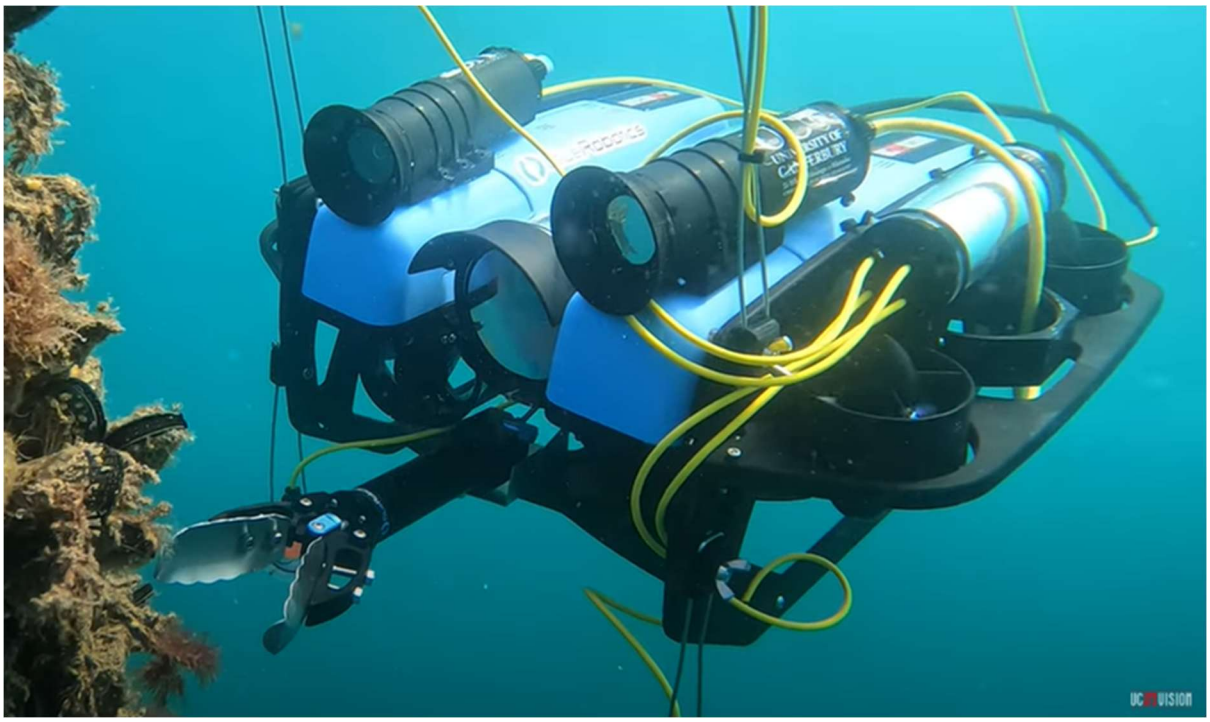


Figure 4: University of Canterbury underwater robot (UC Vision, 2026).

1.2 Underwater computer vision

The robot navigates to the mussel rope and decides what mussel to collect by using computer vision. This process is currently not very robust as it takes the robot approximately five minutes to collect one mussel. Recently, a postgraduate student has completed extensive work on a computer vision algorithm that is able to determine the orientation of mussels on the rope. The plan is to use this to help the robot determine the easiest mussel for it to collect. Ideally, this should reduce the time it takes to collect one mussel, allowing multiple mussels to be collected for inspection. To achieve this, the robot must first know where it is which it does using sensors and camera. Specifically, the stereo camera that is has on the front must

be calibrated so that the robot accurately knows how far away objects are. Underwater environments are also often unpredictable in terms of visibility. Ranges of particles from fine to chunky can cloud video footage, making navigation of these environments tricky. Using detection methods, the impact of this noise should be able to be reduced. This report aims to research, understand, explore, and reduce some of the nuances involved with this.

2 Literature

Camera calibration is an integral first step for any project that uses them. For this underwater application, it is important to first understand how camera calibration functions before applying denoising methods as camera calibration can affect this and vice versa.

2.1 Methods of underwater stereo camera calibration

Stereo vision cameras are important for computer vision applications as they help to mimic aspects of the human eyes. By having two cameras, algorithms can be used to predict distances, angles, shapes and volumes [1]. For an autonomous robot such as the underwater one in this project, this is required to be able to navigate effectively in its environment.

Stereo camera calibration is relatively straightforward in air applications. Most systems achieve this by placing a checkerboard in front of the camera at different angles, orientations, and distances. From this we can gather various properties such as the focal length and optical centre [2]. Since air is a relatively undistorted medium there are few issues with noise and interference when gathering these values to calibrate the stereo cameras. Calibration software is also readily available with services such as Open CV offering stereo camera calibration as an open source project for cameras in air mediums [3].

The physical properties of water and having camera housings is a major challenge for underwater stereo camera calibration. Compared to air, water results in a significantly higher refraction due to light bending as it passes through different media such as water, glass housing, and air inside the cameras [4]. This results in altering the apparent position of objects. Therefore, if stereo calibration usually performed for systems that have air as a medium was applied to an underwater medium, errors would be introduced. Refractive effects can lead to reduced accuracy and cause scale distortions when the stereo camera feeds are reconstructed into 3D [5]. Underwater environments also introduce light attenuation and scattering which reduces image contrast and increases noise. This makes feature detection such as recognising checkerboard corners less reliable [6], [7].

Several approaches have been developed to address these challenges. One of these is the absorption approach, where refractive effects are indirectly incorporated into standard camera calibration parameters such as radial and tangential distortion. In this method, similar techniques to air calibration are performed underwater instead such as using the checkerboard pattern with a distortion model that is adapted to the refractive behaviour of water. This is a relatively simple approach, however may introduce systematic errors due to the limitations in the model of light refraction [2], [5]. Another similar solution is to use a checkerboard pattern that is coloured. This also assists in colour correction, which is another issue, especially in deeper water.

Geometric ray-tracing models have been developed as another method. These explicitly simulate how light travels through multiple refractive interfaces. This requires knowledge of parameters such as refractive indices, housing thickness, and distance between the cameras. Despite improving accuracy, this is a more complex method to implement. This is due to the requirement of a two-stage calibration process, in-air followed by an underwater calibration. This added complexity limits real-world applications, especially when there is limited time for deploying these autonomous underwater vehicles [8], [9], [10].

Another method of calibration is called self-calibration. This works by using calibration fixtures, where multiple images of a known target, such as a checkerboard or a 3D frame, are captured from different angles underwater. The intrinsic and extrinsic parameters are then estimated simultaneously. This approach is advantageous because it does not require precise knowledge of the target geometry beforehand and can be performed in controlled environments such as pools or tanks. However, calibration in open water remains challenging due to environmental instability and limited control over imaging conditions [5], [11].

Regarding the improvement of calibration robustness and efficiency, real-time calibration can also be achieved by looking at individual pixels. For example, real-time calibration evaluation methods for stereo systems based on pixel offset errors between corresponding points in stereo image pairs. This enables continuous monitoring of calibration quality and can detect degradation over time due to environmental factors that come along with being underwater such as temperature changes, vibrations, and optical changes due to increasing depth in the water. Results demonstrate that incorporating real-time error analysis can significantly improve calibration reliability, reducing pixel offset errors by up to 30% [12].

In summary, underwater stereo camera calibration presents significant challenges compared to in-air calibration due to refraction, light attenuation, and environmental variability. A range of techniques has been developed to address these issues, from simplified distortion models to complex geometric corrections and real-time evaluation methods. Reliable depth perception extremely useful for navigation and interacting with a changing environment, the most effective methods use a form of calibration with a checkerboard underwater.

2.2 Methods of underwater image denoising and object tracking

An important metric for image denoising is being able to access how effective this process is in terms of restoring the original image. This is particularly challenging underwater as it is often impossible to compare a filtered image with what it should look like. Therefore, four main metrics have been developed in this field for evaluating underwater image filtering performance. These are peak signal to noise ratio (PSNR), structural similarity index measure (SSIM), underwater image quality measure (UIQM), and underwater colour image quality evaluation (UCIQE) [13], [14].

Both the PSNR and SSIM metrics compared the enhanced image to the original one. PSNR is measured in decibels where a lower value means that the enhanced image differs largely from the original. Similarly, for SSIM, a lower value indicates an image that is significantly different from the original. UIQM measures the sharpness, colourfulness, and contrast of the enhanced

image with a higher value indicating better image quality. UCIQE is a measurement of chroma variation, luminance contrast, and saturation distribution, where again a higher value indicates better image quality [13], [14].

There are several methods for tracking objects of interest across a video. Some popular ones are fast iterative shrinkage-threshold algorithm (FISTA) and segment anything model two and three (SAM2, SAM3) [15], [16]. These algorithms achieving tracking in a similar way; by segmenting objects of interest from the background to focus on. This can be made challenging with changing camera position, changing backgrounds and changing illumination levels which are all increased further in underwater applications [15], [16], [17], [18].

3 Methods and Materials

3.1 Project Objectives

The purpose of this paper was to investigate and develop computer vision techniques to improve the reliability and computational efficiency of underwater mussel rope tracking, image processing, and investigate stereo camera calibration using imagery captured by the University of Canterbury's underwater robot. Underwater imaging presents several challenges, including light attenuation, wavelength-dependent absorption, backscatter, turbidity, and image degradation caused by the optical properties of water. It is necessary to have an understanding of stereo camera calibration to accurately determine where mussels are in relation to the robot to be able to precisely collect them. With this knowledge, the mussel rope can be tracked to apply filtering and denoising algorithms to it. To address these challenges, the project was divided into four main workstreams: investigating underwater image calibration, underwater image enhancement and denoising, mussel rope segmentation, and quantitative performance evaluation.

3.2 Approach

The main work streams were as follows.

3.2.1 Investigating underwater image calibration

The investigation into underwater stereo camera calibration was conducted as a literature review to understand the challenges associated with obtaining accurate depth information for mussel localisation relative to the underwater robot. The investigation began by reviewing conventional in-air stereo camera calibration. Particular attention was given to checkerboard-based methods. The review then examined the factors that complicate calibration underwater, focusing on light refraction at the interfaces between water, the camera housing, and the air inside the housing. The potential effects of refraction on apparent object positions and reconstructed three-dimensional geometry were considered alongside underwater image degradation caused by light attenuation and scattering. Existing underwater calibration approaches were subsequently investigated and compared. The use of coloured calibration patterns and methods for monitoring calibration quality through pixel correspondence errors were also considered. Finally, the reviewed methods were considered in relation to the project's objective of supporting accurate mussel localisation relative to the robot. Their

underlying principles, potential advantages, and implementation challenges were examined to identify considerations relevant to future underwater stereo calibration. No physical calibration experiments or software implementations were performed as part of this investigation as the client stated that camera calibration was already occurring.

3.2.2 Applying denoising and filtering algorithms

The goal of this workflow was to identify algorithms capable of improving the visibility of the underwater camera feed during robot operation. The process began with a review of existing underwater image enhancement methods. Although several algorithms were identified, many were difficult to integrate into the software environment used for this project, including SLURPP [19]. CLAHE and UDCP were selected based on their use in previous research and their suitability for implementation in Python. Both algorithms were implemented and tested on image sequences previously captured by the UC underwater vision robot. Their adjustable parameters were tuned through visual inspection, with image quality metrics also used to assess the resulting images. This allowed the visual effects of different parameter settings to be compared alongside quantitative measurements. However, without clean reference images, these metrics could not establish how accurately the original scene had been restored. Finally, both algorithms were developed as separate modules to support integration with other components of the project.

3.2.3 Segmenting the mussel rope

The initial approach to mussel rope segmentation used SAM2 to identify the rope in images captured by the underwater robot. However, the results were inconsistent, and processing individual images took several seconds, limiting its suitability for real-time operation. An alternative approach used depth data recorded alongside the robot's video frames to generate a mask separating the rope from the background. Initial masks contained substantial noise and holes within the rope region. Median filtering, followed by binary erosion and dilation, was therefore applied to reduce noise and improve the continuity of the segmented region. Segmentation quality was assessed visually rather than against manually labelled ground-truth masks, limiting the ability to quantify its accuracy. Following a request from the client to investigate more effective segmentation, SAM3 was explored as an alternative. Several prompts were tested, with “water” producing the most effective results by attempting to distinguish the surrounding water from the rope. However, its performance varied between videos, indicating that the method was sensitive to changes in the visual content. An offline implementation was also investigated, but unresolved software version compatibility issues prevented its completion within the project timeframe.

3.2.4 Assessing the performance metrics of each algorithm

Python modules were developed to calculate PSNR, SSIM, UIQM, and UCIQE for images processed by the enhancement algorithms. These metrics provided complementary measures of image similarity and underwater image quality, allowing the outputs of different enhancement configurations to be compared. PSNR and SSIM were calculated using the original, unprocessed images as references because clean ground-truth images were unavailable. Consequently, these two metrics measured differences from the input images rather than restoration accuracy and were interpreted alongside the other metrics and visual inspection. Processing performance was then evaluated using recorded underwater video on a computer, measuring frame processing time and the corresponding frames per second

(FPS). Individual algorithms and the combined segmentation and enhancement pipeline were timed to assess the computational cost of integrating multiple processing stages. The target was to achieve a processing rate exceeding the robot camera's 15 FPS frame rate. Average processing speed was considered alongside variation in frame times, since occasional slow frames could prevent consistent real-time operation even when average FPS exceeded the target.

4 Results

An overview of the results can be found in the section below, further results can be found in the git repository linked in Appendix A.

4.1 Assessing the performance of the algorithms

Two main algorithms have been used for filtering and denoising underwater images. These are the CLAHE and UDCP algorithms. An example of these algorithms being applied to an underwater image can be found in Figure 2.



Figure 5: An example of an underwater frame that the algorithms were applied to where the first image is the original, the second uses CLAHE, the third uses UDCP, and the fourth uses both.

To qualitatively compare these algorithms, four different metrics were used; PSNR, SSIM, UIQM, and UCIQE, the results of which are displayed in Table 1 and are developed from the average output of several different images using each algorithm.

Table 1: Qualitive comparison of filtering algorithms.

	PSNR (dB)	SSIM	UIQM	UCIQE
CLAHE	15.26261	0.506407	1.488112	0.323432
UDCP	9.020561	0.447543	1.097053	0.513494
CLAHE and UDCP	11.31433	0.346694	1.389651	0.421734

4.2 Assessing depth segmentation with denoising algorithms performing in real time

To apply these computer vision algorithms to the underwater robot while it is running, they must be able to work in real time. In Table 2, the ability for each algorithm to work in real time is assessed.

Table 2: Real-time performance of algorithms.

	Segmentation with depth	CLAHE and UDCP	CLAHE	UDCP	Segmentation with depth + CLAHE and UDCP
Average time/frame (ms)	26.4	55.9	39	52.1	120.7
Fastest Frame (ms)	18.2	45.5	25.9	42.1	70.5
Slowest Frame (ms)	88.7	163.6	111.1	109.4	160.4
Frames processed	621	621	621	621	621
Average FPS	37.89	17.89	25.65	19.2	8.29

Segmentation was achieved using the depth values provided with videos taken by the robot underwater, the results of which are shown in Figure 3.



Figure 6: A section of underwater mussel rope segmented from the background.

Following this, the aforementioned algorithms, CLAHE and UDCP were applied to the segmented section. This resulted in a clearer section of rope as seen in Figure 4.

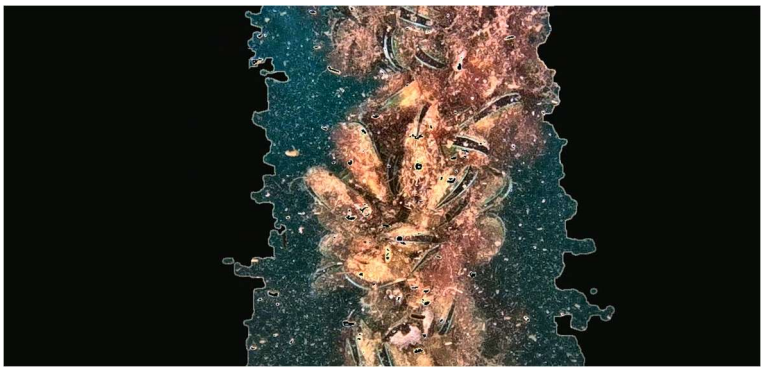


Figure 7: A section of underwater mussel rope segmented from the background with the CLAHE and UDCP algorithms applied.

4.3 SAM performance

Neither of the SAM algorithms were timed to assess their real-time performance as they all took more than 10 seconds to run. Therefore, they could not be used for a real-time application. The enhancement algorithms were not applied to SAM2 as it was discarded before this occurred. They were also not applied to SAM3 as it was only possible to run this online, so the local data was not available to be processed. Figure 5 and 6 both show sections of the mussel rope with the SAM2 segmentation applied. The first figure is one of the most successful sections where the algorithm is working, conversely, the second figure shows an example of the algorithm working poorly.



Figure 8: A section of the underwater mussel rope with the SAM2 algorithm applied.

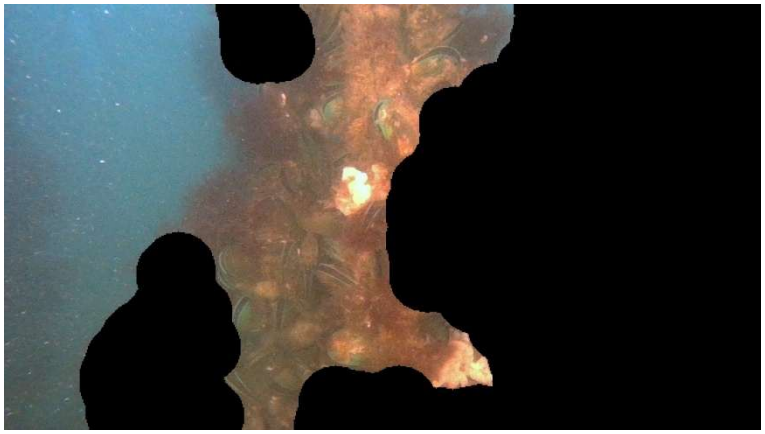


Figure 9: A section of the underwater mussel rope with the SAM2 algorithm applied ineffectively.

Following the limited success of SAM2, later in the project, SAM3 was applied instead. Figure 7 displays an example of this algorithm working extremely effectively. The were many other videos that were put through the SAM3 algorithm, however, in the majority of them, it could not segment the water from the mussel rope as it did not recognise anything in the image.

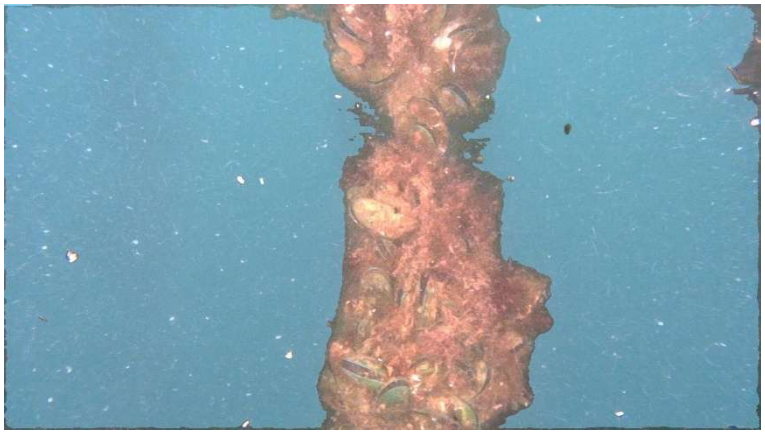


Figure 10: A section of the underwater mussel rope with the SAM3 algorithm applied.

5 Discussion

5.1 Findings

5.1.1 Underwater image enhancement

Both the CLAHE and UDCP algorithms had, on average, a relatively low PSNR and SSIM. This was expected and similar to other research papers [14]. A low PSNR and SSIM means that the enhanced image is significantly different from the original, confirming that these algorithms are having the intended effect of changing the images. However, these metrics alone cannot establish whether the changes represent an improvement, as enhancement can introduce distortions while simultaneously improving visibility. The CLAHE algorithm had the highest UIQM score which is intuitive as this algorithm works on increasing sharpness in the image. The value of 1.48 indicates that productive image enhancement is occurring, however there

is a lot of room for improvement. Inversely, UDCP had the highest score for UCIQE. Since this algorithm focuses on colour correction, this is not unexpected. UIQM and UCIQE scores of 1.390 and 0.422, respectively, alongside lower PSNR and SSIM scores of 11.31 dB and 0.347. These results suggest that the algorithms have complementary effects on image appearance, although their combination does not necessarily produce the best result across all quality metrics. Furthermore, the results for all algorithm combinations are at a reasonable level that is consistent with other similar works [14]. Naturally, some differences are expected that come with the nature of analysing different images. This indicates that the colour quality of the underwater images is being improved with these algorithms. The UIQM and UCIQE results provide additional information about perceived underwater image quality and colour characteristics. The higher UIQM score obtained using CLAHE is consistent with its role in enhancing local contrast, while the higher UCIQE score obtained using UDCP suggests that it produces more favourable colour-related characteristics according to this metric.

5.1.2 Real-time processing performance

The real-time performance results demonstrate that the computational cost of the developed algorithms varies considerably. Depth-based segmentation achieved an average processing time of 26.4 ms per frame and an average frame rate of 37.89 FPS, making it the fastest of the tested methods. CLAHE achieved 25.65 FPS, UDCP achieved 19.20 FPS, and the combined CLAHE and UDCP implementation achieved 17.89 FPS. Each of these methods individually satisfied the nominal requirement of processing at least 15 FPS. However, when depth-based segmentation was combined with both enhancement algorithms, the average frame rate decreased to 8.29 FPS, with an average processing time of 120.7 ms per frame.

For a camera operating at 15 FPS, each frame is acquired approximately every 66.7 ms. Therefore, the average processing time of the combined segmentation and enhancement pipeline exceeds the available processing interval by approximately 54 ms per frame. At 8.29 FPS, the pipeline processes approximately 55% of the camera's nominal frame rate. This difference could result in an accumulating backlog or an increasing delay between the robot's current position and the information displayed to its operator. Alternatively, if older frames are discarded to maintain responsiveness, some visual information may be lost. Both outcomes could affect the reliability of a future real-time tracking system.

The most important statistic to extract from the real-time testing is the average FPS for each algorithm combination. The underwater robot uses a camera that runs at 15 FPS, therefore these algorithms must be processing faster than 15 FPS to be applicable for real-time applications. All algorithm combinations met this threshold except for the combined segmentation with CLAHE and UDCP. This indicates that more code optimising should take place to be able to use this in real-time, specifically with the segmentation and CLAHE parts as they were the slowest. It is also important to ensure that optimisation does not significantly reduce image quality or segmentation reliability, as faster processing alone is insufficient if the resulting information is unsuitable for tracking. As both the SAM methods were far too slow to be applied in real-time, the results therefore support the use of depth-based segmentation as a computationally efficient foundation for the system.

5.1.3 Depth-based mussel rope segmentation

Depth-based segmentation provided a practical alternative to the initial use of SAM2 for extracting the mussel rope from the surrounding underwater environment. The method used the depth information recorded alongside the camera imagery to distinguish the rope region from the background. Although the initial segmentation contained holes and noise, median filtering and morphological operations improved the continuity of the resulting mask. Applying CLAHE and UDCP to the segmented region subsequently improved its visual appearance, as demonstrated in Figures 3 and 4.

A key advantage of this approach is that it uses information already recorded by the ROV rather than relying entirely on image appearance. However, the effectiveness of depth-based segmentation depends on the quality and consistency of the available depth measurements. Missing values, inaccurate depth estimates, and insufficient separation between the rope and the background could result in incomplete or incorrect masks. Morphological filtering can mitigate some of these problems, but it cannot reliably recover information that is absent from the original depth data. Due to a large amount of noise in the depth values, this method was not able to accurately segment the mussel rope to the client's needs. The results demonstrate that the method can produce visually useful segmentation and operate at a suitable average processing rate. However, the segmentation accuracy is poor.

5.1.4 Performance of SAM2 and SAM3

The investigations into SAM2 and SAM3 demonstrated the potential of foundation-model-based segmentation approaches but also highlighted their limitations in the context of real-time underwater robotics. SAM2 produced inconsistent results, with some images showing successful segmentation and others containing substantial errors, as illustrated in Figures 5 and 6. Furthermore, the processing time exceeded 10 seconds per image. At this rate, the algorithm would process fewer than 0.1 frames per second, making it unsuitable for a camera operating at 15 FPS without a fundamentally different deployment or processing strategy.
SAM3 produced an effective segmentation in the example shown in Figure 7, demonstrating that it could identify the mussel rope successfully under some conditions. However, it frequently failed to distinguish the rope from the surrounding water in other videos. This inconsistency is likely a result of the training data not being well suited to this application. Therefore, specific training on a mussel rope application could improve these results.

In their current implementations, the observed processing times and inconsistent outputs make SAM2 and SAM3 less suitable than depth-based segmentation for continuous real-time operation. These findings do not establish that foundation models are inherently unsuitable for underwater robotics. Their performance may improve through model optimisation, reduced input resolution, hardware acceleration, or adaptation to underwater imagery through training.

5.2 Implications

The main practical contribution of this work is the identification of image processing techniques that could improve the usability of underwater robotic imagery while remaining compatible with real-time operation. For the stakeholders involved in developing and operating the mussel-collection robot, this has implications for operator visibility and accessing methods of segmentation.

First, improved image contrast and colour characteristics could make it easier for an operator to interpret the underwater environment. Better visibility may assist with navigating towards the mussel rope, identifying relevant features, and monitoring the robot during mussel collection. This could reduce some of the difficulties associated with operating in turbid water, although the extent of the benefit must be established through operator-based testing rather than inferred solely from image quality metrics.

Second, the segmentation results demonstrated the advantages and disadvantages of different methods which the client found to be extremely useful. For example, the segmented rope region could be used to estimate the rope's image-space position or identify a region of interest for more detailed mussel detection. This could reduce the amount of irrelevant background information that subsequent algorithms must process.

From a broader operational perspective, a reliable underwater perception system could contribute to reducing dependence on expensive, manually operated mussel-collection infrastructure. By improving the information available to the operator and providing a foundation for automated rope tracking, this work supports the longer-term objective of making robotic mussel collection more practical.

5.3 Limitations

Underwater image quality can vary substantially depending on suspended particles, lighting direction, depth, camera-to-object distance, and the optical properties of the water. An algorithm that performs well on one recording may produce significantly different results when these conditions change. For example, CLAHE may increase local contrast in a relatively clear image but amplify visible noise in an image containing substantial backscatter. Similarly, UDCP may produce useful colour and contrast corrections under some conditions while introducing colour distortions under others. The current results do not establish how consistently these effects occur across different deployments.

The segmentation approach is also affected by environmental variation. The depth-based approach had huge performance issues due to a significant amount of noise. Since the depth-based approach relies on recorded depth information, its performance is also constrained by the quality and availability of those measurements.

A further limitation is the use of PSNR, SSIM, UIQM, and UCIQE as the primary measures of enhancement performance. Although these metrics provide complementary information, none independently establishes whether an enhanced image is more useful for mussel tracking or robotic collection.

PSNR measures pixel-level differences relative to a reference image, while SSIM measures structural similarity. When the original underwater image is used as the reference, both metrics can penalise changes that improve visibility because the enhanced image is intentionally different from the original. Conversely, an image that remains similar to the original may achieve a comparatively high score without providing a meaningful improvement in visibility. Therefore, the relatively low PSNR and SSIM values observed in this work should not be interpreted as evidence of successful enhancement on their own.

UIQM and UCIQE are designed to assess aspects of underwater image quality. However, enhancing colour or contrast may improve the visual appearance of the scene while also amplifying noise or introducing artefacts that interfere with subsequent processing.

The performance of depth-based segmentation was primarily assessed through visual inspection of the resulting masks. Figures 3 and 4 demonstrate that the method can extract a useful representation of the rope and that image enhancement can improve its appearance. However, these examples do not quantify how accurately the rope boundaries correspond to the actual rope in the original scene.

Although the individual algorithms achieved average processing rates above the camera's nominal 15 FPS requirement, the combined segmentation and enhancement pipeline achieved only 8.29 FPS. This is a significant limitation because the ultimate application requires the algorithms to operate as part of a complete robotic perception system rather than as isolated components.

The performance measurements also show considerable variation between frames. For example, depth-based segmentation achieved an average processing time of 26.4 ms but had a slowest recorded frame time of 88.7 ms. The combined pipeline ranged from 70.5 ms to 160.4 ms per frame. This variability means that average frame rate alone is insufficient to characterise responsiveness or guarantee that a real-time deadline will be met consistently.
Furthermore, processing time is only one component of end-to-end latency. Camera acquisition, data transfer, buffering, visualisation, communication with the robot, and any downstream tracking or control algorithms may add additional delays. The current measurements do not establish the latency of the complete robotic system under operational conditions.

The SAM2 and SAM3 experiments were limited by inconsistent segmentation results and, in the case of SAM2, substantial processing time. The observed performance demonstrates that these models could successfully segment the rope in selected images but did not reliably do so across the available recordings. This limits the extent to which their results can be generalised to underwater mussel tracking.

5.4 Future work

An updated Gantt chart can be found in Appendix B.

Currently, combining all the algorithms does not work fast enough to be applied in real-time. Code optimisation could improve this with the goal of getting it working in real-time. If this were successful, then the resulting segmented and enhanced image could be overlayed on the original image to improve visibility in the overall video.

SAM3 was not able to be ran offline due to conflicting versions issues. If SAM3 could be run offline, then training on the ROV data could commence which in theory would make this both a fast and reliable method of segmentation.

A final outcome of how the segmentation and denoising will be applied to the ROV footage was not decided. This would involve further communication with the client to figure out what

they would prefer and implementing it. For example, this could be outlining where the mussel rope is in the footage or only applying denoising to the mussel rope.

Finally, parts of this project could be implemented onto the ROV itself. This would enable testing of the ROV's software and assessing its ability to run these algorithms.

6 Conclusions

This project investigated computer vision techniques to improve underwater mussel rope tracking and image processing for the University of Canterbury underwater robot. The first objective was to investigate underwater stereo camera calibration. The literature review identified refraction, light attenuation, and scattering as key challenges that distinguish underwater calibration from conventional in-air methods. Checkerboard-based calibration, distortion models, and geometric ray-tracing approaches were examined to understand their potential for improving depth estimation and mussel localisation. However, calibration was not implemented or experimentally tested, as this work was limited to reviewing existing methods.

The second objective was to improve underwater image visibility using denoising and enhancement algorithms. CLAHE and UDCP were implemented and evaluated using visual inspection and image quality metrics. CLAHE achieved the highest UIQM score of 1.49, while UDCP achieved the highest UCIQE score of 0.51, suggesting that the two methods provide complementary benefits in contrast and colour characteristics. However, because clean reference images were unavailable, the metrics could not establish the accuracy of image restoration.

The third objective was to investigate methods for segmenting the mussel rope. Depth-based segmentation, combined with median filtering and morphological operations, produced visually useful masks and was substantially faster than the initial SAM2 approach. Nevertheless, noisy depth measurements limited segmentation accuracy, and the results were not validated against ground-truth masks. SAM3 also showed potential in selected images but was inconsistent across videos, while offline implementation remained incomplete.

The final objective was to evaluate image quality and computational performance. Individual enhancement algorithms and depth-based segmentation exceeded the camera's 15 FPS frame rate on average when tested on recorded footage using a computer. However, the combined segmentation and enhancement pipeline achieved only 8.29 FPS, demonstrating that further optimisation is required for real-time operation.

Overall, this project established a foundation for improving underwater imagery and investigating mussel rope segmentation. Further work should prioritise segmentation accuracy, optimisation of the combined pipeline, and testing on the robot under operational conditions before the methods can be considered suitable for autonomous mussel sampling.

Declarations

Author cont*ributions: N/A*

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Conflicts of interest statement: N/A

Acknowledgements: Richard Green, Phuong Nguyen, Timothy Rensen.

AI usage: AI tools namely ChatGPT was used to collate references and for help rewriting certain sections of the report after the first draft.

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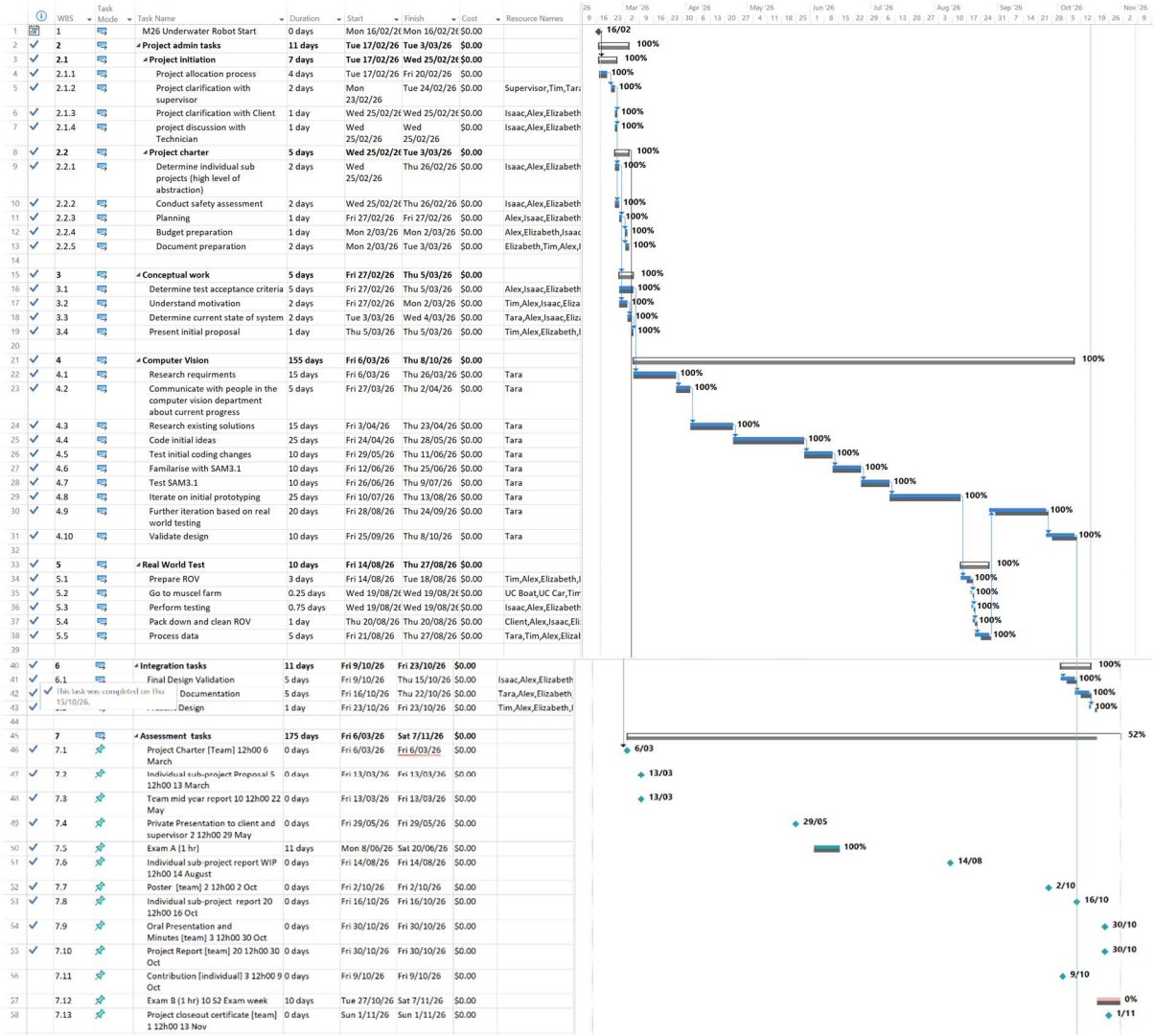
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Appendix A: Git repository link

https://eng-git.canterbury.ac.nz/tst125/fyp_underwater_robot

Appendix B: Project planning

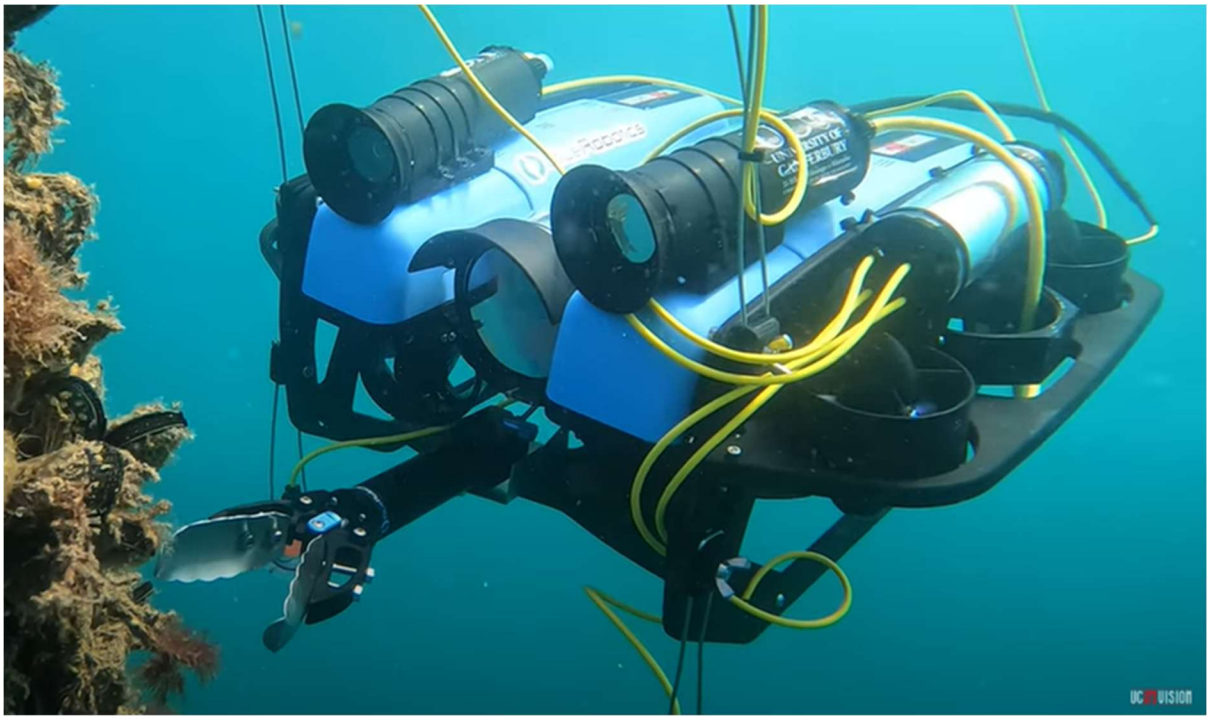


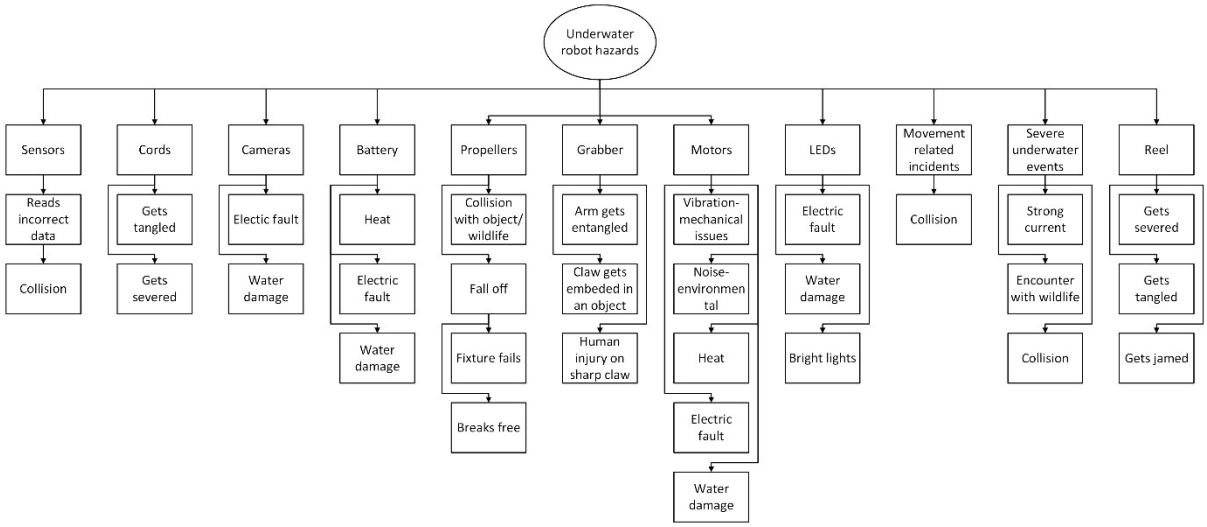
	WBS	Task Mode	Task Name	Duration	Start	Finish	Cost	Resource Names
1	1		M26 Underwater Robot Start	0 days	Mon 16/02/26	Mon 16/02/26	$0.00	
2	**2**		**Project admin tasks**	**11 days**	**Tue 17/02/26**	**Tue 3/03/26**	**$0.00**	
3	**2.1**		**Project initiation**	**7 days**	**Tue 17/02/26**	**Wed 25/02/26**	**$0.00**	
4	2.1.1		Project allocation process	4 days	Tue 17/02/26	Fri 20/02/26	$0.00	
5	2.1.2		Project clarification with supervisor	2 days	Mon 23/02/26	Tue 24/02/26	$0.00	Supervisor,Tim,Tar
6	2.1.3		Project clarification with Client	1 day	Wed 25/02/26	Wed 25/02/26	$0.00	Isaac,Alex,Elizabeth
7	2.1.4		project discussion with Technician	1 day	Wed 25/02/26	Wed 25/02/26	$0.00	Isaac,Alex,Elizabeth
8	**2.2**		**Project charter**	**5 days**	**Wed 25/02/26**	**Tue 3/03/26**	**$0.00**	
9	2.2.1		Determine individual sub projects (high level of abstraction)	2 days	Wed 25/02/26	Thu 26/02/26	$0.00	Isaac,Alex,Elizabeth
10	2.2.2		Conduct safety assessment	2 days	Wed 25/02/26	Thu 26/02/26	$0.00	Isaac,Alex,Elizabeth
11	2.2.3		Planning	1 day	Fri 27/02/26	Fri 27/02/26	$0.00	Alex,Isaac,Elizabeth
12	2.2.4		Budget preparation	1 day	Mon 2/03/26	Mon 2/03/26	$0.00	Alex,Elizabeth,Isaac
13	2.2.5		Document preparation	2 days	Mon 2/03/26	Tue 3/03/26	$0.00	Elizabeth,Tim,Alex,I
14								
15	**3**		**Conceptual work**	**5 days**	**Fri 27/02/26**	**Thu 5/03/26**	**$0.00**	
16	3.1		Determine test acceptance criteria	5 days	Fri 27/02/26	Thu 5/03/26	$0.00	Alex,Isaac,Elizabeth
17	3.2		Understand motivation	2 days	Fri 27/02/26	Mon 2/03/26	$0.00	Tim,Alex,Isaac,Eliza
18	3.3		Determine current state of system	2 days	Tue 3/03/26	Wed 4/03/26	$0.00	Tara,Alex,Isaac,Eliz
19	3.4		Present initial proposal	1 day	Thu 5/03/26	Thu 5/03/26	$0.00	Tim,Alex,Elizabeth,I
20								
21	**4**		**Computer Vision**	**155 days**	**Fri 6/03/26**	**Thu 8/10/26**	**$0.00**	
22	4.1		Research requirments	15 days	Fri 6/03/26	Thu 26/03/26	$0.00	Tara
23	4.2		Communicate with people in the computer vision department about current progress	5 days	Fri 27/03/26	Thu 2/04/26	$0.00	Tara
24	4.3		Research existing solutions	15 days	Fri 3/04/26	Thu 23/04/26	$0.00	Tara
25	4.4		Code initial ideas	25 days	Fri 24/04/26	Thu 28/05/26	$0.00	Tara
26	4.5		Test initial coding changes	10 days	Fri 29/05/26	Thu 11/06/26	$0.00	Tara
27	4.6		Familarise with SAM3.1	10 days	Fri 12/06/26	Thu 25/06/26	$0.00	Tara
28	4.7		Test SAM3.1	10 days	Fri 26/06/26	Thu 9/07/26	$0.00	Tara
29	4.8		Iterate on initial prototyping	25 days	Fri 10/07/26	Thu 13/08/26	$0.00	Tara
30	4.9		Further iteration based on real world testing	20 days	Fri 28/08/26	Thu 24/09/26	$0.00	Tara
31	4.10		Validate design	10 days	Fri 25/09/26	Thu 8/10/26	$0.00	Tara
32								
33	**5**		**Real World Test**	**10 days**	**Fri 14/08/26**	**Thu 27/08/26**	**$0.00**	
34	5.1		Prepare ROV	3 days	Fri 14/08/26	Tue 18/08/26	$0.00	Tim,Alex,Elizabeth,I
35	5.2		Go to muscel farm	0.25 days	Wed 19/08/26	Wed 19/08/26	$0.00	UC Boat,UC Car,Tim
36	5.3		Perform testing	0.75 days	Wed 19/08/26	Wed 19/08/26	$0.00	Isaac,Alex,Elizabeth
37	5.4		Pack down and clean ROV	1 day	Thu 20/08/26	Thu 20/08/26	$0.00	Client,Alex,Isaac,Eli
38	5.5		Process data	5 days	Fri 21/08/26	Thu 27/08/26	$0.00	Tara,Tim,Alex,Elizal
39								
40	**6**		**Integration tasks**	**11 days**	**Fri 9/10/26**	**Fri 23/10/26**	**$0.00**	
41	6.1		Final Design Validation	5 days	Fri 9/10/26	Thu 15/10/26	$0.00	Isaac,Alex,Elizabeth
42			Documentation	5 days	Fri 16/10/26	Thu 22/10/26	$0.00	Tara,Alex,Elizabeth
43			Design	1 day	Fri 23/10/26	Fri 23/10/26	$0.00	Tim,Alex,Elizabeth,I
44								
45	**7**		**Assessment tasks**	**175 days**	**Fri 6/03/26**	**Sat 7/11/26**	**$0.00**	
46	7.1		Project Charter [Team] 12h00 6 March	0 days	Fri 6/03/26	Fri 6/03/26	$0.00	
47	7.2		Individual sub-project Proposal 5 12h00 13 March	0 days	Fri 13/03/26	Fri 13/03/26	$0.00	
48	7.3		Team mid year report 10 12h00 22 May	0 days	Fri 13/03/26	Fri 13/03/26	$0.00	
49	7.4		Private Presentation to client and supervisor 2 12h00 29 May	0 days	Fri 29/05/26	Fri 29/05/26	$0.00	
50	7.5		Exam A (1 hr)	11 days	Mon 8/06/26	Sat 20/06/26	$0.00	
51	7.6		Individual sub-project report WIP 12h00 14 August	0 days	Fri 14/08/26	Fri 14/08/26	$0.00	
52	7.7		Poster [team] 2 12h00 2 Oct	0 days	Fri 2/10/26	Fri 2/10/26	$0.00	
53	7.8		Individual sub-project report 20 12h00 16 Oct	0 days	Fri 16/10/26	Fri 16/10/26	$0.00	
54	7.9		Oral Presentation and Minutes [team] 3 12h00 30 Oct	0 days	Fri 30/10/26	Fri 30/10/26	$0.00	
55	7.10		Project Report [team] 20 12h00 30 Oct	0 days	Fri 30/10/26	Fri 30/10/26	$0.00	
56	7.11		Contribution [individual] 3 12h00 9 Oct	0 days	Fri 9/10/26	Fri 9/10/26	$0.00	
57	7.12		Exam B (1 hr) 10 S2 Exam week	10 days	Tue 27/10/26	Sat 7/11/26	$0.00	
58	7.13		Project closeout certificate [team] 1 12h00 13 Nov	0 days	Sun 1/11/26	Sun 1/11/26	$0.00	

Appendix C: H&S hazard assessment

Authors: Elizabeth Mora, Tim Parker, Alex Pirie, Issac Rowe, Tara Stevens
Date: 06/03/2026

C.1 Architecture of the system





C.2 Risk matrix and scales adopted

The ISO31000 risk method was used[1]. The Consequence and Likelihood scales were used per [2].

Likelihood of the consequence arising	Description	Probability, events per annum	Qualitative likelihood	Likelihood score								
	Annual occurrence	1.00	Almost certain	10	5000	1000	600	300	200	100	50	20
	Has occurred several times in your career	0.6	Likely	6	3000	600	360	180	120	60	30	12
	Might occur once in your career	0.4	Possible	4	2000	400	240	120	80	40	20	8
	Event does occur somewhere from time to time	0.2	Unlikely	2	1000	200	120	60	40	20	10	4
	Heard of something like this happening elsewhere	0.1	Rare	1	500	100	60	30	20	10	5	2
	Theoretically possible but not expected to occur	0.01	Almost incredible	0.1	50	10	6	3	2	1	0.5	0.2

Severity of Adverse health consequences									
Loss of health	500	100	60	30	20	10	5	2
Safety consequence	Catastrophe: multiple deaths occur, would have negative impacts to next generation	Death: People loss of life	Serious harm: Notifiable injurles. Permanent and serious harm	Serious harm: Notifiable injurles. Incident results in serious harm	Moderate incident: Incident occurs and exposure to moderate harm	Minor harm: Incident occurs and minor harm	Minor harm: Incident without harm or with tiny impacts.	No harm: Hazard could be sufficiently managed by the current preventions.
Health consequence	Health disaster: Multiple people affected with high WHODAS scores	WHODAS up to 100 Death or paralysis.	WHODAS up to 60 Half-body, or leg paralysis	WHODAS up to 30 Amputation.	WHODAS Up to 20 Long term impact e.g. hearing loss	WHODAS up to 10 Permanent but not debilitating musculoskeletal injury	WHODAS Up to 5 Temporary effects to human body, healed naturally.	WHODAS up to 2 No harm to human body.
Chemical Toxicity per GHS 7	GHS Category 1: Extremely toxic. Fatal if swallowed, contacted, or inhaled. One drop is fatal	GHS Category 2: Highly toxic. Fatal if swallowed, contacted, or inhaled. One teaspoon is fatal	GHS Category 3: Moderately toxic. Toxic if swallowed, contacted, or inhaled. One tablespoon is fatal	GHS Category 4: Slightly toxic. Harmful if swallowed, contacted, or inhaled. Two cups are fatal	GHS Category 5: Practically non-toxic. May be harmful if swallowed, contacted, or inhaled. One litre is fatal	Level 6: Relatively harmless. Permanent but not debilitating injury. May cause death for sensitive individuals.	Level 7: Temporary harm with no permanent loss of ability or quality of life. Exposure is below LOAEL.	Level 8: No adverse effects expected. Exposure is below NOAEL.

Risk appetite response scale	
1000+	Cessation, immediate intervention required
120+ to 1000	Unacceptable risk. Cease operations until risk is under control
60+ to 120	Urgent treatment required. Operations proceed with caution and ongoing monitoring of risk.
10+ to 60	Implement treatment in a reasonable time period. Continue operations with caution and monitoring
0-10	No special treatment required. Continue the operations with on-going monitoring of the efficacy of existing preventions.

This Chemical Toxicity scale assumes an existing safety system comprising substances that are properly labelled, securely stored, only accessible by people with sufficient training, and appropriate personal protective equipment is provided. If the safety system is not strong, then consider increasing the likelihood of the consequences arising.

[1] ISO 31000:2018 Risk management — Guidelines, https://www.iso.org/standard/65694.html

[2] Pons, D., An integrated industrial risk assessment methodology for accident safety, chronic health, and chemical exposure. New Zealand Journal of Health and Safety Practice, 2024. 1(3) DOI: https://doi.org/10.26686/nzjhsp.v1i3.9601

C.3 Risk Register

Risks of system in its CURRENT STATE, with its existing controls.								Risks of system in its FUTURE STATE after these treatments. Insert any new threats caused by the treatments.			
Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per 'Severity of harm' scale 1. Hazard occurred 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost certain 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment 7 or less No intervention necessary.	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
Sensors	Reads incorrect data	2	3	6	Minimal action required, code alterations	Relevant team member	No	2	3	6	Yes - Further action not necessary
Cords	Gets tangled	3	3	9	Maintain cable management	Relevant team member	Yes	3	2	6	Yes - Further action not necessary
Cords	Gets severed	4	2	8	Active management of tether when operating	Relevant team member	Yes	4	2	8	Yes - Further action not necessary
Cameras	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
Cameras	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
Battery	Overheating	3	2	6	Check battery health/condition prior to operation and monitor power draw while in use	Every handler	No	3	2	6	Yes - Further action not necessary
Battery	Electric Fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
Battery	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
Propellers	Collision with object/ wildlife	3	4	12	Monitor surroundings, Ensure personal safety, before startup inspect propellers	Every handler	Yes	3	3	9	Yes - Further action not necessary
Propellers	Mechanical failure	3	2	6	Inspect propellors prior to use of ROV	Every handler	No	3	2	6	Yes - Further action not necessary
Grabber	Arm get entangled	2	4	8	Monitor surroundings while operating arm	Relevant team member	Yes	2	4	8	Yes - Further action not necessary
Grabber	Claw embeds in drop line	2	4	8	Monitor to ensure grabber arm does not grip the line too tightly	Relevant team member	Yes	2	4	8	Yes - Further action not necessary
Grabber	Human injury because of sharp object	4	4	16	Design cover for grabber, follow proper handling instructions , keep clear during operation	Relevant team member	Yes	3	3	9	Yes - Further action not necessary
Motors	Vibration – mechanical issues	2	4	8	Inspect motors prior to operation	Every handler	No	2	4	8	Yes - Further action not necessary
Motors	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
Motors	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
Motors	Overheating	3	2	6	Monitor power draw from motors while ROV in use	Relevant team member	Yes	3	2	6	Yes - Further action not necessary
Motors	Noise – environmental damage	2	4	8	Ensure motors don't result in propellor cavitation - monitor rpms	Relevant team member	Yes	2	4	8	Yes - Further action not necessary
LEDs	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
LEDs	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
LEDs	Bright lights	3	4	12	Proper PPE, Disable LED when it is out of the water	Every handler, Code	Yes	2	3	6	Yes - Further action not necessary
Controllable Underwater Events	Collision – self inflicted	2	4	8	Monitor surrounding while operating and approach motion control sensibly	Every handler	Yes	2	4	8	Yes - Further action not necessary
Uncontrollable Underwater Events	Strong current	2	3	6	Test ocean currents prior to operation and set current threshold for non-operation	Every handler	Yes	2	3	6	Yes - Further action not necessary
Uncontrollable Underwater Events	Encounter with wildlife	4	4	16	Monitor surroundings and ensure no abundance of wildlife (fish, dolphins etc) in testing area	Every handler	Yes	4	4	16	NO – not much we can do to mitigate this
Uncontrollable Underwater Events	Collision	2	3	6	Monitor surroundings and motion of robot while in operation and actively avoid collisions through piloting robot	Every handler	Yes	2	3	6	Yes - Further action not necessary
Reel	Gets severed	3	2	6	Monitor reel/tether to ensure no snags/sharp objects threaten tether	Relevant team member	Yes	3	2	6	Yes - Further action not necessary
Reel	Tangled	2	3	6	Monitor reel/tether to ensure no tangles generate while in use	Relevant team member	Yes	2	3	6	Yes - Further action not necessary
Reel	Jammed	2	3	6	Monitor reel/tether while in operation and manage tether to ensure no jams	Relevant team member	Yes	2	3	6	Yes - Further action not necessary

Appendix C - Automated Tether Management

ENME408-2026 Individual student report

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

Autonomous Tether Management System for a Remotely Operated Underwater Vehicle

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Document version: 20261016

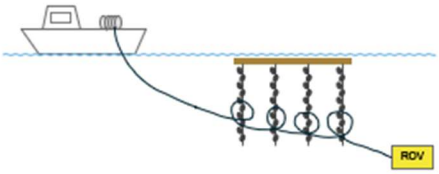
Abstract

Context: Remotely operated underwater vehicles (ROVs) have potential to improve efficiency and reduce labour requirements in mussel-farm inspection and sampling. However, excess tether increases entanglement risk, while insufficient tether restricts vehicle movement and introduces forces that affect control. **Objectives:** This project investigated the feasibility of a proof-of-concept automated tether management system capable of regulating deployed tether length in response to changes in loading. **Method:** The existing manual reel was retrofitted with a motor and belt transmission. Motor current was used as an indirect control input to command retraction, hold or extension through an external microcontroller. Performance was assessed through exploratory bench testing and manually controlled sea trials. **Findings:** The retrofit provided powered operation while retaining the existing tether and electrical passthrough arrangement. Correcting the electronic speed controller's (ESC) zero-speed response resolved the initial oscillation. Qualitative testing demonstrated the required automatic states, manual override and smoother manual operation. Field trials showed reduced thruster power draw during assisted payout and surface retrieval. Longer tether deployments increased water-current loading and reel tangling, while mechanical deterioration prompted mounting-plate and idler redesigns. Drivetrain resistance also contributed to motor current, and ESC failure prevented quantitative assessment of the final controller. **Conclusions:** The results demonstrated practical powered tether handling and automatic control behaviour on the bench. Reliable autonomous operation requires improved cable distribution, quantitative assessment of current-based tension feedback and in-water testing of the integrated system.

Graphical abstract

Problem

- Excess tether increases drag and entanglement risk
- Insufficient tether restricts vehicle motion



Prototype winch

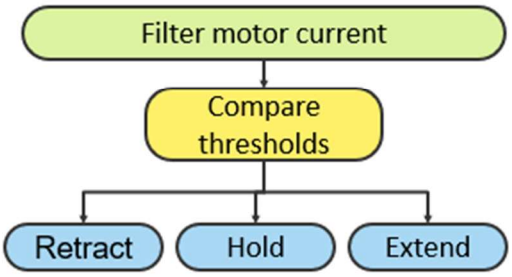
- Existing hand reel retrofitted for powered operation
- Belt drive retains powered and manual operation





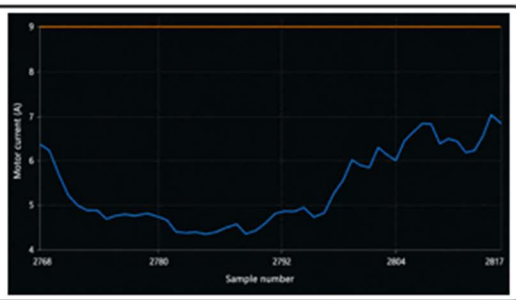
Control system

- Motor current provides indirect feedback on tether loading
- ESP32 provides control logic with manual override



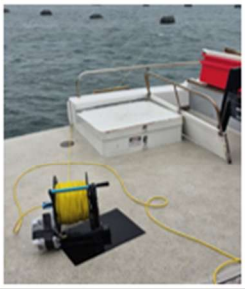
Quantitative findings

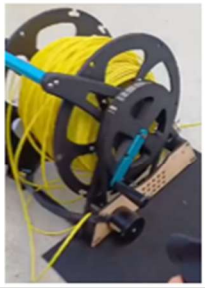
- Free retraction: approximately 4 - 7 A over one revolution.
- Mechanical resistance affects current measurements



Qualitative findings

- Bench: automatic states and manual override demonstrated.
- Field: lower thruster draw observed; tangles interrupted scanning.





Next steps

- Validate autonomous operation in water
- Increase integration into ROVs ecosystem
- Design and implement level winder

1 Introduction

1.1 Context

The aquaculture industry is steadily growing in New Zealand and is considered an important sector for future economic development. Mussel farming represents the largest component of the country's aquaculture production; however, farms require regular inspection and sampling to monitor mussel size and meat condition throughout the growth cycle. Traditionally, these tasks are performed manually using purpose-built vessels that lift mussel lines from the water so that workers can remove samples for inspection. This process is labour-intensive, time-consuming and dependent on operator judgement. The vessels required are also relatively slow to operate and expensive to store and maintain, further increasing the cost of routine sampling operations.

1.2 Tether Management System

The overall project proposes the use of a remotely operated underwater vehicle (ROV) to collect mussel samples directly underwater, allowing a reduced crew on a significantly smaller vessel to perform an equivalent sampling operation. This approach removes the subjective element of operator-selected mussels while reducing sampling time and operational costs.

A key component of the ROV system is the tether cable, which provides electrical power, control signals and data communication between the vehicle and surface vessel. However, the tether introduces significant operational challenges. Excess deployed tether increases hydrodynamic drag and the likelihood of entanglement with the vehicle, mussel ropes and other submerged infrastructure, while insufficient tether can restrict vehicle movement and introduce forces that affect its control accuracy. Manual tether management also requires continuous attention from a trained operator, limiting the number of personnel capable of effectively operating the ROV system.

This sub-project therefore focuses on the development of a proof-of-concept automated tether management system capable of regulating deployed tether length in response to changes in tether loading. The work investigates a suitable method for estimating tether tension, the retrofit of the existing manual tether reel for powered operation, and integration of the motor, sensing and control system. Controlled bench testing and in-water trials are used to evaluate the feasibility of the approach and identify the requirements for further development towards a commercially applicable tether management system.

1 Literature

2.1 Underwater robotics in aquaculture

Underwater robots are used for monitoring and intervention tasks in environments that are difficult or hazardous for human divers. Remotely operated vehicles (ROVs) and autonomous underwater vehicles (AUVs) are commonly deployed for subsea inspection, scientific research and infrastructure monitoring due to their ability to collect data and perform tasks while being

remotely controlled or operating autonomously [1]. Advances in sensing, navigation and propulsion technologies have significantly improved the capabilities of these systems, enabling their use across a wide range of marine applications, including offshore engineering, environmental monitoring and aquaculture operations [1].

Within the aquaculture industry, underwater robots are becoming increasingly common for inspection and data collection. These systems can be used to monitor fish cages, inspect mooring systems and collect information about the farm environment and infrastructure [2], [3]. Robotic inspection can reduce operational overheads, equipment requirements and labour demand while allowing inspections to be performed more frequently than traditional methods [4]. Aquaculture has also been identified as a key pathway for increasing sustainable food production while remaining within environmental limits [5]. As the industry expands, continued technological development is therefore likely to increase the use of automated monitoring and robotic systems. Many underwater robotic platforms remain tethered, however, introducing additional design and operational challenges associated with tether behaviour and management.

2.2 Tethered underwater vehicles

Remotely operated vehicles commonly use an umbilical tether to provide communication and, where required, electrical power between the vehicle and surface operator [6]. Tethered systems enable real-time control and transmission of sensor data such as live video, while maintaining a physical connection to the surface.

The tether can, however, significantly influence vehicle dynamics and manoeuvrability. It behaves as a flexible cable subjected to tension and hydrodynamic forces, which can affect vehicle motion and stability [7]. Numerical studies have combined hydrodynamic modelling with vehicle control models to investigate these interactions [7], [8]. Considering the tether and vehicle together provides a stronger basis for assessing control performance than modelling the vehicle alone. However, these studies primarily examine vehicle response rather than the deployment and retrieval of cable. They therefore support the need to account for tether forces but do not directly establish a suitable winch control strategy for this application.

2.3 Tether management challenges

Effective tether management requires sufficient cable to permit unrestricted vehicle motion while avoiding unnecessary deployed length. Excess tether increases hydrodynamic drag and the likelihood of entanglement with the vehicle, seabed or submerged infrastructure [7]. This risk is particularly relevant in mussel farms, where ropes and suspended farming equipment create a dense underwater environment.

Conversely, insufficient tether can restrict movement and introduce tension forces that interfere with vehicle control [9]. Tether loading can also vary due to vehicle motion, hydrodynamic drag, ambient current and changes in cable configuration. As a result, determining when tether should be extended or retracted is not straightforward and requires a suitable control input or sensing method.

2.4 Existing tether management systems

The simplest approach to tether management is manual operation, where an operator monitors the vehicle and releases or retrieves tether as required. Although mechanically simple and adaptable to changing conditions, this requires continuous supervision and limits the overall level of system automation.

Several automated approaches have been investigated. Tortorici et al. [10] developed a compliant buoy–ballast arrangement with a flex sensor providing feedback to a surface cable feeder. Water-tank testing demonstrated its ability to maintain a partially stretched tether while limiting unwanted pulling on the ROV. Sensing near the vehicle is an advantage because it captures the local tether condition. However, the additional submerged components would require assessment for use among mussel ropes, and the reported tank tests do not establish performance in that environment.

Another approach combines an autonomous surface vehicle with an ROV through a tether management and docking system [11]. Laboratory and at-sea trials provide evidence that the concept can operate beyond controlled conditions. However, it requires an additional robotic platform and coordination between the surface vehicle, ROV and tether controller. This adds considerable complexity compared with a reel mounted on an existing crewed vessel.

Commercial systems also provide powered tether handling. The Nautilus system incorporates electric motors, braking, tether tensioning, a level-wind mechanism and an external control interface [12]. These features address both cable movement and its distribution on the drum. However, the manufacturer's description alone does not establish its suitability for regulating tether loading during mussel-farm scanning. Compatibility with the existing tether and vehicle would still need to be confirmed.

2.5 Research gap

The reviewed literature demonstrates methods for modelling tether–vehicle interaction and automating tether handling. However, these approaches do not directly establish a suitable solution for the existing reel and small ROV used in this project. The application requires a compact system that limits excess deployed tether without introducing forces that restrict vehicle movement.

Mussel farms also present entanglement risks associated with closely spaced suspended ropes and infrastructure. A practical system must therefore accommodate both tether loading and the vehicle's movement through this environment. This project investigates whether a powered retrofit of the existing reel, using motor current as an indirect control input, can provide a feasible basis for automated tether management. Bench testing and field observations are used to assess the approach and identify requirements for further development.

3 Methods and Materials

3.1 Project Objectives

The objective of this sub-project is to develop and demonstrate a proof-of-concept automated tether management system for a remotely operated underwater vehicle used for underwater inspection and sampling. The system will regulate deployed tether length by detecting changes in tether tension. This will be achieved by: (i) evaluating a suitable method for estimating tether tension; (ii) designing and constructing a mechanical winch retrofit compatible with the existing tether system; (iii) integrating the selected sensing method, motor, controller, and control logic; and (iv) validating the integrated system through controlled bench testing and in-water trials. The outcomes will determine the feasibility of automated tether management and identify requirements for further development towards commercialisation.

3.2 Approach

3.2.1 Tension-sensing method

The compliant buoy–ballast arrangement [10] was initially preferred because it provides feedback near the ROV while limiting forces transmitted to the vehicle. However, the additional submerged components, sensing and communication could not be developed within the project timeframe.

Motor current was therefore selected as an indirect measure of winch loading, using the measurement available through the electronic speed controller. This allowed the concept to be investigated without modifying the ROV or adding submerged sensors. The approach assumed that changes in tether tension would produce detectable changes in current. However, current also reflected drivetrain resistance and acceleration, so it was treated as a control input rather than a calibrated tension measurement.

3.2.2 Mechanical winch development

Commercial systems and powered winches were investigated but presented cost, capacity or integration constraints. Retrofitting the existing manual reel retained compatibility with the 150–300m tether and electrical passthrough arrangement, reducing development requirements.

The client-supplied skateboard motor drove the reel through a GT5M timing belt. A 15-tooth motor pulley and 100-tooth reel pulley provided a 6.67:1 speed reduction. The belt accommodated minor misalignment and allowed changes to the reduction ratio, while an adjustable idler provided tension. The initial mounting plate was laser-cut from MDF, and the reel pulley and spacers were additively manufactured. All printed components used PLA with 25% honeycomb infill, five wall loops, and four top and four bottom shell layers. The existing handle was retained for manual winding.

Mechanical revisions included an aluminium mounting plate, stainless steel fasteners, a metal torque-transfer adapter and an idler with two S6000-2RS bearings on a 10 mm shoulder bolt.

This targeted durability in the drive and handle connection, and idler friction. Mechanical drawings, component dimensions and print specifications are provided in Appendix B.

3.2.3 System integration

A VESC 4.12-based ESC controlled the motor from the existing nominal 18 V battery supply. The final motor-current limit was set to 4 A, based on in-water operation showing that this was sufficient for tether handling without pulling the reel from its position on the boat. This restricted available torque and reduced the risk of excessive retrieval force. Earlier bench tests used a different configuration.

Control logic was initially implemented within the ESC before being transferred to an ESP32 communicating through UART. The final software filtered measured motor-current magnitude and compared it against the threshold boundaries. Low current commanded retraction, high current commanded extension, and intermediate current commanded hold. Hysteresis and a minimum time between state changes were incorporated to reduce rapid switching. Automatic movement was limited to 100 motor rpm, with speed ramping to soften transitions. Hold commanded zero speed, with the ESC configuration intended to maintain the reel stationary.

The hand controller provided manual operation and remote stopping of automated movement. Throttle input outside the neutral deadband disabled automatic control until the ESP32 was restarted. Manual inputs were scaled to a maximum of 500 motor rpm in either direction and ramped to reduce sensitivity. Returning the throttle to neutral then commanded zero speed. Loss of current feedback during automatic operation commanded hold, while loss of hand-controller communication during manual operation issued a zero-speed command. These functions did not disconnect electrical power. Appendix C contains the code repository, C.1 contains the initial bench-test code and C.2 contains the final controller.

3.2.4 System testing

Exploratory bench testing examined behaviour during state transitions and whether current changes could reliably indicate changes in tether tension. Current and thresholds were monitored using the Arduino Serial Plotter while the tether was manually restrained or allowed to retract freely.

Restrained testing used a lower threshold of 5 A to observe the retract-to-hold transition. During unrestricted retraction, this threshold was increased to 9 A to maintain retraction while observing current over one drum revolution. The unrestricted test was repeated with the anti-free-spool brake tensioners removed to investigate their contribution to reel resistance. No calibrated loads were applied, so testing provided a qualitative assessment.

In-water testing explored tether handling during descent, seabed travel, mussel-rope scanning, resurfacing and retrieval. The original hand controller was used to adjust payout and retrieval in response to vehicle movement. Attention was given to tether loading, slack accumulation and cable distribution. No direct tension measurements were taken, and autonomous control was not tested in water. The project hazard assessment is provided in Appendix A.

Following field use, the assembly was inspected for deterioration, binding and damage. Revised components were checked for fit, alignment, clearance and free idler rotation.

4 Results

4.1 Initial powered reel retrofit

The mechanical retrofit was successfully manufactured and assembled onto the existing hand-operated tether reel, as shown in Figure 1. The completed prototype incorporates the motor, belt transmission, adjustable tensioner and electronics enclosure while retaining the existing tether and electrical passthrough arrangement. Following installation of the timing belt, powered operation of the reel was successfully demonstrated in both directions.

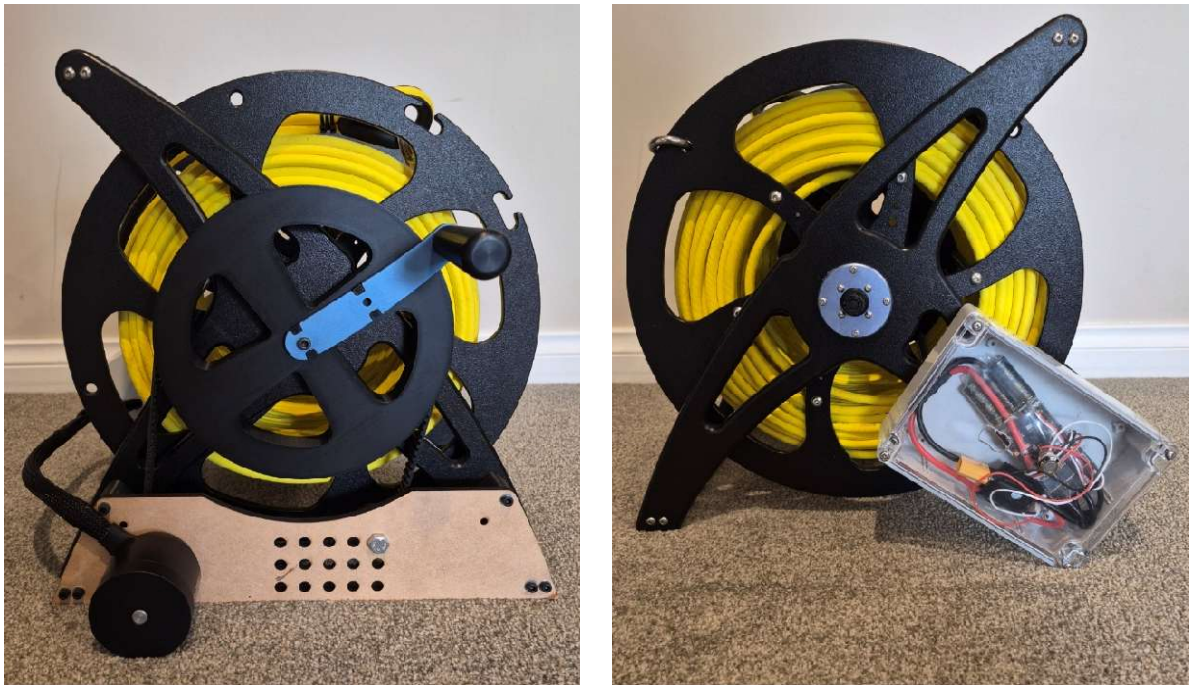


Figure 1. Initial tether reel retrofit showing the drive assembly and electronics enclosure.

The completed assembly demonstrated that the existing tether reel could be converted to powered operation while retaining its original functionality. The motor and transmission were able to drive the reel as intended, providing the mechanical platform required for integration and testing of the automated control system.

4.2 Initial controller behaviour

Initial testing of the VESC-based control system demonstrated commands for the required three operating states of tether extension, retraction and hold using motor current as the control input. The initial implementation used current commands to control the motor; however, rapid transitions between control states resulted in inconsistent motor behaviour. Changing the output to velocity commands produced smoother motor response through the ESC's built-in acceleration ramping.

Despite this improvement, intermittent failures remained within the VESC scripting environment. The script would occasionally stop operating unexpectedly, while sporadic transitions between control states were also observed. These issues prevented reliable operation of the automated control system and led to the control logic being transferred to an external ESP32 communicating with the VESC through UART. The ESP32 control code used to generate the motor-current data presented in Figures 2 and 3 is provided in Appendix C.1.

Preliminary testing of the ESP32 implementation was conducted while monitoring motor current and the control thresholds using the Arduino Serial Plotter. With the tether restrained and the lower threshold set to 5 A, the current repeatedly increased above the threshold before falling below it again. This produced the pronounced oscillation shown in Figure 2, with fluctuations of approximately 6 A.

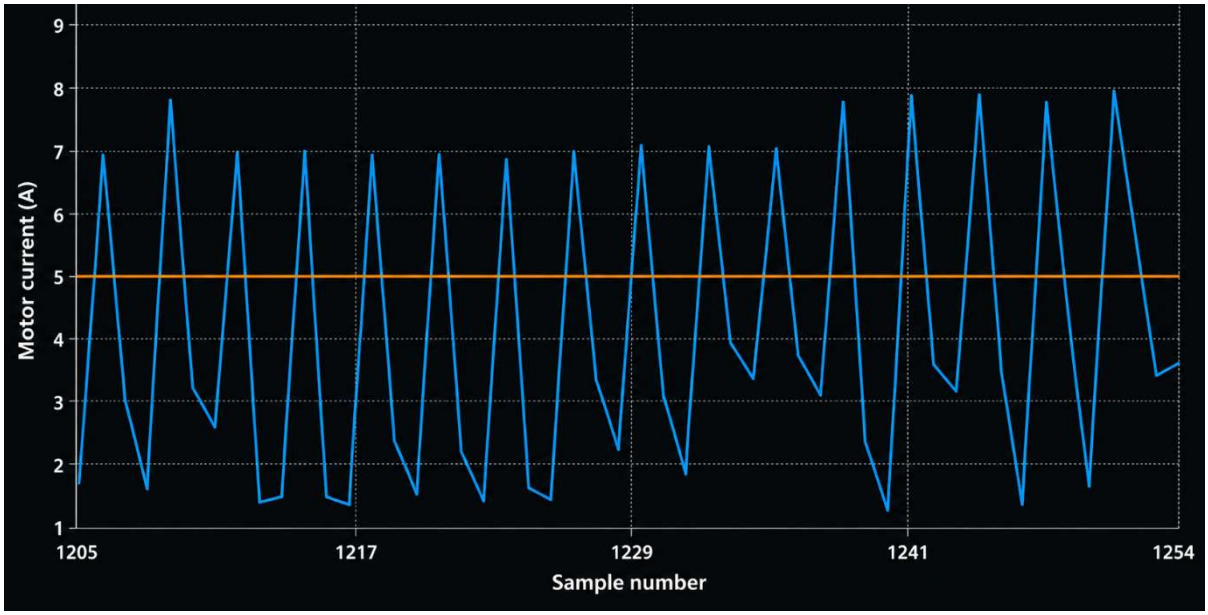


Figure 2. Motor current during restrained tether testing with a lower control threshold of 5 A.

A second test was conducted with the tether free to retract and the lower threshold deliberately increased to 9 A to prevent transitions from the retract to hold state. Under continuous retraction, the large oscillations were no longer observed and the current varied more gradually between approximately 4 and 7 A, as shown in Figure 3.

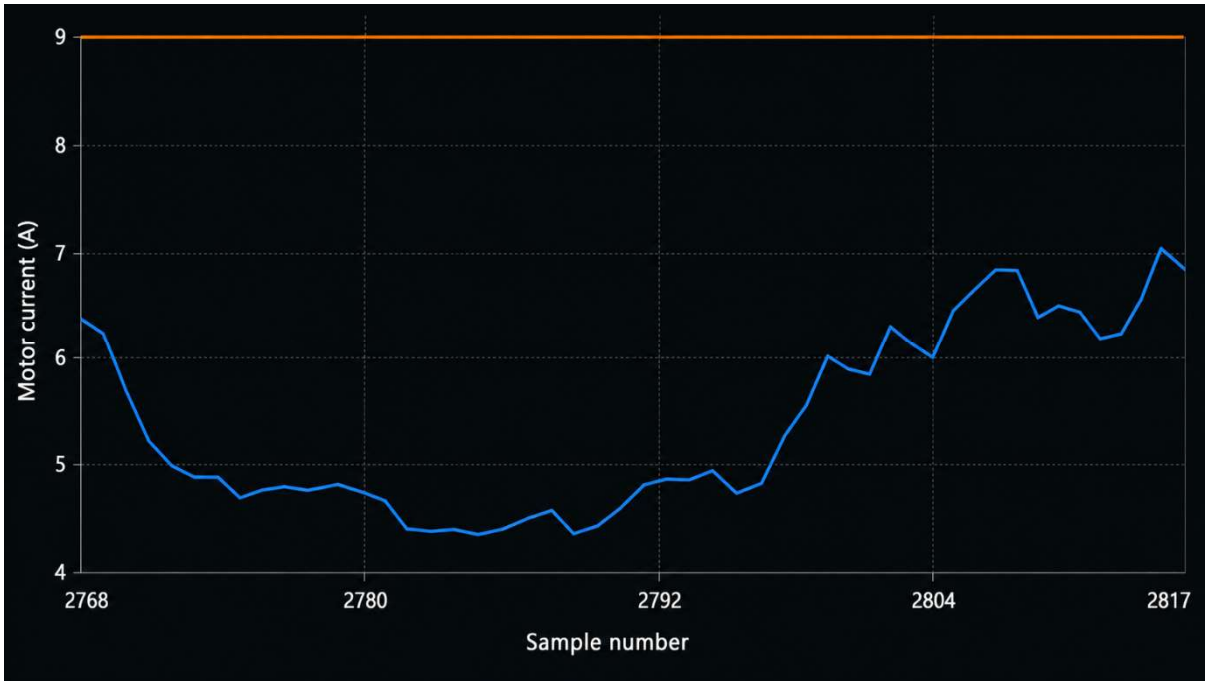


Figure 3. Motor current during unrestricted tether retraction with a lower control threshold of 9 A.

Removing the anti-free-spool brake tensioners lowered the current required for rotation, but a similar pattern of variation remained. The brake therefore contributed to the load but did not account for all the variation during unrestricted retraction. These tests did not establish a calibrated relationship between motor current and tether tension.

4.3 In-water tether handling

4.3.1 Tether behaviour during vehicle movement

Initial in-water testing at Lyttelton Marina provided an understanding of tether behaviour during representative ROV operation. Despite being neutrally buoyant, the tether followed the ROV through the water column rather than returning directly towards the surface. This behaviour was most apparent when the ROV descended, travelled along the seabed, and subsequently resurfaced. During tether retrieval, the reel could pull the ROV back underwater rather than recovering the submerged tether towards the surface.

The trial demonstrated that tether behaviour depended on ROV motion and operating conditions. These observations highlighted the influence of different operating conditions on tether loading and indicated that separate control parameter profiles may be required for different operational tasks.

4.3.2 Mussel-farm scanning

Before commercial sea-trial operation, the external buck converter failed to provide a stable 5 V supply to the ESP32. The fault had been masked while the laptop supplied USB power. The ESP32 was therefore bypassed, and the hand-controller receiver was connected directly to the ESC. This restored the original manual arrangement, without automated control or the ESP32-based scaling and ramping.

The powered reel was used under manual control during mussel-rope scanning. The ROV could pull tether from the reel without motor assistance, but tether resistance affected vehicle stability and increased the thruster power draw observed on the live display. Applying approximately 0.2-0.5 A of motor current assisted payout by overcoming reel resistance. Payout faster than the vehicle required produced loose cable and tangles on the drum, as shown in Figure 4(a).

Once the ROV surfaced, powered retrieval pulled it back across the surface towards the vessel, reducing the need for thruster-powered return. Lower thruster power draw was observed on the live display, and fewer battery swaps were reported during operation. No power histories or battery-use records were collected, so the energy saving was not quantified. The 4 A limit selected for the final ESC configuration was sufficient for the mussel-farm handling conditions, with the reel remaining in position on the deck.



Figure 4. Field trials: (a) tether tangling on the reel; (b) powered reel during mussel-farm scanning.

4.3.3 Seabed scanning

Seabed scanning required substantially more deployed tether, with approximately 100-150 m recovered during retrieval. Tangles on the reel were more frequent and interrupted scanning. Cable accumulated unevenly on the drum, requiring manual guidance during winding. Higher retrieval tension could also draw the tether into underlying turns and cause it to become trapped.

Water currents had a greater influence on the longer deployed tether than during mussel-farm scanning. Greater motor current was required for handling, and at 4 A the motor struggled to retrieve the tether after a transect and stalled on several occasions. Conversely, when resisting tension was too low, water currents pulled additional tether from the reel and deployed more cable than the vehicle required. Tether resistance could also impede or stall the ROV during scanning.

Powered surface retrieval remained useful for returning the ROV to the vessel with reduced thruster use. However, the reel still required an operator to manage cable distribution and tangles. Neither the magnitude of water-current loading nor its effect on retrieval performance was measured.

4.4 Final controller behaviour and electrical failure

After the sea trials, the ESP32 was reintroduced and powered from the ESC's 5 V supply. Further investigation identified that the ESC treated a zero-rpm command as an inactive output under its original speed-controller settings. During restrained-tether testing, current increased until the controller entered hold. The motor then stopped applying drive, current fell below the lower threshold, and retraction resumed. This repeated sequence accounted for the observed retract-to-hold oscillation.

Changing the ESC speed-control dead zone from its default of 900 electrical rpm to zero enabled active zero-speed control. In qualitative bench testing, the reel remained stationary in hold and the repeated oscillation was no longer observed. Retraction, hold, and extension, responded as intended when tether loading was changed manually. Current filtering, hysteresis and a minimum state duration were also included in the final implementation, although their individual effects were not measured separately.

Manual input successfully overrode automatic operation, which remained disabled until the ESP32 was restarted. Scaling and ramping the hand-controller commands made manual movement smoother and less sensitive. The abrupt payout and stopping behaviour that had produced loose loops and tangles on the reel was no longer observed during these bench checks. Communication timeouts were implemented in the final code, but no separate test results were recorded for these functions. The final code is provided in Appendix C.2.

The ESC subsequently failed before quantitative recordings of the final controller could be collected. Its indicator lights extinguished, and the connected ESP32 and hand-controller receiver lost power. Approximately 16 V remained at the ESC input, but its 5 V output was absent. The ESP32 operated when powered independently through USB. A replacement ESC was ordered, but it did not arrive in time for quantitative testing before submission.

4.5 Mechanical deterioration and redesign

Inspection after the sea trials found that the MDF mounting plate had swollen and several steel components had begun to corrode. The old, printed idler was found bound to its running sleeve during disassembly. The printed handle-engagement tabs on the reel pulley also broke during manual retrieval. These failures limited the durability of the initial prototype.

The waterjet-cut aluminium mounting plates and revised idler were installed on the reel, shown in Figure 5. The plate was made in two pieces to accommodate the workshop cutter's size limit. The replacement idler used two bearings on a 10 mm shoulder bolt, with 2 mm-thick M10 washers as spacers and an M10 spring washer applying axial preload to the bearing inner races. The M8 threaded end was positioned in the mounting-plate adjustment slot.

The new idler in Figure 6 rotated freely when secured, and its position could be adjusted and fixed to set belt tension. Its rotational resistance was higher than that of the original idler before binding but was more consistent. The revised construction removed direct sliding contact between the printed pulley and its sleeve. It had not undergone further sea trials, so its long-term durability was not established.



Figure 5. Revised reel assembly with the installed aluminium mounting plates and bearing-supported idler.

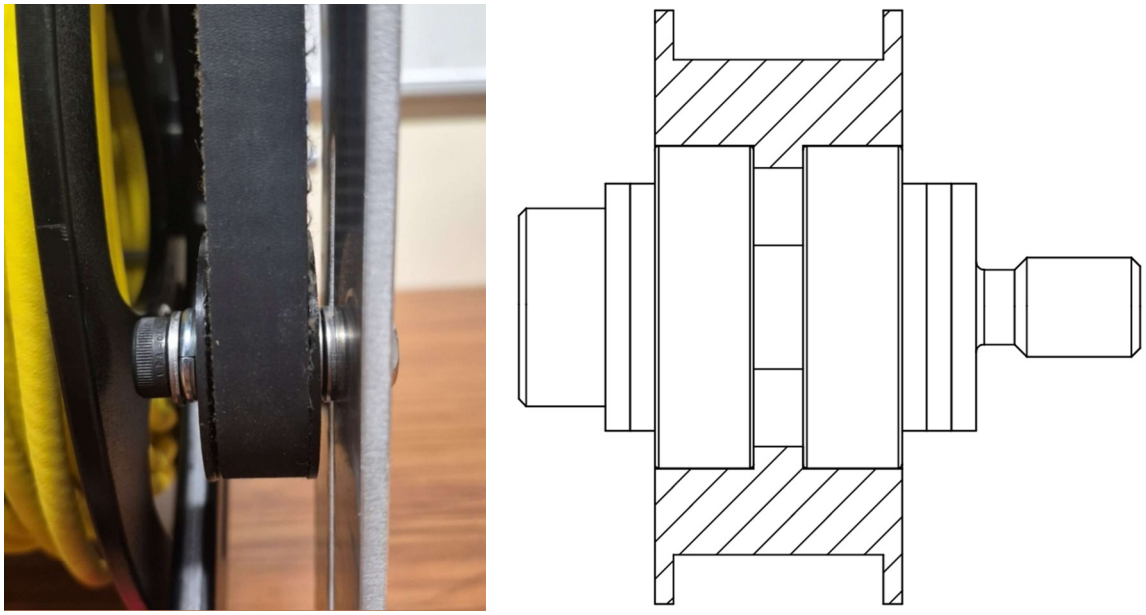


Figure 6. Redesigned idler: (a) installed belt-tension adjustment; (b) bearing, shoulder-bolt and washer arrangement.

A metal torque-transfer adapter was designed to replace the failed printed handle connection and transmit rotation between the pulley and reel, as shown in Figure 7. The drawing was supplied to the workshop, but manufacture and installation were not complete at submission. The component drawings and specifications are provided in Appendix B.

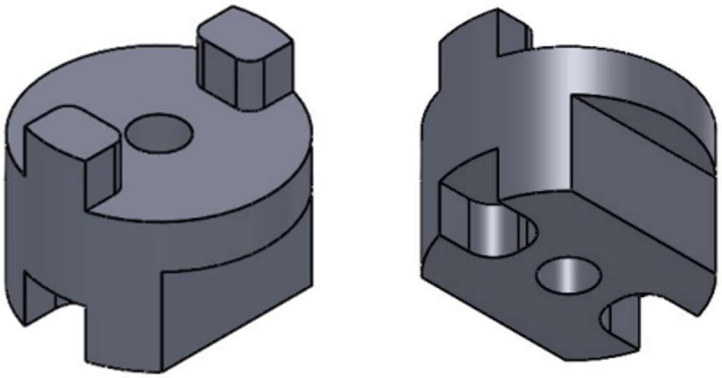


Figure 7. CAD model of the proposed metal torque-transfer adapter

Inspection of the client-supplied motor found exposed windings without a visible protective coating, raising concerns about corrosion and suitability for long-term marine use. The revised mechanical assembly was completed, but the replacement ESC had not arrived for powered testing of that assembly before submission.

5 Discussion

5.1 Findings

The results demonstrate that the existing tether reel provides a viable mechanical platform for powered tether management while retaining the original tether and electrical passthrough arrangement. Powered payout and retrieval were demonstrated during both bench testing and field operation. However, the field trials also showed that reliable tether management depends on cable distribution, mechanical durability and the interaction between the tether and ROV, alongside the control system.

The initial VESC-based controller demonstrated the required extend, retract and hold commands, but limitations within the scripting environment prevented reliable operation. Transferring the control logic to an external ESP32 provided greater flexibility while retaining motor-current measurement and velocity control through the VESC. Further testing identified that the ESC's response to a zero-rpm command caused the repeated switching between retract and hold. Changing the speed-control dead zone enabled active hold and resolved the oscillation during qualitative testing. This showed that the observed instability arose from the implementation of the hold state rather than establishing that motor-current feedback was unsuitable.

The remaining current variation during unrestricted retrieval indicated that the baseline load required to rotate the reel was not constant. Removing the brake tensioners reduced the current but did not eliminate this variation. Motor current therefore reflected both tether loading and mechanical resistance within the reel and drivetrain. Although the final controller

responded as intended during qualitative testing, this did not establish a reliable relationship between current and tether tension. Filtering reduced the influence of short-term fluctuations in the control logic but could not distinguish between sustained mechanical resistance and sustained tether loading.

The in-water trials further showed that tether behaviour depended on vehicle motion, deployed length and environmental conditions. The longer tether used during seabed scanning was more affected by water currents, with insufficient resistance allowing excess cable to be pulled from the reel and higher loading making retrieval difficult. Together with the reel tangling, these findings indicate that separate control parameters and improved cable distribution may be required for different operational tasks.

5.2 Implications

Powered tether handling provided a practical benefit during the commercial trials. Assisting payout reduced the resistance that the ROV had to overcome, while powered surface retrieval reduced the need for the vehicle to return using its thrusters. The lower power draw observed on the live display suggests that this could increase operating time between battery changes. However, the extent of this benefit could not be quantified without recorded power and battery-use data.

The use of an external ESP32 has important implications for further development. Separating the control logic from the VESC avoided reliance on the scripting environment and allowed current filtering, state timing and manual override to be incorporated. Scaling and ramping the hand-controller inputs also improved manual operation by reducing abrupt reel movement. The additional interfaces available on the ESP32 provide a practical pathway for incorporating direct tension sensing or vehicle control information.

The field trials demonstrated that a powered reel still requires manual supervision when cable is not distributed evenly across the drum. A level-wind mechanism would therefore be necessary to reduce operator involvement and address the tangles that interrupted scanning. This remains a mechanical requirement even if the automatic controller responds correctly to tether loading.

The mechanical revisions addressed the deterioration and failures observed during field use. The aluminium mounting plates improved resistance to moisture, while the bearing-supported idler removed direct sliding contact between the printed pulley and its sleeve. Although the new idler had greater rotational resistance, this was more consistent. The revised assembly still requires further testing before improvements in long-term reliability can be confirmed.

5.3 Limitations

The primary limitation is that the relationship between motor current and tether tension was not quantitatively characterised. The bench tests used manually applied resistance without calibrated loads or an independent tension measurement. Consequently, the results demonstrate that current responds to changes in loading, but do not establish its accuracy or

repeatability as an indirect measure of tension. Drivetrain resistance, acceleration and changes in the effective winding radius may also influence this relationship.

Qualitative testing of the final controller demonstrated the intended retract, hold and extend behaviour, manual override and smoother manual operation. However, ESC failure prevented quantitative recordings of the final implementation. The reduction in oscillation was therefore observed but could not be assessed through measures such as settling time or current variation. The individual effects of filtering, hysteresis and state timing were not measured separately, and the communication timeouts were not separately tested.

The in-water trials used the original hand controller after the external power-supply fault prevented ESP32 operation. They therefore assessed powered tether handling rather than autonomous control. The trials formed part of time-limited commercial operations, and recording control and power-consumption data was outside their scope. Observations of vehicle stability, thruster power draw and battery use could not be supported by controlled comparisons. Differences between mussel-rope and seabed scanning also involved different tether lengths and environmental conditions, so the contribution of each factor could not be isolated.

The revised mechanical assembly had not undergone further powered or in-water testing. The new idler's consistent resistance and successful adjustment therefore demonstrate its assembly functionality, but not its long-term durability. The metal torque-transfer adapter had also not been installed or tested. Inspection of the motor raised concerns about corrosion protection while operating in marine environments.

5.4 Future work

Following completion of the current proof-of-concept, future development should investigate alternative methods of improving tether-state estimation and control. Direct tether-tension sensing could provide a more accurate measure of loading than motor current, while sensing at both ends of the tether could provide greater understanding of the forces acting between the winch and ROV. The buoy–ballast arrangement considered during the initial design phase should also be reconsidered as a means of maintaining a more predictable tether load and simplifying the tether–vehicle interaction.

Greater integration with the ROV control system should also be investigated. Incorporating vehicle motion and thruster-control information could help distinguish vehicle-induced changes in tether loading from environmental disturbances. Combined with a tether dynamics model, ROV position data and measurements of deployed tether length, this could enable prediction of the required cable length and allow the winch to operate proactively rather than responding only after changes in loading occur.

A level-wind mechanism should be incorporated to distribute tether evenly across the drum during retrieval. This would address the cable accumulation and tangling observed during the sea trials and reduce the need for manual guidance. Together with improved sensing and control, this would support further development towards unattended tether handling.

A further extension should investigate incorporating ROV deployment and recovery into the tether management system. Developing the winch and supporting structure to safely lift the ROV from the water could reduce manual handling and simplify vehicle recovery. This would require further investigation of winch capacity, tether loading, structural requirements and appropriate safety measures.

6 Conclusions

The project established a proof-of-concept platform for automated tether management of the ROV. The existing manual reel was successfully converted to powered operation while retaining the original tether and electrical passthrough arrangement. Field trials demonstrated that powered payout and retrieval could assist vehicle operation, with reduced thruster power draw observed during assisted payout and surface recovery.

Transferring the control logic from the VESC scripting environment to an external ESP32 allowed the controller to be refined and manual override incorporated. The initial retract-to-hold oscillation was traced to the ESC treating a zero-rpm command as an inactive output. Changing the speed-control dead zone enabled active hold and resolved the oscillation during qualitative bench testing. The final implementation demonstrated the required retract, hold and extend behaviour, while scaling and ramping the hand-controller commands produced smoother manual operation.

Motor current responded to changes in tether and drivetrain loading, supporting its use as an indirect control input. However, variation in mechanical resistance meant that its reliability as a measure of tether tension was not established. ESC failure prevented quantitative assessment of the final controller, and the manually controlled sea trials did not validate autonomous operation in water.

The field trials also demonstrated that tether handling requirements varied with deployed length and environmental conditions. Longer deployments increased the influence of water currents, while uneven winding and reel tangling interrupted scanning. Mechanical deterioration led to revised aluminium mounting plates and a bearing-supported idler, although the revised assembly had not undergone further powered or in-water testing.

Overall, the work demonstrated practical powered tether handling and the intended automatic control behaviour on the bench. Further development requires improved cable distribution, quantitative assessment of tension feedback and in-water testing of the integrated system before reliable autonomous operation can be established.

Declarations

Author contributions: Isaac Rowe: Conceptualisation, Investigation, Methodology, Software, Validation and Writing.

Funding: This work was partly funded by Richard Green and the University of Canterbury Machine Vision Department.

Conflict of interest statement: The authors declare that there are no financial conflicts of interest regarding this work.

Acknowledgements: The author wishes to thank Richard Green, Tim Rensen and Yilei Zhang.

AI usage: AI tools, namely ChatGPT6.1 (OpenAI) and Copilot (Microsoft) were used to assist with research, broadening knowledge and clarifying human written text. Grammarly was used for spell-checking, grammar and improving the readability of the original text. All AI outputs were verified, and all ideas, opinions and analysis in this report are my own.

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Appendix A - H&S hazard assessment

Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per 'Severity of harm' scale 1. Hazard occurred 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost certain 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment 7 or less No intervention necessary.	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
Prototype build	Workshop equipment	5	4	20	Only operate machinery after training from a workshop technician, wear PPE, use machine gaurds	Myself, Workshop technicians	Yes	3	3	9	Yes - Further action not neccessary
Prototype build	Electricution	4	3	12	Ensure system is grounded at all times with a fused connection to supply	Myself	Yes	2	2	4	Yes - Further action not neccessary
Prototype build	Electrical fire	3	2	6	Monitor power draw and ensure system is grounded	Myself	Yes	2	2	4	Yes - Further action not neccessary
Testing	Tether snapping	3	2	6	Monitor reel/tether to ensure no snags/sharp objects threaten teather	Myself	Yes	3	2	6	Yes - Further action not neccessary
Testing	Short circuit from water ingress	4	3	12	Test water resistance of components before exposing to a wet enviroment and ensure system is grounded	Myself	Yes	3	2	6	Yes - Further action not neccessary
Testing	Human Entanglement	5	3	15	Guard around rotating reel, maintain safe distance during operation, emergency stop accessible to operator	Myself	Yes	3	1	3	Yes - Further action not neccessary
Tether Reel	Tangled	2	3	6	Monitor reel/tether to ensure no tangles generate while in use	Myself	Yes	2	3	6	Yes - Further action not neccessary
Tether Reel	Tether tension overload	3	3	9	Load sensor/strain gauge, automatic shutoff	Myself	Yes	2	2	4	Yes - Further action not neccessary
Tether Reel	Reel motor runaway	3	2	6	Software limits, Emergency stop	Myself	Yes	2	2	4	Yes - Further action not neccessary
Tether Reel	Jammed	2	3	6	Monitor reel/tether while in operation and manage tether to ensure no jams	Myself	Yes	2	3	6	Yes - Further action not neccessary
Battery	Overheating	3	2	6	Check battery health/condition prior to operation and monitor power draw while in use	Every handler	No	3	2	6	Yes - Further action not neccessary
Battery	Electric Fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not neccessary
Battery	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not neccessary
Propellers	Collision with object/ wildlife	3	4	12	Monitor surroundings, Ensure personal safety, before startup inspect propellers	Every handler	Yes	3	3	9	No - System only operated by trained personal
Propellers	Mechanical failure	3	2	6	Inspect propellors prior to use of ROV	Every handler	No	3	2	6	Yes - Further action not neccessary

Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per 'Severity of harm' scale 1. Hazard occurred 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost certain 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment 7 or less No intervention necessary.	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
Motors	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not neccessary
Motors	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not neccessary
Motors	Overheating	3	2	6	Monitor power draw from motors while ROV in use	Every handler	Yes	3	2	6	Yes - Further action not neccessary
Controllable Underwater Events	Collision – self inflicted	2	4	8	Montitor surrounding while operating and approach motion controll sensibly	Every handler	Yes	1	4	4	Yes - Further action not neccessary
Uncontrollable Underwater Events	Strong current	2	3	6	Test ocean currents prior to operation and set current threshhold for non-operation	Every handler	Yes	2	3	6	Yes - Further action not neccessary
Uncontrollable Underwater Events	Encounter with wildlife	4	4	16	Do not operate ROV in areas with marine mammals present, Immediate shutdown if wildlife approaches	Every handler	Yes	3	4	12	Yes - Will require constant monitoring
Uncontrollable Underwater Events	Collision	2	3	6	Monitor surroundings and motion of robot while in operation and actively avoid collisions through piloting robot	Every handler	Yes	2	3	6	Yes - Further action not neccessary

Appendix B - Mechanical drawings and component specifications

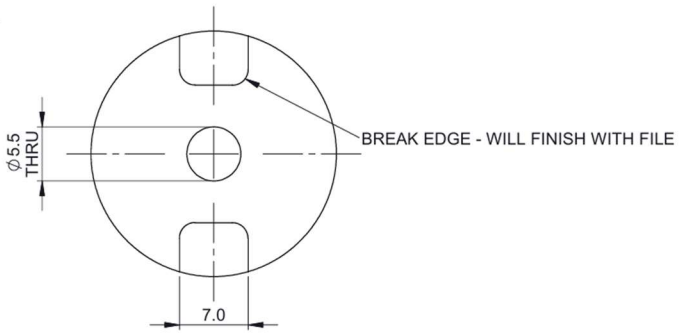
The following drawings document the mechanical components of the tether reel retrofit described in Section 3.2.2. Each drawing is presented on a separate landscape page for readability. All printed components used PLA with 25% honeycomb infill, five wall loops, and four top and four bottom shell layers.

Table B1. Drawing index

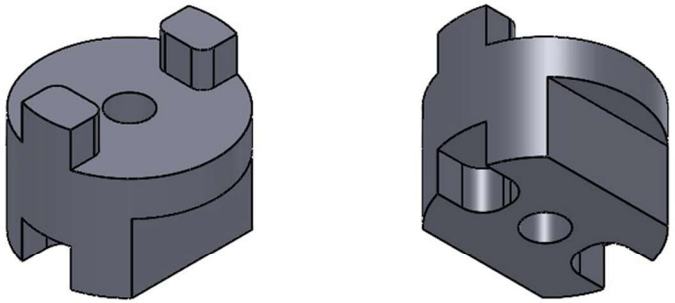
Drawing number	Component	Page
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2001	Mounting plates	102
3001	Idler pulley	103
4001	Reel pulley	104
5001	Mounting spacers	105

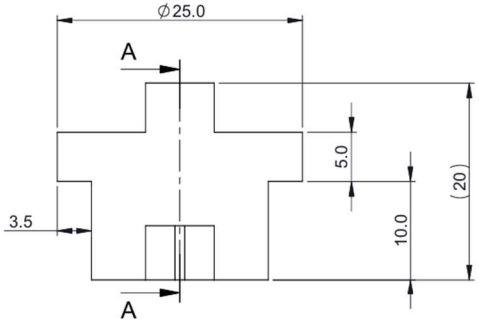
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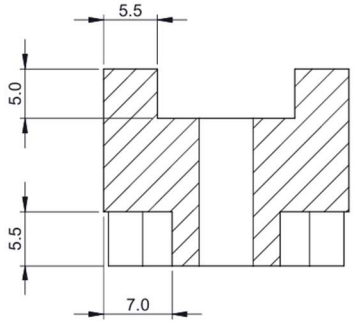


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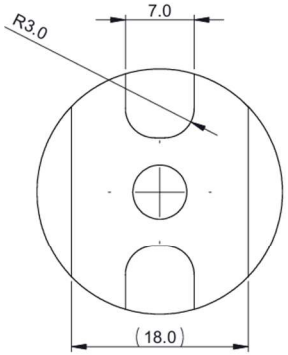




FRONT



SECTION A-A



BOTTOM

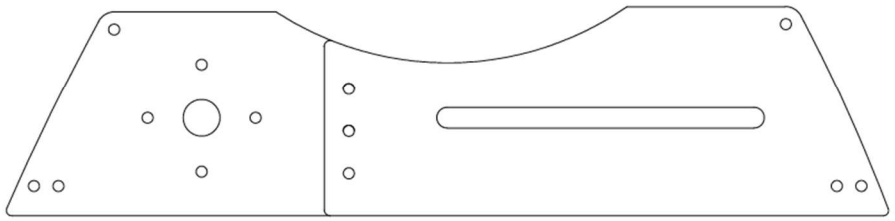
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MATERIAL: ALUMINIUM BAR		PULLEY CORE	
FINISH: AS RECEIVED BREAK ALL EDGES		DESIGN: ISAAC ROWE	DWG NO.: 1001
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		REV: REV1	

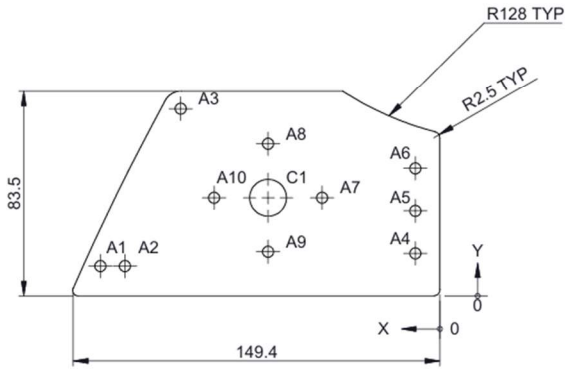
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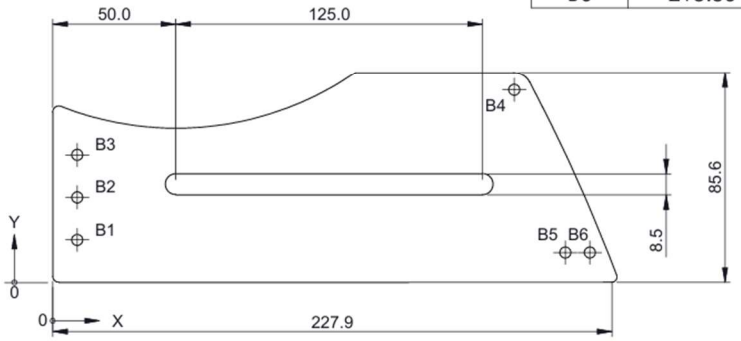
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A4	10.01	17.32	∅4.5 THRU
A5	10	34.64	∅4.5 THRU
A6	10	51.95	∅4.5 THRU
A7	48	40	∅4.5 THRU
A8	70	62	∅4.5 THRU
A9	70.01	18	∅4.5 THRU
A10	92	40	∅4.5 THRU
C1	70	40	∅15.0 THRU

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B1	10	17.33	∅4.5 THRU
B2	10	34.66	∅4.5 THRU
B3	10.01	51.98	∅4.5 THRU
B4	188.07	78.51	∅4.5 THRU
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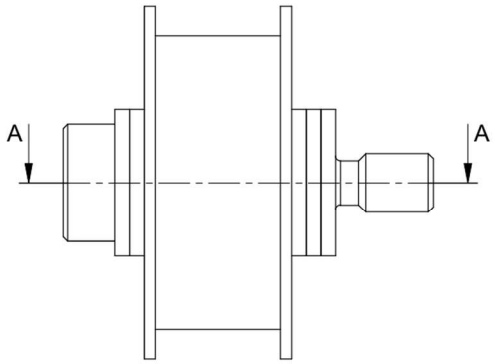


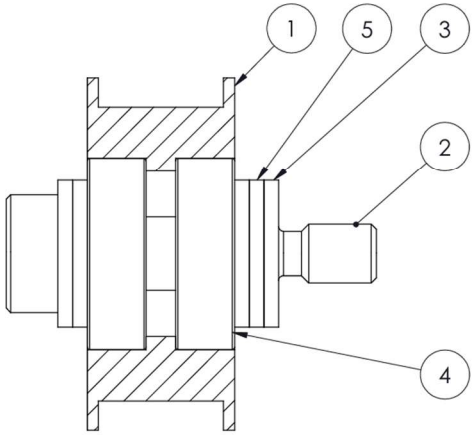
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MATERIAL: 4mm ALUMINIUM PLATE		MOUNTING PLATES	
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			REV: REV1

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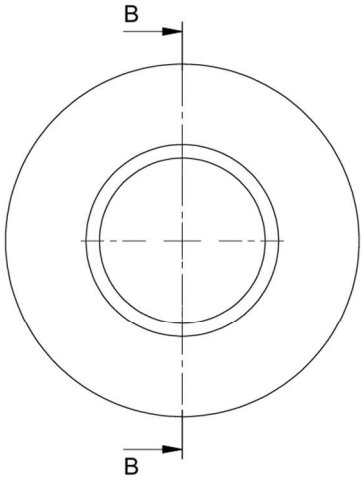
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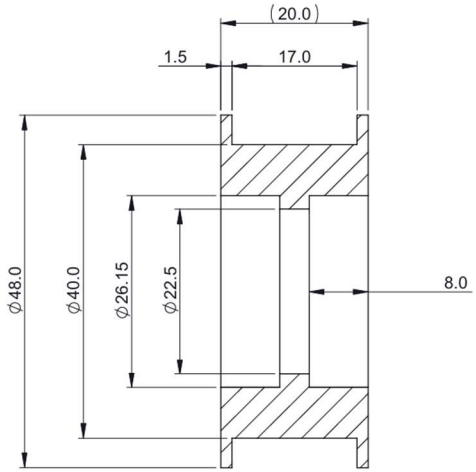
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1	IDLER PULLEY	PLA	1
2	10X30 SHOULDER BOLT	GRADE 12.9 HIGH-TENSILE STEEL	1
3	10x19x2 HEAVY FLAT WASHER 316 STAINLESS	316 STAINLESS	4
4	S6000-2RS	316 STAINLESS, Si3N4	2
5	M10 SPRING WASHER	316 STAINLESS	1





SECTION A-A





SECTION B-B

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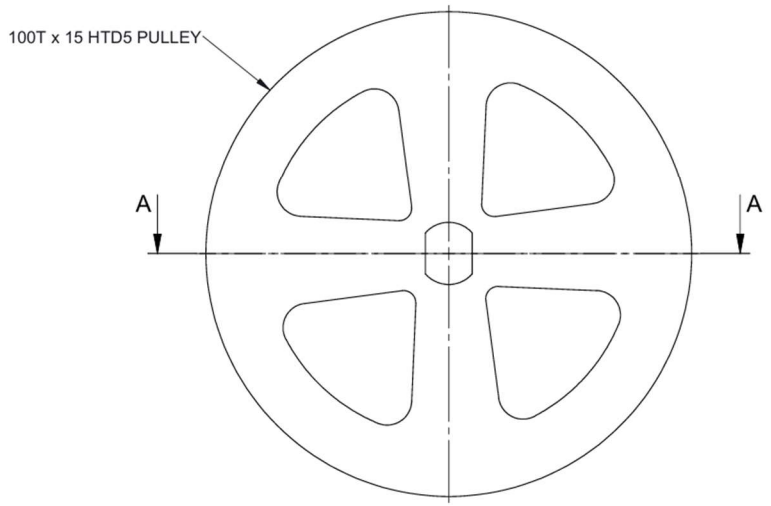
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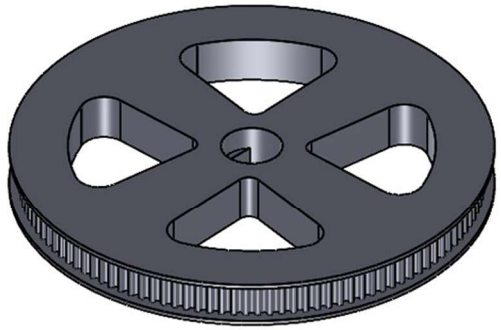
IDLER PULLEY

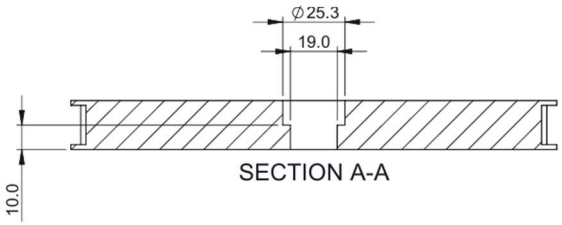
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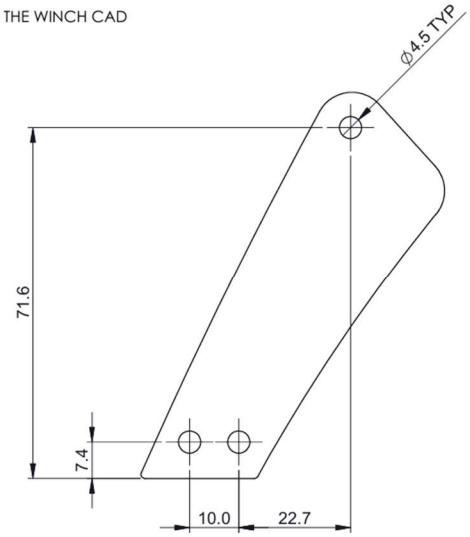


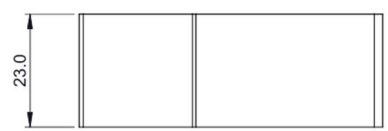


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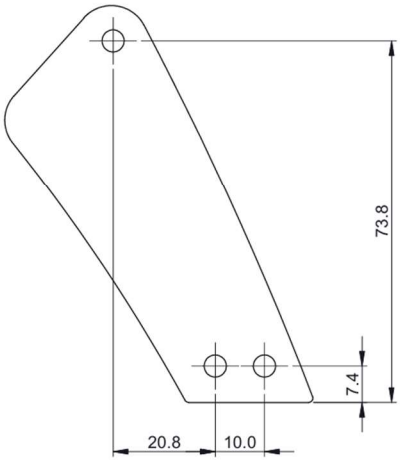
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Appendix C - Bench-test control code

C.1 Bench-test control code

The following code was used for the initial bench tests reported in Section 4.2, including the motor-current traces in Figures 2 and 3. It was subsequently revised; the final implementation is provided below. The code was written with assistance from ChatGPT.

Code can be found in the Git repository (Code_1.0.ino): https://eng-git.canterbury.ac.nz/iro35/m26-tether-management

C.2 Final control code

The following code is the final implementation of the ESP32-based tether management controller. It includes motor-current filtering, hysteresis, speed ramping, manual override and communication timeouts. Qualitative bench testing demonstrated the intended behaviour, but ESC failure prevented quantitative data collection. The code was not validated through in-water testing. The ESC motor-current limit and zero-speed holding configuration were set separately. The code was written with assistance from ChatGPT.

Code can be found in the Git repository (Code_2.0.ino): https://eng-git.canterbury.ac.nz/iro35/m26-tether-management

Appendix D - Mussel Detachment Force

ENME408-2026 Individual Student Report

Force Required to Detach Mussels from their Environment

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Document version: 20261015mod

Abstract

Context: An autonomous underwater drone is being developed for commercialization by UC Vision for the purpose of collecting mussels from mussel lines. The purpose of this is to limit the cost and labour involved with manually collecting mussels for testing as they grow. This process is necessary to test the quality/size/health of growing samples on the mussel line for the mussel farming industry. **Objectives:** The purpose of this paper is to identify the force required for a mussel, specifically green mussels, to be successfully pulled off the material it is attached to using tensile force. **Method:** The method for achieving this was to use a load cell with a handle and claw to test the tensile force required to remove mussels from their environment. This was done in three locations around the Christchurch area to test detachment forces for 1) mussels growing individually 2) clustered together with other mussels. **Results:** There is a clear relationship between a complex growing environment and higher detachment force with the mean forces being 40.13 N for isolated mussels and 65.87 for clustered mussels. This was influenced by substrate type and presence of biofouling around samples. **Conclusions:** The results provide realistic field-based detachment force measurements that can aid in the design of the underwater drone's claw to successfully remove mussels under commercial farming conditions.

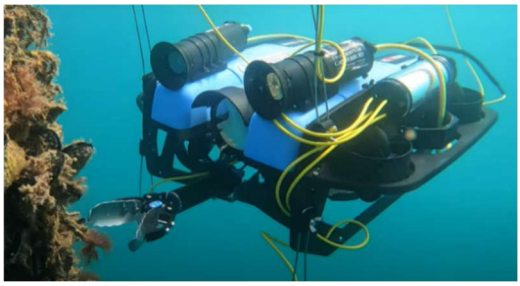
Keywords: Mussel; Byssal

Graphical Abstract



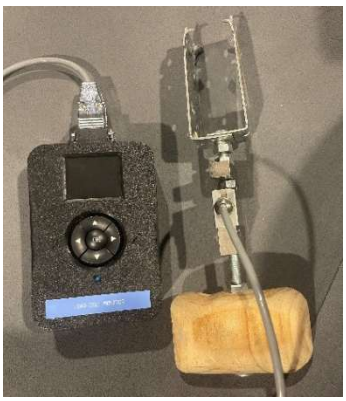
Manual mussel testing is expensive and labour intensive

The switch to UC Vision's Underwater Drone automatically collects mussels for testing



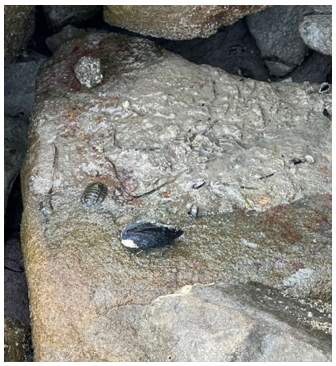
Claw needs sufficient force to detach mussels, this information will impact the claw design

Digital force reader and load cell claw with handle for environmental testing to determine force to detach mussels



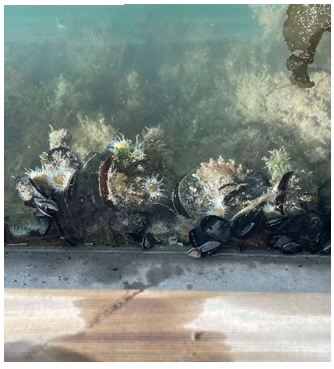
Testing Locations

Beachville Road



Isolated mussels with mean peak detachment force of 40.13 N

Lyttelton Marina



Clustered mussels with mean peak detachment force of 64.17 N

Pigeon Bay



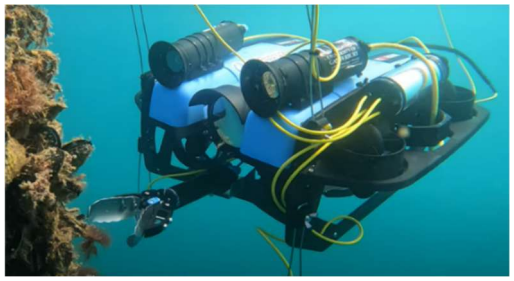
Clustered mussels with mean peak detachment force of 67.58 N

Clustered mussels generally require more force for detachment compared to individual

1 Introduction

1.1 Context

Mussel collection is a vital aspect of the mussel farming industry as it allows farmers to assess the quality, health, and growth of samples during their lifetime. As well as this, the sampling process can help determine the best harvesting time for the highest yield, directly impacting the profit of mussel farming companies. Traditional collection methods are often labour-intensive, time consuming, and costly because they require multiple staff and specialised harvesting vessels to access and remove the mussels. To combat the human labour and cost of this process, an underwater drone, see Figure 1, is being developed by UC Vision to autonomously identify and collect mussels for testing. The drone uses computer vision to locate correctly orientated mussels for collection and then uses a mechanical claw to detach the samples and store them in a net until the drone resurfaces. These samples are then tested by farmers to validate mussel health, growth, and overall quality. Automating this process



saves time, money, reduces the reliance on labour-intensive sampling methods, and improves the consistency of sample collection.

Figure 1: UC Vision's Underwater Drone for Mussel Collection. Image Public domain https://ucvision.org.nz/aquaculture/

1.2 Area under examination

A key issue with the autonomous collection of mussels is detaching them from the mussel lines they grow on without severe damage being done to the outer shell. This process currently requires three degrees of freedom from the mechanical arm and the correct growing orientation of the mussel so the claw can properly grab it. The force required to detach the sample is key to successful collection.

As this industry expands, the push for automated systems, to make the collection process more efficient and safer, become more important. Manual collection of mussels is labour heavy work in which 74.4% of physical postures during collection pose risk of musculoskeletal injuries [1]. Especially in ocean withdrawal systems, 54% of the postures of workers detaching mussels had immediate effect on their musculoskeletal systems [1]. This research supports the need for an automated response to help the growth of the mussel farming industry progress safely. As well as the benefit to physical labour, the automated system can save time and money during collection systems for mussel companies.

This report aims to identify the force required for a mechanical arm to detach a mussel from its line, taking into consideration the surrounding biofouling and effects of individual vs clustered mussel attachment.

2 Literature Review

2.1 Background

Mussels are commercially grown in marine aquaculture using mussel line systems, where individual mussels attach to a line via a collection of byssal threads commonly referred to as the 'beard'. These threads are produced by glands in the mussel's foot, which release small amounts of adhesive protein, in turn, forming a plaque that anchors the threads to the mussel line. The individual threads converge at the byssal root, where they are controlled by the byssal retractor muscles that regulates attachment to a substrate [2]. This phenomenon allows the mussel to remain attached to a material in exposed environments. Understanding the mechanical force required to detach mussels from their lines is important for designing an automated collection system. The detachment force depends on multiple factors including the collection orientation, the material of the aquaculture line, and the presence of surrounding biofouling or neighbouring mussels. This literature review examines research on mussel attachment strength and byssal mechanics to determine the factors influencing attachment and removal.

2.2 Influence of Substrate Material

The material which the mussel attaches itself to plays a significant role in determining the strength of the attachment. In aquaculture systems, mussels commonly grow on rope or plastic composites designed to encourage settlement. Research on substrate materials shows that the interaction between the byssal adhesive and the surface of the substrate influences attachment strength.

Brenner et al. investigated the attachment properties of blue mussels (*Mytilus edulis*) on different artificial collector substrates [3]. These experiments demonstrate that the force required to detach mussels varied depending on the type of material used as the substrate. Mussels attached more strongly to rough materials that allowed byssal threads to intertwine with the substrate, whereas smoother surfaces had weaker attachment forces. This finding is relevant for mussel lines, which are typically made of synthetic fibres such as polypropylene or polyethylene that provide numerous attachment points for byssal threads. Similar findings were reported by Protopopescu et al. who examined the attachment strength of blue mussels on different rope designs in a controlled setting. This experiment identified that complex ropes with larger surface areas resulted in higher mussel settlement populations compared to smoother ropes [4]. The structure of rope also allows multiple threads to attach in different directions, increasing the force of attachment. This research shows that mussels grown on aquatic ropes may require greater detachment forces than those attached to smooth laboratory surfaces.

2.3 Influence of Biofouling and Mussel Clusters

In both natural and aquaculture environments, mussels rarely grow in isolation. They often form clusters, where neighbouring mussels are connected by linking byssal threads and surrounded by other debris such as shells and algae. The presence of neighbouring mussels can significantly influence detachment forces. Additional byssal threads may link adjacent mussels together, meaning that the applied force may be distributed across multiple mussels rather than a single attachment point [5]. This interconnected structure increases the overall resistance to detachment. These environmental factors highlight the importance of considering realistic aquaculture conditions when estimating detachment forces. Laboratory experiments that isolate individual mussels may underestimate the extent of force required to detach compared to commercial longline systems.

2.5 Orientation of Detachment

The orientation of a mussel relative to the tensile force from a collection claw is an important aspect of determining the magnitude of force required for removal. Research on byssal attachment mechanics indicates that mussels have certain modes of byssal failure, including thread rupture, adhesive plaque failure, and root failure [9]. The orientation of the applied force influences which of these failure modes occur. When an applied force acts perpendicular to the mussel, plaque detachment is more likely because the adhesive between the plaque and the surface is placed in tensile stress. In contrast, when the force acts parallel to the mussel, the load is transmitted along the byssal threads, placing them in tension and increasing the likelihood of thread rupture rather than plaque detachment [9]. This is relevant to the present study because the orientation of the applied load may significantly influence the measured peak detachment force. Therefore, the direction of pull used during testing should reflect, as closely as possible, the expected loading direction of the autonomous claw.

2.4 Methodology Comparison

Previous studies have used several techniques to quantify mussel detachment force, these methods were considered when developing this studies' approach. One study measured the attachment strength of mussels directly on natural rocky substrates by tying fishing line around individual mussel bases and connecting the line to a spring scale. The force was applied by pulling perpendicular to the substrate until detachment occurred [6]. Their primary objective was to investigate how attachment strength varied with mussel size, habitat, and season, and how this influenced the probability of mussels being dislodged by waves. Their method was successful however it was reported that mussels located in clusters were harder to access and thus were not recorded. Zardi, G., McQuaid, C., & Nicastro, K. et al. used a similar approach but drilled a small hole through the mussel shell and inserted a fishhook connected by fishing line to a recording spring scale [7]. Force was applied perpendicular to the rock surface. Their study examined seasonal changes in attachment strength and its relationship with wave exposure, however the technique was successful in determining force of detachment. This method also allowed any mussel to be detached due to the base not having to be accessible. A more recent study used a digital force gauge and custom made forceps to grip submerged mussels and, again, pull them perpendicular to their substrate. This method measured the detachment force of the whole mussel rather than the strength of individual byssal threads [8]. Whilst this method was successful in detaching the samples, the purpose of the study was to assess the effects of global warming on attachment. These studies demonstrate that portable detachment testing is an established method for assessing mussel attachment strength, but due to slight variation in research objectives, these methods have been adapted to suit the outcomes being investigated. A portable load-cell system will be used, to closely simulate the claw of the current drone's claw. Most importantly, this study aims to find the force an autonomous collection mechanism would need to generate to remove mussels from realistic harvesting environments including clustered mussels and biofouling.

3 Methods and Materials

3.1 Research Objectives

The purpose of this report was to develop a system to predict the force needed by a mechanical claw to detach mussels from their natural environment. Rather than determining a single detachment force, this research investigated how detachment force varied with environmental and biological factors including sampling location, substrate type, surrounding biofouling, and mussel clustering. The resulting data was analysed to identify the factors associated with detachment force and to determine an appropriate force range for the mechanical claw to ensure successful collection. This research took place in multiple different field locations to understand the attachment properties of a range of mussel samples.

3.2 Approach

3.2.1 Measurement of Mussel Detachment Force

All mussel test subjects were detached from their substrate using a portable load cell and digital force reader. This method allowed the mussels to be detached in their natural orientation and environment. As well as this, field testing allowed the mussels to have limited exposure to the air which could potentially impact their attachment strength. Previous research has shown that the mechanical properties of byssal threads may change after prolonged removal from seawater [10]. To successfully collect data, a handle and claw were designed to aid the pulling of the mussel and were attached to either side of the load cell. This design closely simulated the mechanical grabber of UC vision's underwater drone. Before testing each sample, the load cell was zeroed to ensure proper testing results. The load cell reader recorded the maximum tensile force experienced during the detachment of each sample.

For each test, the claw was positioned around the shell of the selected mussel and tightened with wire before pulling off. Care was taken to avoid damaging the shell, disturbing the mussel before testing, or placing an initial load on the byssal attachment. Once the mussel had been secured, tensile force was applied steadily until the mussel was fully detached. The maximum tensile force represented the total force required to separate the mussel from its existing attachment, rather than the strength of an individual byssal thread. The number of byssal threads could not be recorded due to the complexities of attachment of the samples.

3.2.2 Field Testing Locations

Mussel samples were taken from three locations to gain a range of data across different environments. Figure 2 shows the testing locations, Beachville Road in Sumner, Lyttleton Marina, and Pigeon Bay, all in the Canterbury region. Beachville Road samples were collected from rocks, Lyttleton Marina samples were attached to the concrete marina, and the Pigeon Bay samples were attached to an industrial mussel farm and were growing on a rope substrate.



Figure 2: Summary of testing locations from left to right Beachville Road, Lyttelton Marina, and Pigeon Bay. Image Public domain *https://www.google.com/maps*

3.2.3 Factors Recorded for Comparison

For each mussel, the testing location, peak detachment force, substrate type, presence of biofouling, surrounding obstructions, and whether the mussel was isolated or located within a cluster were recorded. It was also noted whether the mussel was attached directly to the main substrate, to neighbouring mussel shells, or to a combination of surfaces. These factors were recorded (Microsoft Excel, 2024) to determine how different attachment environments influenced the force required to remove the mussel. The claw ensured that the orientation of the mussel was the same for each sample and the force applied was acting perpendicular to the byssal thread's attachment. The results can be used to estimate the force required by the claw to detach mussels in future applications using the drone.

4 Results

4.1 Design of the Mussel Collector

A portable mussel detachment testing mechanism was developed using a load cell, handle, and gripping claw. A stainless steel, S-Beam load cell with a rated capacity of 20kg was used for this application with an IP rating of 67. The claw was designed to reproduce the loading conditions of the drone's current claw, a controlled tensile load applied to the mussel shell until detachment. The geometry of the claw ensured that contact was applied to both sides of the shell to minimise slip. This was made out of stainless steel, chosen to resist the constant exposure to saltwater. The handle was fashioned out of plywood and shaped for user interface. Both the handle and claw were attached via stainless steel nuts and bolts. Figure 3 shows the completed testing mechanism.

The entire design was waterproof up to 1m of submersion and resistant to corrosion which allowed the mussels to be tested in their natural environments. Initial trials with a 10kg weight, attached to the load cell, confirmed that it was calibrated correctly and accurate to two decimal points of force, measured in Newtons (N). The claw was successfully able to detach mussels in the three testing locations.

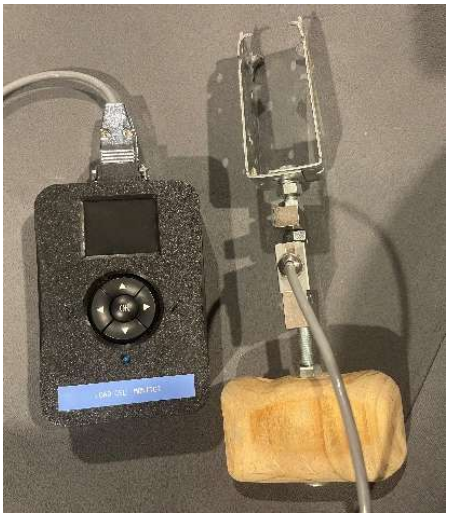


Figure 3: Completed load cell claw, handle, and digital force reader

4.2 Completion of Testing and Preliminary Results

A total of 23 mussel samples were recorded across the three field environments and a table of results, with observations, can be seen in Appendix C. Eleven single mussels attached to rock were tested at Beachville Road, six clustered mussels attached to a concrete substrate were tested at Lyttelton Marina, and six clustered farm mussels were tested at Pigeon Bay. The mean forces were calculated to be 40.13 N for Beachville Road, 64.17 N for Lyttelton Marina, and 67.58 N for Pigeon Bay. The highest individual detachment force was 105.32 N at Pigeon Bay, closely followed by 104.68 N at Lyttelton Marina. The maximum test load was roughly 54% of the load cells capacity, well within the rated value. Relative to Beachville Road, the mean detachment force increased by approximately 60% at Lyttelton Marina and 68% at Pigeon Bay. The variation between measurements shows that the attachment substrate did not completely determine the force required for removal but is a major consideration. Some isolated rock mussels required greater force than some clustered mussels due to complex byssal attachments. The results therefore suggest that detachment force is affected by a combination of substrate, clustering, biofouling, and byssal attachment arrangement.

4.3 Impact on Autonomous Drone Claw Design

The maximum measured force at Pigeon Bay (105.32 N) gives a lower bound for the necessary force used by UC Vision's drone. The claw design should be designed to combat complex biofouling and should not rely on the mean force of attachment alone.

The results also show that designing around the mean detachment force would be insufficient. Although the mean values ranged from 40.13 N to 67.58 N depending on environment, individual samples exceeded 100 N. This indicates that the autonomous claw should be designed using the upper range of the measured forces rather than the average. This information is useful for future iterations of the claw design to reliably detach mussels under different field conditions.

5 Discussion

5.1 Findings

5.1.1 Clustered vs non clustered with attachment complexities

Results from these tests have demonstrated a clear relationship between the amount of force required to detach a sample and the environmental complexities of its location. A clear comparison can be made between the difference in the clustered samples vs non clustered mussels. Table 1 highlights the main observations from the field-testing results.

Table 1: Environment of the mussel sample compared to the mean force, range, and standard deviation

Environment	Number of tests	Mean force (N)	Range (N)	Standard deviation (N)
Rock substrate/ predominantly isolated mussels	11	40.13	15.35–80.25	16.65
Clustered mussels	12	65.87	37.37–105.32	22.42

Mussel tests that were found individually on a rock substrate had mean peak detachment force of 40.13 N, compared with 65.87 N for mussels removed from clustered environments. This represents an increase of approximately 64% in mean detachment force from the clustered environment. The observations recorded during testing suggest that attachment complexity is a major contributor to the required detachment force.

Most isolated mussels attached directly to rock required relatively moderate forces for removal. Tests 1, 2, 4, 6, 7, 8, and 9 were samples with no visible biofouling and produced detachment forces between 25.39 N and 42.51 N. Test 5 was an outlier example where the mussel was attached between two different rocks and required 53.68 N to remove. The most notable result within the rock substrate tests was Test 3, which required 80.25 N. Unlike the majority of the isolated samples, this mussel was attached both to the rock and closely to another mussel, allowing for a more complex combination of byssal attachment. These results suggest that the detachment force can be directly related to the complexity of the surface it is attached to. Test 10 produced the lowest recorded force of 15.35 N and was attached to a relatively smooth section of rock with no visible biofouling.

Similarly, within the clustered samples, Test 19 required only 37.37 N because the mussel was primarily attached to the shell of another mussel rather than being strongly connected to the underlying rope. Test 12, which had comparatively little biofouling, also produced a relatively low clustered detachment force of 40.28 N. The highest detachment forces were recorded from mussels that were attached to dense clusters with surrounding mussels and were also connected directly to their substrate. Test 16, found on the marina, required 104.68 N, while Test 23, attached to a rope, required the maximum recorded force of 105.32 N. Both samples were heavily surrounded by biofouling and neighbouring mussels. Test 21, which was also observed to be directly attached to the rope, required 83.46 N to remove. In comparison, Tests 18, 20, and 22 were removed from clusters but may not have been attached directly to

the rope and produced lower forces of 50.80 N, 61.48 N, and 67.04 N respectively. These results agree with previous research which highlights the impact of more complex clustered arrangements adding to the expected detachment force. Figure 4 demonstrates the difference in mean detachment force from the clearly isolated mussels and the clustered mussels.

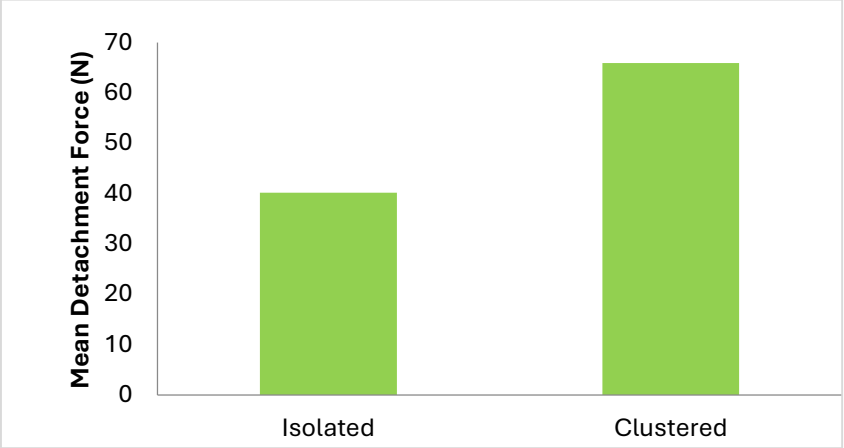


Figure 4: Mean Detachment Force comparison between isolated and clustered Mussels

5.1.2 Effects of Biofouling

The effects of increased amounts of biofouling appear to be connected to the increase of the required detachment force of the samples. This aspect, however, cannot be explicitly distinguished from the mussel's cluster size, attachment type, and its substrate throughout this experiment. As observed during testing, mussels surrounded by more biofouling generally required a greater detachment force than isolated mussels with little or no biofouling. The addition of biofouling to the detachment of the mussel either impacted the experiment directly, by adding to the surrounding physical material of the sample, or indirectly by being present in clusters where mussels have had greater opportunities to form byssal attachments on multiple surfaces.

Greater variability was also observed within the clustered samples. The standard deviation increased from 16.65 N for the rock substrate tests to 22.42 N for the Marina/Pigeon Bay tests. This is consistent with the field observations, as clustered mussels differed considerably in their position, level of biofouling, connection to neighbouring mussels and whether they were attached directly to the rope. The force variability becomes an important consideration for the underwater drone as designing for the mean average force will not necessarily be adequate for the environment.

5.2 Implications

UC Vision's Underwater drone has the potential to provide significant benefits to the mussel farming industry by improving the efficiency, safety, and consistency of harvesting. This research provides UC Vision with an initial force recommendation that the claw design should be able to overcome under realistic field conditions. The results outline a mean force of 67.58 N from a mussel farm, which should be taken as a very lower bound when designing the claw. The claw should be designed to overcome the highest forces measured during these initial tests, to be able to consistently detach samples. More importantly, the findings from this study highlight the importance of designing to operate effectively in environments with high levels of biofouling and dense mussel clusters. The results show that detachment force can

vary considerably depending on factors such as attachment surface, clustering, and biofouling. The wider implications for the mussel industry would reduce reliance on manual labour, improving worker safety, and increase the efficiency of mussel collection before harvesting.

5.3 Limitations

There were multiple limitations to this research which need to be considered for future research and interpreting the results. The field based experiments introduced measurement variability which could not be controlled as precisely as under laboratory conditions. Factors such as pulling direction, claw placement, and position of the mussel may have impacted the peak detachment force. Due to this, some variation between results many have been caused by the testing method rather than the environment.

The relatively small and unequal sample sizes at each location also limit the extent to which the results can be generalised. Because of this, outlying force measurements may have skewed the results one way or the other and may have contributed inaccurately to the mean force calculations. In addition, substrate type, biofouling, and clustering sizes varied simultaneously between locations, meaning that differences in detachment force cannot be singerly attributed to one factor. As well as this, there is natural biological variation between each mussel which contributes to the overall uncertainties of the results.

There were also limitations associated with the test apparatus. Although the load cell was suitable for the measured force range, the manually operated claw meant that force was not necessarily applied at a perfectly constant angle or rate between tests. Future testing could reduce this by using a fixed test frame or motorised pulling system that controls the pulling direction and rate. Despite these limitations, the field based method provided useful measurements under realistic environmental conditions. Future work should use a larger sample size and more controlled loading conditions to better isolate the factors affecting detachment force.

6 Conclusion

Mussel collection is a vital part of the mussel industry which supports and validates the growth and quality of samples during their lifetime. This process is key to successful mussel farming and typically a costly and labour-intensive aspect of successful testing. UC Vision's underwater drone is combatting this by automatously collecting samples from farms which requires a collection claw with sufficient force to detach the mussels. This research outlines the detachment force of mussel samples in different field environments and identifies the main factors influencing those forces, including clustering, substrate attachment, and biofouling.

As explored, the Pigeon Bay mussel farm samples recorded the highest mean force of 67.58 N and the highest individual force of 105.32 N. The lowest mean force, 40.13 N was recorded at Beachville road indicating that the drone's design should base force values of the higher, mussel farm data. Based on these results, a clear relationship between the complexity of the attachment and the detachment force can be made, however there are discrepancies when quantifying the effects of biofouling, size of cluster, and byssal attachment. In general, the

research completed provides important information that can be incorporated into future design decisions for the underwater drone, helping to ensure that the claw can constantly detach sample.

The purpose of this report was to develop a system to predict the force needed by a mechanical claw to detach mussels from their natural environment. Rather than determining a single detachment force, this research investigated how detachment force varied with environmental and biological factors including sampling location, substrate type, surrounding biofouling, and mussel clustering. The resulting data was analysed to identify the factors associated with detachment force and to determine an appropriate force range for the mechanical arm and pusher system for successful collection.

Declarations

Funding: This work was partly funded by The University of Canterbury.

Conflict of interest statement: The authors declare that there are no financial conflicts of interest regarding this work.

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AI usage: AI tools namely were used for citing references, method development, developing results table, and editing (Grammarly).

Author contributions: The author was in charge of conceptualization of the research, testing, compiling results, writing report, and editing the document. Oscar Torres and Jullian Phillips provided advise on how to complete the methodology and provided the loadcell for testing.

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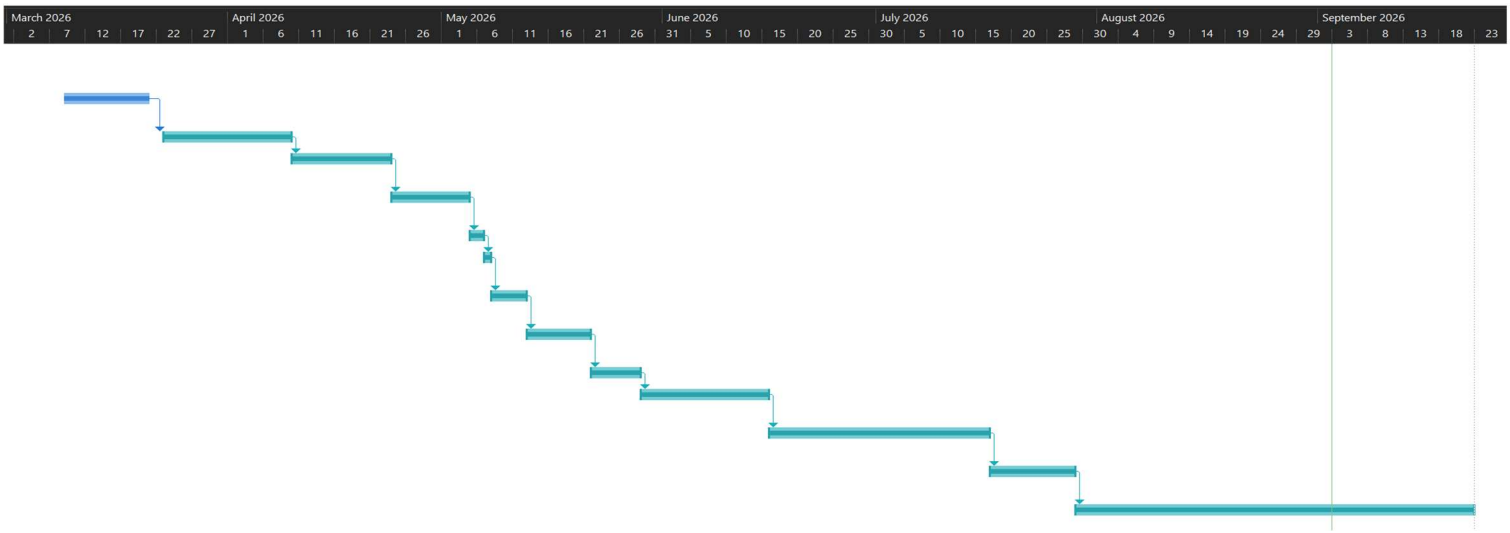
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Appendix A: Project Planning Gantt Chart

Task Mode	Task Name	Duration	Start	Finish	Predecessors	Baseline6 Cost	Add New Column
	Research/Sub-project proposal	10 days	Mon 9/03/26	Fri 20/03/26		$0.00	
	Literature review	14 days	Mon 23/03/26	Thu 9/04/26	3	$0.00	
	Devlope methods to test strength of Mussels	10 days	Fri 10/04/26	Thu 23/04/26	4	$0.00	
	Procure parts to build Musssel collector	7 days	Fri 24/04/26	Mon 4/05/26	5	$10.00	
	Build testing device	2 days	Tue 5/05/26	Wed 6/05/26	6	$0.00	
	Initial testing of Grabber design	1 day	Thu 7/05/26	Thu 7/05/26	7	$0.00	
	Write out testing method	3 days	Fri 8/05/26	Tue 12/05/26	8	$0.00	
	Testing in field - indivdual mussels	7 days	Wed 13/05/26	Thu 21/05/26	9	$0.00	
	Initial results write up	5 days	Fri 22/05/26	Thu 28/05/26	10	$0.00	
	Planning for clustered mussel testing	12 days	Fri 29/05/26	Mon 15/06/26	11	$0.00	
	Testing in field - clustered mussels	23 days	Tue 16/06/26	Thu 16/07/26	12	$0.00	
	Results write up/WIP Report	8 days	Fri 17/07/26	Tue 28/07/26	13	$0.00	
	Final project write up/Disscussion and	40 days	Wed 29/07/26	Tue 22/09/26	14	$0.00	



Appendix B: Health and Safety Matrix

Risks of system in its CURRENT STATE, with its existing controls.								Risks of system in its FUTURE STATE after these treatments. Insert any new threats caused by the treatments.			
Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per "Severity of harm" scale 1. Hazard 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment 7 or less No intervention necessary.	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
In field testing risks											
Sensors	Reads incorrect data	2	3	6	Minimal action required, code alterations	Relevant team member	No	2	3	6	Yes - Further action not necessary
Cords	Gets tangled	3	3	9	Maintain cable management	Relevant team member	Yes	3	2	6	Yes - Further action not necessary
Cords	Gets severed	4	2	8	Active management of tether when operating	Relevant team member	Yes	4	2	8	Yes - Further action not necessary
Cameras	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
Cameras	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
Battery	Overheating	3	2	6	Check battery health/condition prior to operation and monitor power draw while in use	Every handler	No	3	2	6	Yes - Further action not necessary
Battery	Electric Fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
Battery	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
Propellers	Collision with object/ wildlife	3	4	12	Monitor surroundings, Ensure personal safety, before startup inspect propellers	Every handler	Yes	3	3	9	Yes - Further action not necessary
Propellers	Mechanical failure	3	2	6	Inspect propellors prior to use of ROV	Every handler	No	3	2	6	Yes - Further action not necessary
Grabber	Arm get entangled	2	4	8	Monitor surroundings while operating arm	Relevant team member	Yes	2	4	8	Yes - Further action not necessary
Grabber	Claw embeds in drop line	2	4	8	Monitor to ensure grabber arm does not grip the line too tightly	Relevant team member	Yes	2	4	8	Yes - Further action not necessary
Grabber	Human injury because of sharp object	4	4	16	Design cover for grabber, follow proper handling instructions , keep clear during operation	Relevant team member	Yes	3	3	9	Yes - Further action not necessary
Motors	Vibration – mechanical issues	2	4	8	Inspect motors prior to operation	Every handler	No	2	4	8	Yes - Further action not necessary
Motors	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
Motors	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
Motors	Overheating	3	2	6	Monitor power draw from motors while ROV in use	Relevant team member	Yes	3	2	6	Yes - Further action not necessary
Motors	Noise – environmental damage	2	4	8	Ensure motors don't result in propellor cavtation - monitor rpms	Relevant team member	Yes	2	4	8	Yes - Further action not necessary
LEDs	Electric fault	2	2	4	Ensure electrical rules followed when wiring	Every handler	No	2	2	4	Yes - Further action not necessary
LEDs	Water damage	2	2	4	Ensure ROV is closed fully and seals are intact	Every handler	No	2	2	4	Yes - Further action not necessary
LEDs	Bright lights	3	4	12	Proper PPE, Disable LED when it is out of the water	Every handler, Code	Yes	2	3	6	Yes - Further action not necessary
Controllable Underwater	Collision – self inflicted	2	4	8	Monitor surrounding while operating and approach motion controll sensibly	Every handler	Yes	2	4	8	Yes - Further action not necessary
Uncontrollable Underwater	Strong current	2	3	6	Test ocean currents prior to operation and set current threshhold for non-operation	Every handler	Yes	2	3	6	Yes - Further action not necessary
Uncontrollable Underwater	Encounter with wildlife	4	4	16	Monitor surroundings and ensure no abundance of wildlife (fish, dolphins etc) in testing area	Every handler	Yes	4	4	16	NO – not much we can do to mitigate this
Uncontrollable Underwater	Collision	2	3	6	Monitor surroundings and motion of robot while in operation and actively avoid collisions through piloting robot	Every handler	Yes	2	3	6	Yes - Further action not necessary
Reel	Gets severed	3	2	6	Monitor reel/tether to ensure no snags/sharp objects threaten teather	Relevant team member	Yes	3	2	6	Yes - Further action not necessary
Reel	Tangled	2	3	6	Monitor reel/tether to ensure no tangles generate while in use	Relevant team member	Yes	2	3	6	Yes - Further action not necessary
Reel	Jammed	2	3	6	Monitor reel/tether while in operation and manage tether to ensure no jams	Relevant team member	Yes	2	3	6	Yes - Further action not necessary
Lab testing risks											Yes - Further action not necessary
Heavy machinary	Malfunction	2	2	4	Moniter machinery during use	Operator of machine	Yes	2	2	4	Yes - Further action not necessary
Heavy machinary	Jammed	2	3	6	Moniter machinery during use	Operator of machine	Yes	2	3	6	Yes - Further action not necessary
Mussels	Sharp	3	3	9	Where cut gloves to avoid getting cut	Handler of	Yes	3	2	6	Yes - Further action not necessary
Clamp	Heavy/stuck limbs	4	3	12	Stand awar from the machine during use	Operator of machine	Yes	3	2	6	Yes - Further action not necessary

Appendix C: Detailed mussel detachment results

Test	Peak detachment force (N)	Attachment environment	Observed attachment condition
Test 1 – Beachville Road	38.53	Rock substrate – isolated	Mussel positioned between two rocks but readily accessible for gripping.
Test 2 – Beachville Road	42.51	Rock substrate – isolated	Single mussel attached directly to the rock and surrounding substrate.
Test 3 – Beachville Road	80.25	Rock substrate – mixed attachment	Mussel attached to both the rock and a neighbouring mussel, resulting in more byssal threads contributing to attachment. Greater attachment complexity than most rock-substrate trials.
Test 4 – Beachville Road	38.28	Rock substrate – isolated	Single mussel attached directly to the rock and surrounding substrate.
Test 5 – Beachville Road	53.68	Rock substrate – multiple surfaces	Single mussel attached across two different surfaces.
Test 6 – Beachville Road	25.39	Rock substrate – isolated	Single mussel attached directly to rock with no visible biofouling.
Test 7 – Beachville Road	34.62	Rock substrate – isolated	Single mussel attached directly to rock with no visible biofouling.
Test 8 – Beachville Road	36.70	Rock substrate – isolated	Single mussel attached directly to rock with no visible biofouling.
Test 9 – Beachville Road	31.57	Rock substrate – isolated	Single mussel attached directly to rock with no visible biofouling.
Test 10 – Beachville Road	15.35	Rock substrate – isolated	Mussel attached to a comparatively smooth section of rock with no visible biofouling. Relatively smooth attachment surface.
Test 11 – Beachville Road	44.60	Rock substrate – isolated	Single mussel attached directly to rock with no visible biofouling.
Test 12 – Lyttelton Marina	40.28	Mussel cluster	Attached within a cluster with relatively low levels of biofouling. Lower biofouling than most clustered trials.
Test 13 – Lyttelton Marina	51.83	Mussel cluster	Attached within a cluster with greater surrounding biofouling.
Test 14 – Lyttelton Marina	62.51	Mussel cluster	Attached within a cluster with substantial surrounding biofouling.
Test 15 – Lyttelton Marina	73.18	Large mussel cluster	Attached within a larger cluster with substantial biofouling.
Test 16 – Lyttelton Marina	104.68	Dense mussel cluster	Heavily surrounded by biofouling and deeply integrated with neighbouring clustered mussels. Strongly integrated into surrounding cluster.
Test 17 – Lyttelton Marina	52.51	Mussel cluster	Attached within a cluster with substantial surrounding biofouling.
Test 18 – Pigeon Bay	50.80	Mussel cluster	Removed from a bio fouled cluster; direct attachment to the rope was uncertain. Possible indirect rope attachment.
Test 19 – Pigeon Bay	37.37	Mussel cluster – shell attachment	Mussel attached primarily to the shell of another mussel within a larger cluster rather than strongly to the supporting substrate. Attachment was mainly to another mussel shell.
Test 20 – Pigeon Bay	61.48	Mussel cluster	Removed from a biofouled cluster; direct attachment to the rope was uncertain. Possible indirect rope attachment.
Test 21 – Pigeon Bay	83.46	Cluster – rope attachment	Clustered and biofouled mussel pulled directly from its attachment to the rope. Confirmed direct rope attachment.
Test 22 – Pigeon Bay	67.04	Mussel cluster	Removed from a biofouled cluster; direct attachment to the rope was uncertain. Possible indirect rope attachment.
Test 23 – Pigeon Bay	105.32	Dense cluster – rope attachment	Heavily surrounded by biofouling, deeply integrated with neighbouring mussels, and pulled directly from the rope. Confirmed direct rope attachment and dense biofouling.

Appendix E – Underwater Robot Arm Control

ENME408-2026 Individual student report

Characterisation and Control of an Underwater Robot Arm-Simulator

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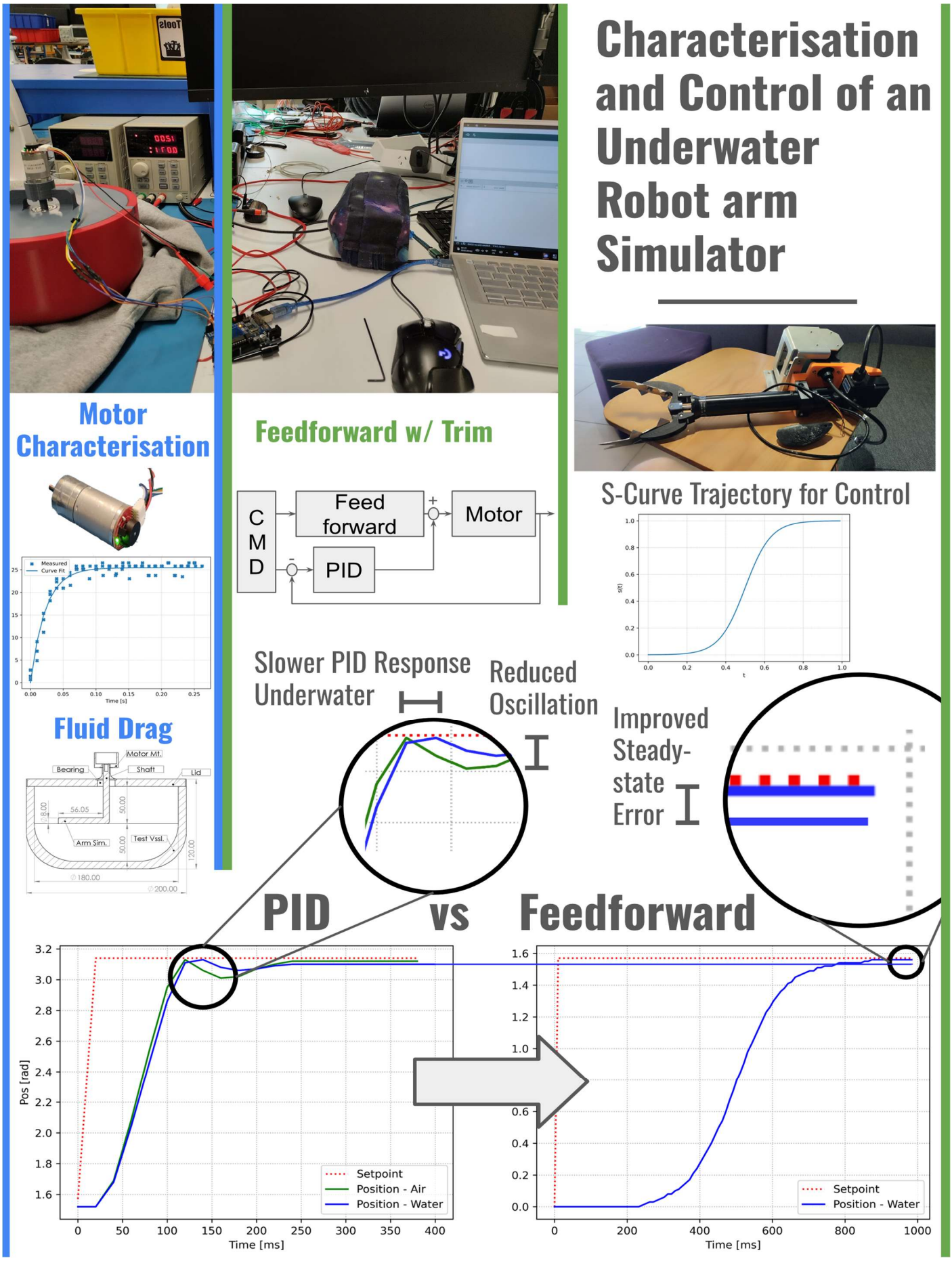
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Abstract

Context: Control over servomotors and robotic arms is critical to industry, including in less-frequent underwater applications for remotely operated vehicles or similar. **Objectives:** The purpose of this research is to characterise a DC motor and the fluid drag acting on an attached robotic arm-simulator, to quantify the impact of submerging a standard PID controller in water, and to construct a suitable model of the system dynamics for a "feedforward with trim" control architecture. **Method:** The motor is characterised through a combination of calculation and measurement; the drag acting on the arm is characterised through experiment with a controlled body of water, based on a measured steady-state angular velocity and current draw. A feedforward control equation is derived from first-principles and combined with a conventional PID control equation; logistic S-curves are used to generate control trajectories. **Results:** A conventional air-tuned PID controller is shown to perform to slower response times and increased steady state error when underwater; the feedforward controller developed is shown to improve on accuracy and tunability compared to the conventional PID architecture. **Conclusions:** The methodology developed has been validated, and the feedforward architecture presented here could find use in underwater robotics where extreme precision is demanded; a conventional PID controller remains a viable alternative where simplicity is preferred instead.

Keywords: Servomotors; DC Motor; Robotic arm; Underwater; Fluid dynamics; PID; Feedforward

Graphical Abstract



1 Introduction

1.1 Context

Servo motors and robotic arms, as present in the mussel-grabbing arm of the underwater Remotely Operated Vehicle (ROV) this research topic developed from, are commonplace in industry. The control of these is clearly of significant importance – response times need to be fast to maximise the frequency at which tasks can be performed, yet the responses also need to be accurate to achieve commanded positions; furthermore, the maximum overshoot of responses generally needs to be restricted to prevent collisions with the larger system the arm is situated within. It is commonplace to use a PID controller in such applications, and this is often based on simplified models of the system response – including, of particular note to an underwater application, the assumption of no air resistance.

In instances such as mussel sampling – where drag from the surrounding water *is* significant – establishing accurate control of a robotic arm *underwater* is critical to success of the project. As motivation for this work, an improved control model for an ROV's onboard robotic arm would be expected to increase sampling precision, reduce sampling time, and ease manoeuvring strain on the ROV – expanding mission profiles and generating the operator additional revenue.

1.2 Area under examination

This project began with the robotic arm shown, detached from its host ROV, in Figure 1.



Figure 1: The mussel grabber arm – work credited to UWAI robotics

The arm itself has three degrees of degrees of freedom (DoF), though the client was concerned primarily with but one of them; DoF 1 in Figure 2 below.

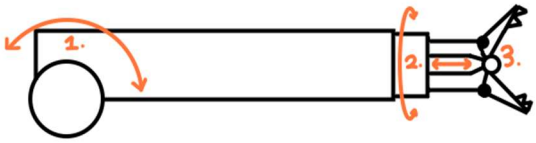


Figure 11: A highly simplified representation of the grabber arm, highlighting its three degrees of freedom

The client reported erratic movement of the arm underwater, and was interested in improving the PID control model for it – which lead to the research presented here. The aims of this research are thus threefold: to study the effects of submerging a PID controller in water, to characterise the servomotor, and to thus develop an improved controller for the underwater robot arm.

As the broader project progressed, independent of this work, it became clear that a multi-DoF robotic arm was not the best solution for grabbing mussels, and the system was redesigned to eliminate the arm entirely. The research, however, remains plainly relevant to other similar projects – such as deep sea sampling, for instance – and was instead continued with a new DC motor and a scaled-down "arm-simulator", as shown in Figure **Error! Reference source not found.**.

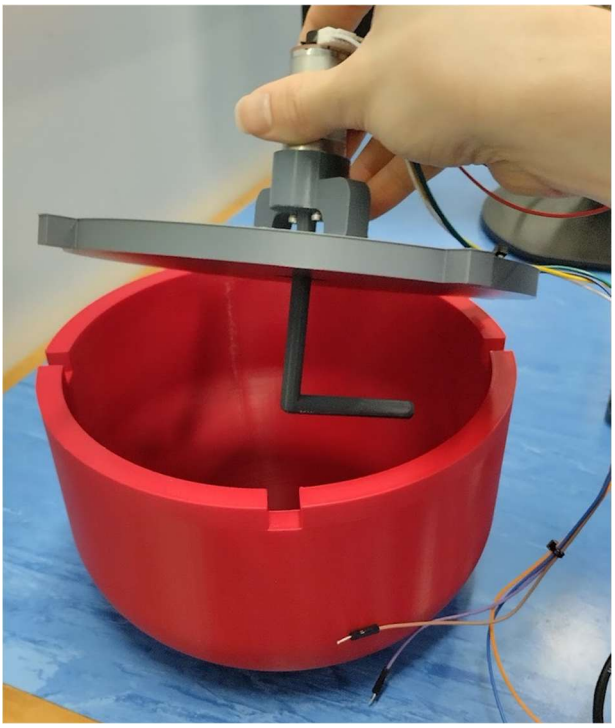


Figure 3: The arm-simulator and associated test vessel used in this report

The large red bowl can be filled with water or alternately left empty, to simulate either air or water conditions – details in later sections.

2 Literature

2.1 Servo Motor and Arm Dynamics

PID controllers for servo motors are common and well-studied systems ([1], [2]). In general, the key characteristics to be aware of when controlling a servo based on a DC motor are as follows: armature wire resistance (R_a), armature wire inductance (L_a), motor torque constant (K_t), rotor inertia (J_t), and rotor friction (D) ([3], [4]). Though sometimes available in datasheets or similar, these characteristics often need to be determined experimentally. The resistance and inductance of the armature wire can be measured with a suitable multimeter/LCR meter – though it is expected that armature inductance will be negligible and only of transient impact, such that it can be assumed zero throughout calculation ([4]). Meanwhile, the torque constant, motor friction, and rotor inertia can be estimated experimentally by applying certain known voltage conditions and measuring the resulting angular speed and current draw ([4]). This will be discussed in more detail, including relevant equations and voltage conditions, in later sections.

Many commercial robot arms require complicated inverse-kinematics control with multi-DoF control systems ([5]). In this case, the primary concern is the precise control only of the lateral swinging motion of the arm. This allows for a much simpler model, wherein the entire arm is modelled as a single mass, and can thus be treated as a rigid extension of the motor's rotor to determine the total inertia of the system experimentally – convenient due to the potentially complex shape and non-uniform density of the arm.

2.2 Underwater Dynamics

The robot arm moving underwater will develop drag. As the scaled-down arm-simulator is relatively slender (Figure **Error! Reference source not found.**), this is most closely related to the problem of drag over a cylinder, which is well-studied in a large variety of scenarios ([6], [7]). Though close in form in the simulator, it is unlikely a cylinder model will sufficiently capture the shape of any real robotic arm – and thus a key concern to determining the fluid drag seems to be determining its drag coefficient. In the general case, determining this analytically will be impractical, given the potentially complex shape of the arm, and it seems similarly impractical to determine with the air-based wind tunnel facilities available on the University of Canterbury campus; it would be difficult to match Reynold's number between the different fluids. A potential solution to this would be mounting the entire robot arm structure to a dynamometer, and measuring the drag experienced in water experimentally, similar to the process used for determining forces on human limbs during swimming in [8]. This would mitigate the need to determine a drag coefficient, and would allow for testing over a range of potential motions and positions of the smaller auxiliary parts of the arm. However, acquiring a dynamometer capable of taking the precise torque readings necessary in the scaled-down simulator model seems prohibitively expensive and similarly impractical.

A potential alternative to an experimental measurement would be constructing a CAD model of the arm, and estimating the drag over it with Computational Fluid Dynamics (CFD) software, in an approach akin to [9] – though they conclude that, while effective at determining drag on complex static models, the method fails to account for the nuances of human (or, in this case, robotic) movement; the method may not be suitable here.

On balance, the best approach appears to be an extension of the characterisation methods described in [4] – at steady state while submerged in water, everything else accounted for, the torque developed by the motor will precisely balance the drag developed over the rotating arm-simulator. Knowing the torque developed by the motor, a lumped "drag constant" can be measured for the arm-simulator, independent of the arm's geometry, allowing for experimental characterisation of complex arm shapes in much the same manner as using a dynamometer.

2.3 Control

Regarding the control of robotic arms underwater, some literature on the topic exists ([10], [11]) but generally seems more interested in the mechanical design of their arms, rather than in developing precise control models that account for fluid drag. Though the area has more-likely-than-not already been explored in commercial applications, this leaves a gap in the literature this research can potentially address: what does a controller dedicated to controlling a robotic arm *underwater* look like?

An issue with a typical PID controller in this scenario is that drag over the arm-simulator is expected to be proportional to the arm's velocity *squared* ([6], or as can be seen in the data gathered in [7] at slight extension of their conclusions). A standard PID feedback controller cannot incorporate a non-linear term like this without linearising it relative to a reference velocity, which comes at cost of accuracy over varying velocities. A solution is a feedforward controller. Knowing at this point quite precisely the characteristics of the motor and submerged arm-simulator, a "feedforward with trim" architecture (as described in [12, p. 2524]) can be used to calculate a theoretically optimal control signal – accounting for fluid drag – and feed it forward to command the motor directly. For robustness, a typical PID feedback controller "trims" the minor deviations between command and response, which result from unpredictable disturbances and an inevitably imperfect model of the system.

Feedforward controllers have been used in underwater motor control before – [13] presents a technique wherein discretised artificial intelligence "learns" the necessary feedforward parameters for underwater motor control, for instance, though [14] attempted to replicate the results and struggled to demonstrate the same improvement over conventional parameter identification techniques. It seems apt to constrain this research to the more conventional physics-based parameter identification techniques.

3 Methods and Materials

3.1 Research Objectives

This research aims, first, to develop a suitably comprehensive model of submerged DC-motor system dynamics; second, to characterise the impact submerging a robotic arm in water has on a no-air-resistance PID controller; and third, to go on to develop an improved "feedforward with trim" control architecture to improve manoeuvring of the arm.

A model is constructed in two stages: First, the non-linear dynamics of the motor and arm in air; second, the dynamics of the system when submerged, accounting for ambient hydrodynamic drag. A majority of the novelty lies in the latter.

3.2 Approach

3.2.1 Motor Dynamics

A DC motor with position feedback is required – in this case, the 280 RPM variant of the GA25-370 DC motor, including an integrated quadrature encoder for position feedback, was used. This motor includes an internal 1:21.3 gearbox; this research treats the entire motor-gearbox system as a black-box and characterises relative to gearbox output. The motor was characterised from first electrical principles to determine the armature wire resistance (R_a), armature wire inductance (L_a), motor torque constant (K_t), rotor inertia (J_t), and rotor friction (D). R_a and L_a were measured directly with an LCR meter; K_t was calculated from the steady-state relation of armature current (I_a) and angular velocity (ω_a); J_t and D were determined using a non-linear least-squares parameter-fit on step-response data. Throughout, I_a was measured using the current-draw readout on a KORAD bench power supply, while ω_a was measured using the motor's integrated quadrature encoder. This procedure ultimately develops a voltage (V_a) to torque (τ_a) motor model for use in the feedforward controller.

3.2.2 Underwater Dynamics

The most promising approach to determining fluid drag on the arm is through empirical experimentation, as determined from the literature review. A suitably large test chamber, including a motor mount and arm-simulator, was designed and produced using additive manufacturing techniques. A bearing was used to reduce parasitic friction in the system while containing splashes, and the relatively large size of the chamber reduced fluidic boundary effects. A contained test chamber was preferred over a larger body of open water for many reasons, including protection of electronics, compatibility with lab environments, and elimination of any drowning risk.

To describe fluid drag, a lumped "drag constant" (C_d) was derived from first principles. C_d accumulates information about the frontal area of the arm, the fluid density and viscosity, and the actual underlying drag coefficient into a single convenient constant that relates the square of angular velocity to drag-induced torque. Submerging the arm-simulator in water, a constant voltage was applied, and after waiting several seconds for the system to reach steady state, armature current (I_a) and angular velocity (ω_a) were measured to calculate C_d.

3.2.3 PID and Feedforward Control

A conventional PID controller was developed and tuned for operation with the arm-simulator in air. Step-response data was recorded, both in air and water, and compared to analyse the impact the additional fluid drag had on the response. This accounts for the second research objective.

Continuing to the feedforward controller, the previously-developed motor model was extended to include internal motor friction (D), total system inertia (J_t), and fluid drag (C_d). Accounting for acceleration beyond the steady state, this now represents a generalised motor model – meeting the first research objective – which can be inverted to find the necessary feedforward control voltage (V_{ff}) to achieve a commanded angular velocity (ω_c) and angular acceleration ($\dot{\omega}_c$). A standard PID controller was selected for the trim component of the controller, which describes a feedback control voltage (V_{fb}) in terms of error in commanded position (θ_c). This feedback control accounts for unpredictable disturbances in the system and any inevitable inaccuracies of the model, resulting in a robust control architecture.

Precise position control over the motor was thus established. Once suitable values of K_i, K_p, and K_d were determined through a combination of experiment and control theory, the motor controller needed only to generate a trajectory of commanded position (θ_c) that moved smoothly between the arm's starting position and a desired final position. This trajectory can be differentiated to obtain similar trajectories for ω_c and $\dot{\omega}_c$. Summing V_{ff} and V_{fb} results in a final equation describing the voltage at each timestep – in terms of θ_c, ω_c, $\dot{\omega}_c$, and motor parameters – that should be applied to manoeuvre the arm along the desired trajectory. This model was tested and compared to the results of the conventional PID controller, meeting the third objective of this research.

4 Results

4.1 Derivation of Motor Dynamics

Consider a general motor. The first step of characterising it is to measure the resistance and inductance of the motor's armature wire, using a suitable multimeter and LCR meter. Following this, the torque constant is then calculated by supplying the motor with a constant voltage, and measuring the angular speed and current draw. The torque constant can thus be determined as in Equation 1 [4].

$$V_a = I_a R_a + K_t \omega_a$$

Equation 1: Determining torque constant (K_t) from a known armature voltage (V_a), armature current (I_a), armature resistance (R_a), and angular velocity (ω_a)

The friction constant and inertia of the motor (including internal gears, output shaft, and arm-simulator) can then be determined by applying a step input to the motor and measuring the resulting step response in angular velocity. Following the method outlined in [4], under the assumption of negligible armature inductance ($L_a = 0$), the transfer function of the motor can be closely modelled by Equation 2, where a and b are defined as in Equations 3 and 4.

$$\frac{\Omega_a}{V_a} = \frac{a}{s+b}$$

Equation 2: Motor transfer function from armature voltage (V_a) to angular velocity (Ω_a)

$$a = \frac{K_t}{J_t R_a}$$

Equation 3: Parameter a in the motor transfer function, in terms of torque constant (K_t), total inertia (J_t), and armature resistance (R_a)

$$b = \frac{DR_a + K_t^2}{J_t R_a}$$

Equation 4: Parameter b in the motor transfer function, in terms of friction constant (D), armature resistance (R_a), torque constant (K_t), and total inertia (J_t)

Standard Laplace analysis of this transfer function shows that the time domain response to a step of amplitude V_a is modelled by Equation 5.

$$\omega_a = \frac{aV_a}{b}(1 - e^{-bt})$$

Equation 5: Time domain step response of the motor

At this point, a non-linear least-squares curve-fit can be applied to experimental step response data to identify parameters a and b. From this result, Equations 3 and 4 can then be solved for the system's total inertia, J_t, and friction constant, D, respectively.

Again neglecting armature inductance, this results in the overall voltage (V_a) to torque (τ_a) motor model shown in Equation 6.

$$V_a = \frac{R_a}{K_t}\tau_a + K_t\omega_a$$

Equation 6: Motor model, given armature resistance (R_a), torque constant (K_t), and total inertia of the arm/rotor (J_t)

4.2 Test Chamber

The test chamber that was developed to contain the underwater experiments is presented in Figure **Error! Reference source not found.**.

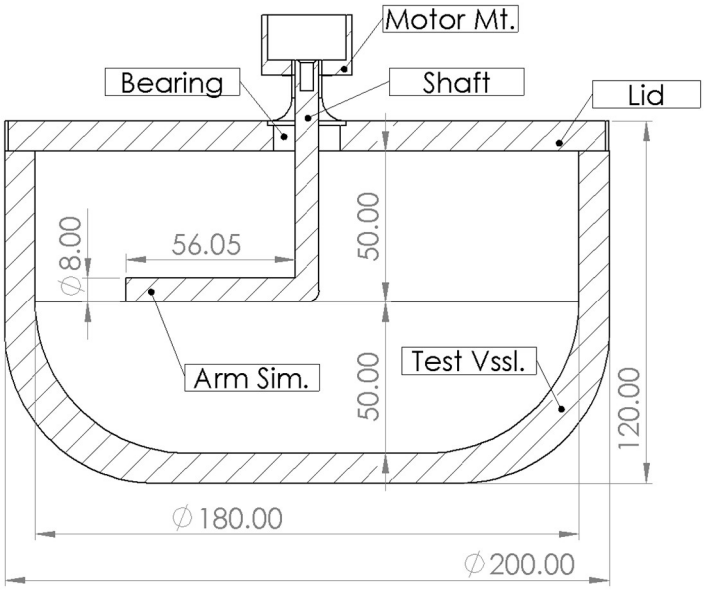


Figure 4: Motor test chamber and arm-simulator

This was produced in two 3d-printed pieces using PLA filament – desirable for its affordability and convenience. The arm-simulator itself has an approximately cylindrical profile of 8 mm diameter and 60 mm length (from shaft centre), suspended approximately 50 mm above the base of the test chamber. The chamber has a height of 100 mm and an internal radius of 90 mm – a fair margin larger than the arm, which helps to mitigate fluidic boundary effects. The completed build is shown in Figure **Error! Reference source not found.**.

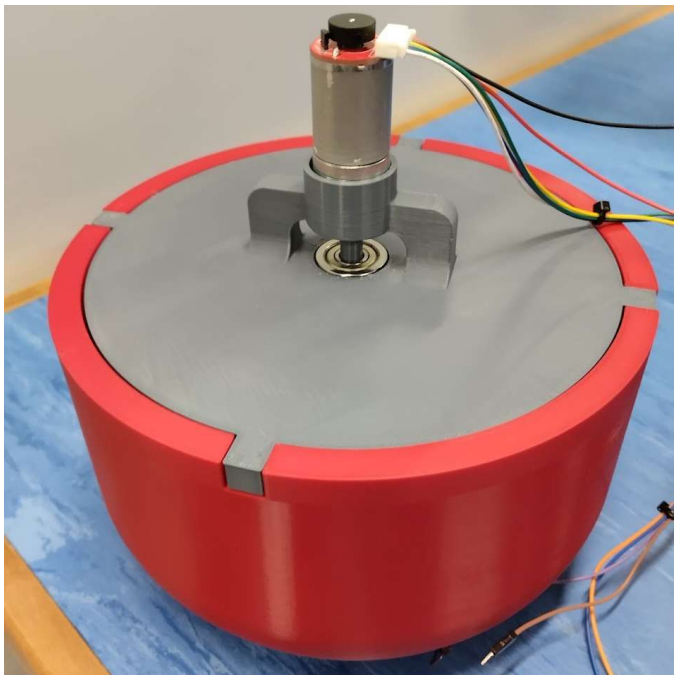


Figure 5: The completed test chamber

4.3 Derivation of Underwater Dynamics

Armed with the test chamber and arm-simulator, consider again a general motor. At steady state, with the arm-simulator submerged in water, the motor applies torque according to Equation 7.

$$\tau_a = K_t I_a$$

Equation 7: The torque applied by the motor; the definition of torque constant (K_t)

At steady state, this balances a torque as described by Equation 8.

$$\tau_b = D\omega_a + C_d\omega_a^2$$

Equation 8: The torque balanced by the motor, in terms of friction constant (D), angular velocity of arm (ω_a), and a lumped "drag constant" (C_d)

The arm-simulator is submerged and the motor is supplied constant voltage. After several seconds, the motor will have settled into steady state: angular velocity (ω_a) and current draw (I_a) can be measured, and, equating Equations 7 and 8, C_d can thus be calculated.

4.4 Derivation of Feedforward Controller

When underwater and *not* at steady-state, the total torque the motor must balance is described by Equation 9.

$$\tau_b = J_t\dot{\omega}_a + D\omega_a + C_d\omega_a^2$$

Equation 9: The total torque balanced by the motor in terms of total inertia (J_t), angular acceleration ($\dot{\omega}_a$), friction constant (D), angular velocity (ω_a), and drag constant (C_d)

Returning to Equation 6, substituting Equation 9 as the applied torque, the feedforward voltage required to achieve a commanded angular velocity and angular acceleration is described by Equation 10.

$$V_{ff} = \frac{R_a}{K_t}(J_t\dot{\omega}_c + D\omega_c + C_d\omega_c^2) + K_t\omega_c$$

Equation 10: The feedforward voltage (V_{ff}) necessary to achieve a commanded angular acceleration ($\dot{\omega}_c$) and angular velocity (ω_c), in terms of motor parameters

This feedforward voltage is combined with a feedback voltage from a standard PID controller, as described by Equation 11, to calculate the overall control signal at any given instant in time.

$$V_{fb} = K_p e + K_i \int e\,dt + K_d\frac{de}{dt}$$

Equation 11: The feedback voltage in terms of tuneable PID control parameters and error (e) in commanded position (θ_c; $e = \theta_a - \theta_c$)

Summing V_{ff} and V_{fb}, the voltage applied to the motor at any given time instant is thus as in Equation 12.

$$V_a = \frac{R_a}{K_t}(J_t\dot{\omega}_c + D\omega_c + C_d\omega_c^2) + K_t\omega_c + K_p(\theta_a - \theta_c) + K_i\int(\theta_a - \theta_c)\,dt + K_d\frac{d(\theta_a - \theta_c)}{dt}$$

Equation 12: The control voltage (V_a) in terms of commanded position/velocity/acceleration trajectories (θ_c, ω_c, $\dot{\omega}_c$), tuned PID control parameters, and motor parameters

A standard logistic S-curve (Equation 13; Figure **Error! Reference source not found.**) was selected to generate the position trajectory (θ_c); its derivatives generate ω_c and $\dot{\omega}_c$ (Equations 14, 15).

$$\theta_c = s(t) = \frac{\theta_t - \theta_0}{1 + e^{-k(t-t_0)}} + \theta_0$$

Equation 13: A logistic S-curve, defining commanded position trajectory (θ_c) over time (t), in terms of final target position (θ_t), starting position (θ_0), logistic growth rate (k), and the midpoint of the response (t_0)

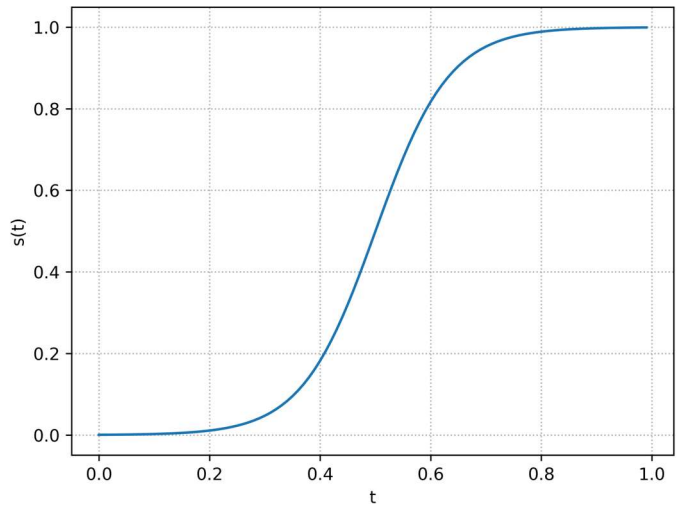


Figure 6: An example S-curve, for reference, as may be generated by Equation 13

$$\omega_c = k(s(t) - \theta_0)\left(1 - \frac{s(t) - \theta_0}{\theta_t - \theta_0}\right)$$

Equation 14: Commanded velocity trajectory (ω_c) over time (t), in terms of final target position (θ_t), starting position (θ_0), logistic growth rate (k), midpoint of the response (t_0), and original logistic S-curve function ($s(t)$; Equation 13)

$$\dot{\omega}_c = k^2(s(t) - \theta_0)\left(1 - \frac{s(t) - \theta_0}{\theta_t - \theta_0}\right)\left(1 - 2\frac{s(t) - \theta_0}{\theta_t - \theta_0}\right)$$

Equation 15: Commanded acceleration trajectory ($\dot{\omega}_c$) over time (t), in terms of final target position (θ_t), starting position (θ_0), logistic growth rate (k), midpoint of the response (t_0), and original logistic S-curve function ($s(t)$; Equation 13)

This formulation of an S-curve was selected for its smoothness over several derivatives, and for the fine control it affords over the system response. The logistic growth rate, k, determines the "steepness" of the S-curve (refer to Figure **Error! Reference source not found.**); the midpoint of the response, t_0, defines the time at which the S-curve is halfway between the starting position and final target position: effectively, t_0 represents half the response time of controller. As one of the original client complaints was that their arm moved erratically underwater, this fine control over the speed of the motor response is desirable to similar underwater robot projects.

In sum, this now entirely defines the motor controller: suitable values are selected for K_i, K_p, K_d, k, and t_0 to tune the motor response; θ_c, ω_c, and $\dot{\omega}_c$ are generated from Equations 13–15, based on the measured current position and a desired target position; and, at each instant in time, these trajectories are substituted into Equation 12 alongside the updated current position to generate the control voltage, V_a.

4.5 System Dynamics

Two GA25-370 DC motors were used throughout this research. The first of these to be purchased was found to have a fault in one of its quadrature encoder channels, rendering it capable only of degraded speed measurement, not position; the second was purchased to replace the first. In the interim between the arrival of the first and the second, the first motor was characterised to validate and trial the previously-described methodology. At conclusion of this process, the system parameters, including C_d for the arm-simulator, are presented in Table 3 below.

Table 3: Motor parameters

Parameter	Symbol	Value	Unit
Armature Resistance	R_a	7.06E+00	Ω
Armature Inductance	L_a	7.41E-03	h
Torque Constant	K_t	3.84E-01	$\frac{V}{rad/s}$
Total Inertia	J_t	0.00120	kgm^2
Friction Constant	D	0.000581	$\frac{N \cdot m}{(rad/s)}$
Lumped Drag Constant	C_d	1.53E-05	$\frac{N \cdot m}{(rad/s)^2}$

Of note is the small armature inductance – only 7.4 mh measured at 1 KHz. This validates the assumption of negligible inductance throughout the method. K_t was measured and averaged across several voltages – the motor's remaining quadrature encoder channel measured angular velocity; a bench power supply with a current draw readout was used to measure current. K_t was highly consistent across each voltage trialled.

In regard to J_t and D, the motor's step response in angular velocity was recorded across 1, 5, 8, 10, and 12 V, and Python's scipy.optimize.curve_fit module was used to fit parameters

(Equation 5) to each of these trials simultaneously. The complete code used can be found in Appendix C, and a plot of the 12 V trial against the ultimate curve fit is presented in Figure **Error! Reference source not found.**.

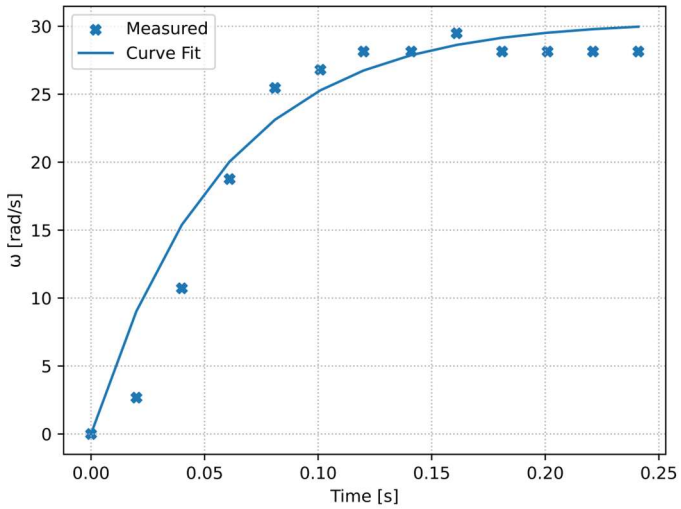


Figure 7: Motor step response, as measured for a 12 V step

The curve-fit generally appears fairly accurate, though quantisation and low density of data – mostly as a result of missing the second channel of the quadrature encoder – are clearly hampering this result.

In similar manner, filling the test chamber with water, the steady-state current draw and angular velocity were measured and averaged over 8, 10, and 12 V to determine C_d. As the cylindrical profile of the arm-simulator used here is actually relatively simple, a theoretical analysis *is* possible – this yields a rough estimate of lumped drag constant (C_d) at around 1.30 × 10^{-5} (calculation shown in Appendix D). Some deviation is expected due to inevitably imperfect modelling, potential boundary effects, 3D-printing not producing a perfectly smooth cylinder, and so on, but this result is well in line with the experimentally measured value of 1.53 × 10^{-5}, as reported above. This agreement validates the methodology presented here and lends confidence to applying this method to an arm of more complex geometry.

With this in mind, upon the arrival of the second motor, it was decided to continue the research with a more complex, irregularly shaped arm-simulator, to better represent a real-world underwater robot arm, as would be used in a real-world robotics project. This is shown in Figure **Error! Reference source not found.**.

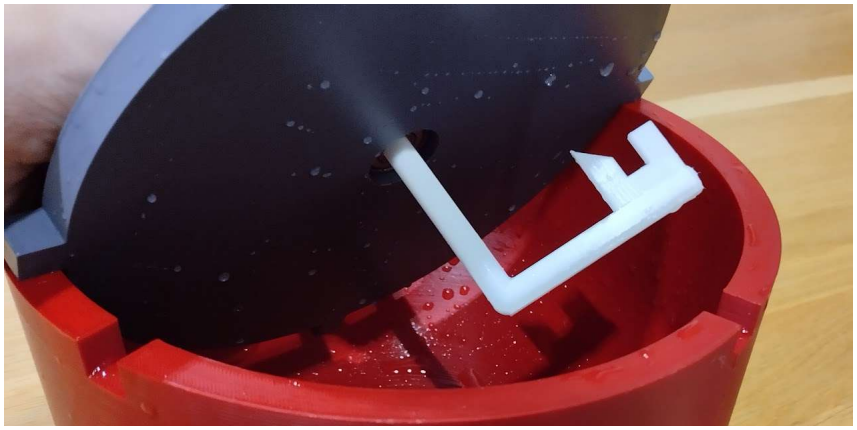


Figure 8: A redesigned arm-simulator of arbitrary irregular shape

Repeating the above motor characterisation process, more-or-less as previously, the relevant parameters of the new motor and arm are presented in table 4.

Table 4: Motor parameters revisited

Parameter	Symbol	Value	Unit
Armature Resistance	R_a	7.30E+00	Ω
Armature Inductance	L_a	7.42E-03	h
Torque Constant	K_t	3.70E-01	$\frac{V}{rad/s}$
Total Inertia	J_t	4.52E-04	kgm^2
Friction Constant	D	0.00115	$\frac{N \cdot m}{(rad/s)}$
Lumped Drag Constant	C_d	4.23E-04	$\frac{N \cdot m}{(rad/s)^2}$

These results are largely in-line with the previous motor and arm-simulator. Of note is the significant increase to C_d – as was expected – and, curiously, the decrease in J_T. The latter may be due to manufacturing variations in the motor gearbox and axle overwhelming the minor increase in lightweight plastic along the arm, though is perhaps more likely due to limitations of the experimental methodology. In attempt to mitigate this, the motor step response was recorded *three* times at each of 1, 5, 8, 10, and 12 V – a plot of the 10 V trials against the ultimate curve fit is presented in Figure **Error! Reference source not found.** – and C_D was averaged over a broader range of trials at 1, 3, 5, 6, 8, 10, 11, and 12 V. These values were selected to provide coverage over a wide range of operating conditions.

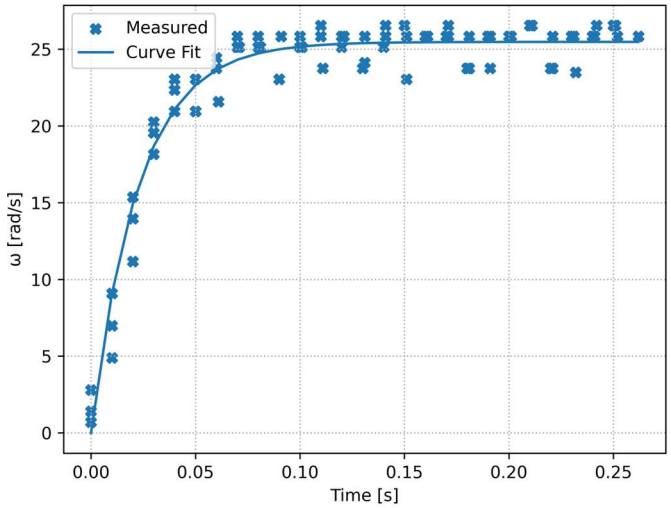


Figure 9: Motor step response, as measured for a 10 V step

The curve-fit here appears much more accurate than that of Figure **Error! Reference source not found.**, working on denser and more finely-quantised data as a result of conducting more trials, increasing sampling frequency, and making use of the full quadrature encoder.

4.6 PID Controller

To complete the second research objective, a standard PID controller was developed and manually tuned for controlling arm position in air. The controller was updated at 50 Hz, approximating the derivative of error using the error from the previous timestep, and approximating the integral of error using a Riemann sum over the last fifty errors – the full code, including values of K_i, K_p, and K_d, is presented in Appendix E. In air, the controller performed well: the controller's response time tended to be around 250 ms, with a typical steady-state error of 1–2 degrees. A plot of the system response to a 90-degree step command, in radians, is presented in Figure **Error! Reference source not found.** for reference.

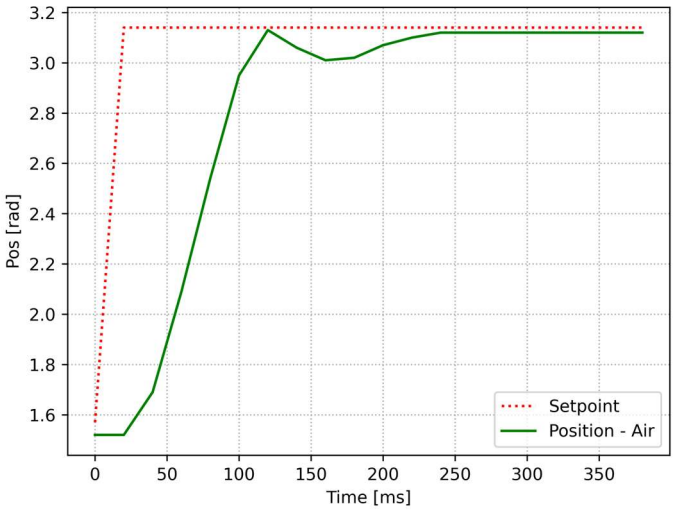


Figure 10: PID response to a 90-degree step command

Submerging the arm in water, the performance worsens. Response time is marginally slower and steady-state error consistently higher, if still typically in a 1–2 degree range. Figure **Error! Reference source not found.** compares the underwater response to an equivalent 90-degree step command to that of Figure **Error! Reference source not found.**.

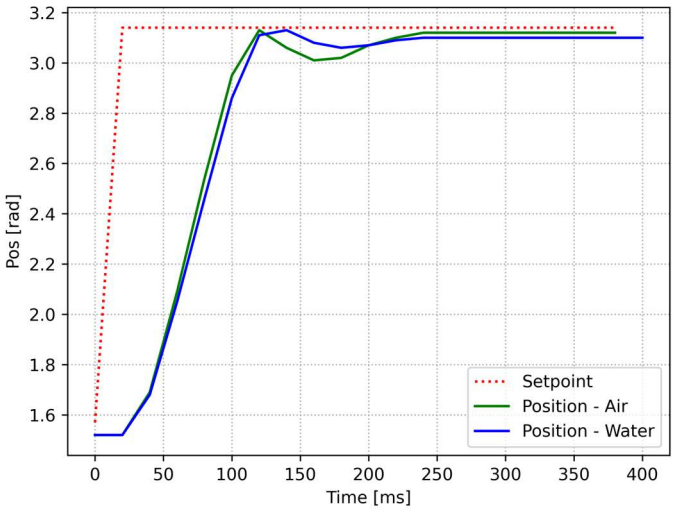


Figure 11: Step response comparison between air and water

This result completes the second objective of this research – submerging an air-tuned PID controller in water slows response time and worsens steady state error.

4.7 Feedforward Controller

Armed with a good understanding of the motor dynamics, and a functional PID controller, the full feedforward-with-trim controller, as described in Equation 12, was implemented – Appendix F. The controller performed well underwater, reducing typical steady state error to less than 1 degree: a notable improvement in accuracy compared to the PID controller. The

minor oscillations observed in Figures **Error! Reference source not found.** and **Error! Reference source not found.** have also been eliminated.

A benefit of having selected S-curves to generate the control trajectory is that the response time is immediately adjustable through t_0 – the client complaint at onset of this project was that their arm swung too fast for safe operation around humans, and so a slower response time of 1 second was selected here, to be representative of this kind of real-world use case.

A plot of the controller response to the same 90-degree step command as before is presented in Figure **Error! Reference source not found.**. Several additional plots of performance, responding to varying step commands, are also presented in Appendix G for reference and confidence in results.

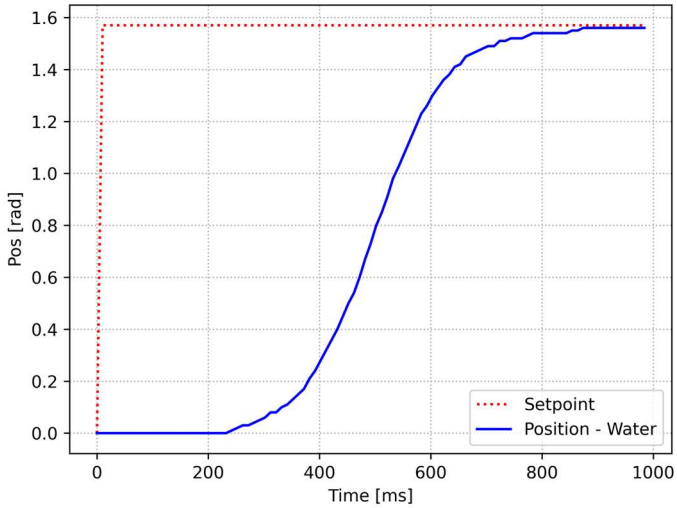


Figure 12: Feedforward response to a 90-degree step

As the feedforward controller already closely follows the intended trajectory, the PID trim component of the controller has to react much more "aggressively" to error than the original PID controller: significantly higher values of K_i and K_p were used. The trajectory closely follows the commanded S-curve – the two are plotted on the same axes in Figure **Error! Reference source not found.**.

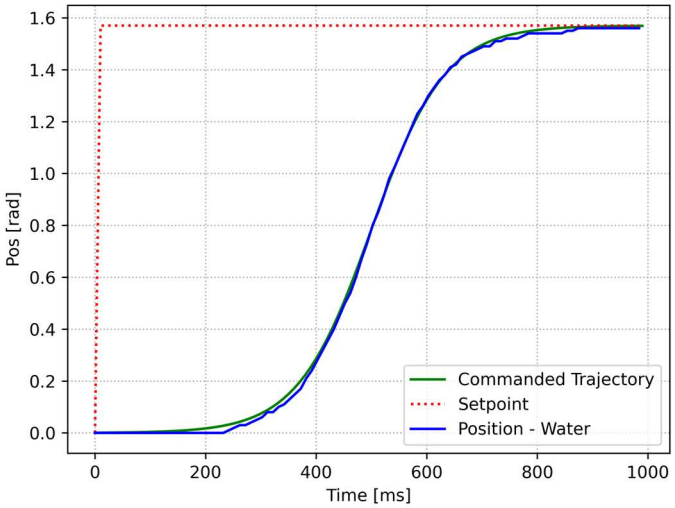


Figure 13: Feedforward response compared to command

Though noisier, the overall shape of the angular velocity profile also closely follows the commanded angular velocity curve – Figure **Error! Reference source not found.**.

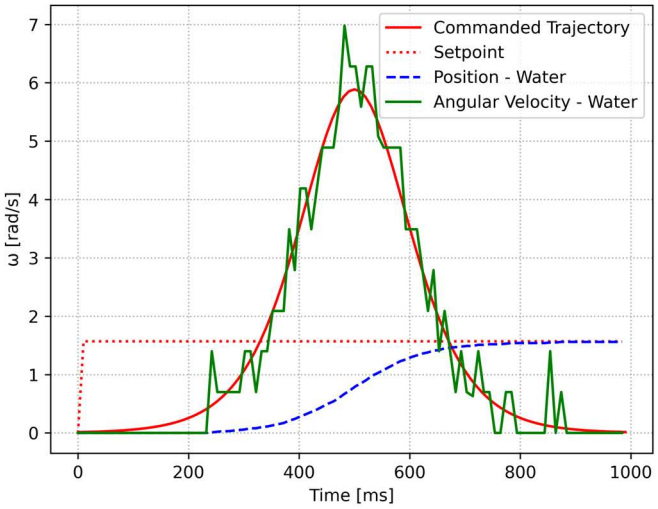


Figure 14: Feedforward velocity profile in a 90-degree step

The close alignment between command and response builds confidence in the controller's functioning. It also bears merit to plot controller effort throughout the step response – Figure **Error! Reference source not found.**.

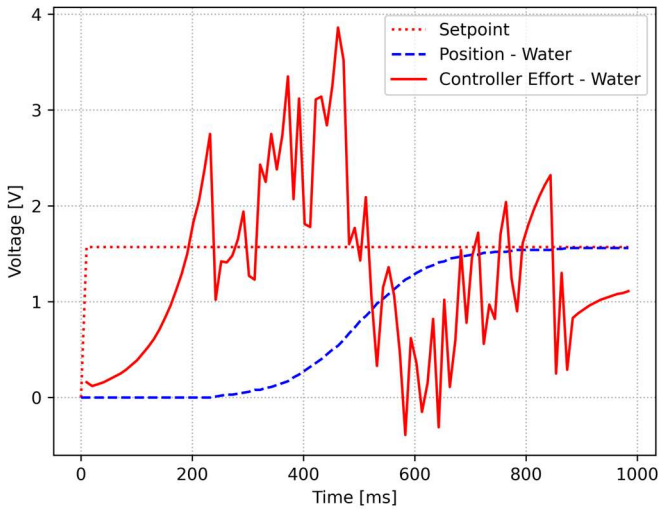


Figure 15: Feedforward controller effort in a 90-degree step

As the motor is accelerated through the front half of the response, voltage rises alongside angular velocity to combat increasing fluid drag; at the halfway point, voltage drops significantly, briefly dipping negative to decelerate the arm as it approaches its setpoint. The fluctuations throughout are a result of the PID trim component of the controller combating both randomness in the real world and an inevitably imperfect model of the system.

Overall, the feedforward controller that was developed is highly accurate, tuneable, and performs to better precision than the conventional PID controller underwater: the third objective of this research is met. The methodology used to develop this controller is low-cost, generalised, repeatable, and supports the adoption of this form of feedforward controller throughout the underwater robotics industry, including in mussel-sampling robots.

5 Discussion

5.1 Findings

The primary finding of this research is that a feedforward-with-trim control architecture has been shown to be reliable, precise, tuneable, and more effective than a conventional PID controller at operating underwater robot arms, consistently achieving positional steady-state error of less than 1 degree. That the method was able to produce a highly accurate controller with an irregularly-shaped arm-simulator lends confidence to its use for real-world underwater robot arms, which will not closely match the simple cylinder model of the initial characterisation trials.

Beyond the controller itself, a principle secondary finding is an inexpensive, generalised, repeatable methodology for characterising both an unknown DC motor and an irregularly shaped robot arm intended for underwater operation. While the method would need to be scaled up for a larger commercial robot arm, the equipment that was used in this research was highly conventional lab equipment – not expensive bespoke equipment or software such as the dynamometer of [8] or the CFD suite of [9].

Furthermore, as seen in section 4.6, this research finds that a conventional PID controller tuned for operation in air performs worse when operating the arm-simulator underwater. This is not a surprising result – the additional fluid drag slows response time and increases steady-state error – but the PID controller is evidently still relatively effective, even in the unconventional conditions. The reduction in response time is minimal, and the steady state error remains bounded within 1–2 degrees: this will be sufficient for many underwater robotics applications and may be preferred to the feedforward controller architecture presented here for the simplicity of using the conventional solution.

5.2 Implications

This research has shown that, in situations where highly accurate position control is required of an underwater robotic arm, such as a mussel-sampling robot, a feedforward-with-trim controller architecture may be preferable to a standard PID controller. The controller developed here allows fine adjustment of response speed, high positional accuracy, reduced oscillation, and minimal overshoot. For an underwater sampling robot, the improvement in performance afforded by this controller architecture could expand mission profiles and produce additional revenue for an operator by improving the robot's sampling precision, frequency, and consistency.

Beyond this, the research has shown that a conventional PID controller also performs adequately underwater, and will remain a suitable choice for many underwater robotics projects where simplicity is preferred over extremely precise control. The motor characterisation methodology shown here, for its low cost and general utility, may also inform related characterisation projects in future, particularly those studying the impacts of fluid drag on a motor's output.

5.3 Limitations

The principle limitation of this research is that the accuracy of the motor characterisation is indeterminate. Though a highly effective feedforward controller was produced, the PID trim component of the controller was tuned to respond very "aggressively" to error, which may have been compensating for large deviations between the measured motor parameters and their actual real-world values. The second arm-simulator that was trialled had more plastic mass; that the measured inertia (J_t) *decreased* from the first arm-simulator implies inaccuracy in the characterisation technique. The apparent "stickiness" at low controller effort that is observed in Figure **Error! Reference source not found.** suggests also that rotor friction (D) was underestimated; the controller ought to be commanding a higher voltage here to compensate.

C_d is also suspect – though the value measured for the first arm-simulator closely aligned with theory, the measured value of C_d for both arm-simulators varied significantly across angular velocity, and even came back negative on occasion – neither of these observations, in theory, should be possible. An average of C_d was taken across a broad range of voltage trials (and hence angular velocities) to help mitigate this, but this variation suggests higher-order fluid dynamics effects – such as skin drag or boundary effects within the test chamber – were having a significant impact on the measurement of C_d.

Beyond this, it is unclear how effectively a conventional PID controller purposefully tuned for underwater operation would perform in comparison to the feedforward controller presented here. Doing this tuning may improve the performance of the PID controller sufficiently to preclude the usefulness of a feedforward controller architecture in real-world underwater robotics.

5.4 Future work

To address the most major limitation of the project, future work should begin by trialling a feedforward controller constructed using the above methodology *without* an associated PID controller for trim – in steady conditions, the positional accuracy of this will quantify to what degree the trim component of the controller is compensating for inaccurate motor parameters.

If the effect is indeed significant, the characterisation could be improved by the use of more accurate equipment – a quadrature encoder of higher pulses-per-revolution would reduce quantisation error, for instance, or a more precise, dedicated measure of current would lend confidence to those parameters dependent on measuring current. Alternately, using a larger, heavier arm-simulator would be expected to help J_t and C_d "break through the noise"; would make them more detectable on the calibre of equipment used in this research. Using a significantly larger test chamber to contain the water would also reduce what impact boundary effects have had on C_d.

Sidestepping these problems entirely, future work could also look to replace the physics-based derivation of motor parameters with a more generalised scheme – [13] presents an approach to developing feedforward controller parameters with deterministic artificial intelligence, for instance.

In a smaller note, future work could, of course, also look to further refine controller tuning parameters – K_i, K_p, K_d, k, and t_0 – though one could spend an eternity doing this and never be satisfied optimal parameters have been found; other workstreams are likely to yield more significant results.

Any of this future work would be expected to produce a more accurate controller with more confidence in that accuracy, which would in turn be useful to the same underwater robotics projects this research is concerned with.

6 Conclusions

The objectives of this research were threefold: to develop a suitably comprehensive model of the system dynamics, to characterise the impact submerging a robotic arm in water has on a no-air-resistance PID controller, and to develop a corrected "feedforward with trim" control system to manoeuvre the arm. When it transpired that the use of the real-world underwater robot arm from the final year project conducted in parallel to this research would not be possible, a scaled-down "arm simulator" was designed and built to continue the research.

The first objective of this research was accomplished in sections 4.1–4.5 – a generalised, repeatable methodology for characterising the non-linear and underwater dynamics of a DC motor and attached robotic arm was developed, trialled, and compared to theory. Though an effective controller was produced, the accuracy of this characterisation is indeterminate – future work should look to quantify accuracy by trialling a blind feedforward controller, and could then look to improving accuracy by using more precise equipment, breaking through noise with larger motors and arms, or foregoing a physics-based approach entirely in favour of more general system identification techniques.

The second objective of this research was accomplished in section 4.6 – having developed a PID controller tuned for operation in air, submerging the arm-simulator in water was shown to slow response time and increase steady state error of the controller. The impact, however, was minor enough that a conventional PID controller remains broadly suitable for use in underwater robotics, and may be preferred in projects where simplicity is desired over extremely precise control.

The third and final objective of this research was accomplished in section 4.7 – a robust, accurate, highly tuneable feedforward-with-trim controller was developed and shown to perform better underwater than the conventional PID controller. This controller architecture may find use in underwater robots that demand extreme precision in their robotic arms, whether for mussel sampling or otherwise, and would be expected to deliver increased revenue to robot operators for the improved performance and expanded mission profiles this architecture affords.

Declarations

Author Contribution: Tim Parker: conceptualisation, resources, data curation, software, formal analysis, visualisation, investigation, writing – original draft, methodology, writing – review & editing.

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Conflict of Interest: The author declares that there are no financial conflicts of interest regarding this work.

AI: Generative AI was not used in the preparation of any body text in this report, though Google's *Gemini* models were used in a limited capacity to aid in formula transcription, code ideation, and general research throughout the project.

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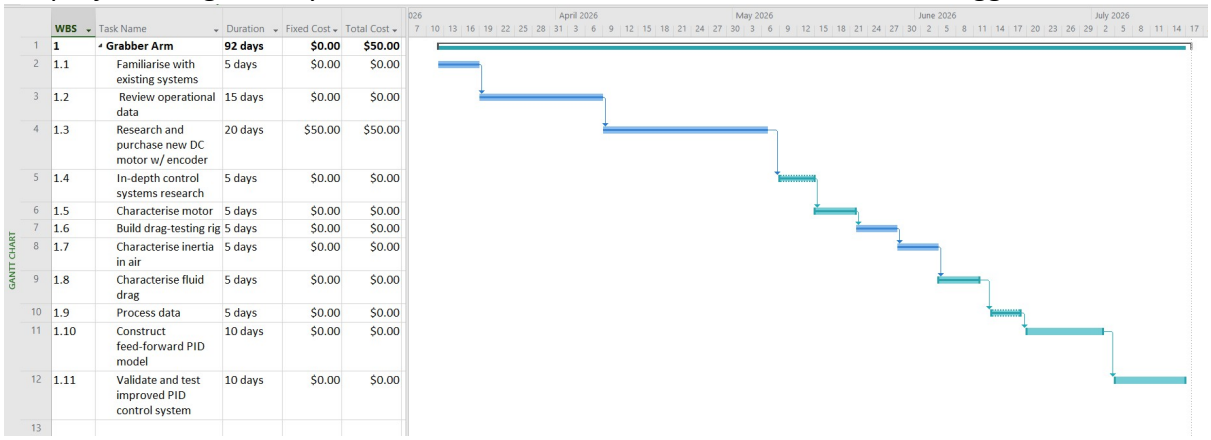
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Appendix A: Project planning

Note that this timeline predicts the project should have been completed in mid-July! Due to ongoing work on the practical, client-driven half of the final year project, work on this subproject has generally not been continuous, as the Gantt chart would suggest.



GANTT CHART

	WBS	Task Name	Duration	Fixed Cost	Total Cost
1	**1**	**Grabber Arm**	**92 days**	**$0.00**	**$50.00**
2	1.1	Familiarise with existing systems	5 days	$0.00	$0.00
3	1.2	Review operational data	15 days	$0.00	$0.00
4	1.3	Research and purchase new DC motor w/ encoder	20 days	$50.00	$50.00
5	1.4	In-depth control systems research	5 days	$0.00	$0.00
6	1.5	Characterise motor	5 days	$0.00	$0.00
7	1.6	Build drag-testing rig	5 days	$0.00	$0.00
8	1.7	Characterise inertia in air	5 days	$0.00	$0.00
9	1.8	Characterise fluid drag	5 days	$0.00	$0.00
10	1.9	Process data	5 days	$0.00	$0.00
11	1.10	Construct feed-forward PID model	10 days	$0.00	$0.00
12	1.11	Validate and test improved PID control system	10 days	$0.00	$0.00
13					

Appendix B: H&S hazard assessment

Authors: Tim Parker
Date: 20/03/2026

A.1 Architecture of the system



Area	***Risk***
Gripper attachment	Cut risk on sharp metal edges Finger trapped during operation
Main arm	Injury due to collision during swinging motion Arm becomes entangled in and damages wires; electrocution risk
Electronics	Electrocution risk when operating in water
Testing Environment	Risk of drowning

A.2 Risk matrix and scales adopted

The ISO31000 risk method was used[3]. The Consequence and Likelihood scales were used per [4].

Likelihood of the consequence arising	Description	Probability, events per annum	Qualitative likelihood	*Likelihood score*							
	Annual occurrence	*1.00*	*Almost certain*	*10*	5000	1000	600	300	200	100	50	20
	Has occurred several times in your career	*0.6*	*Likely*	*6*	3000	600	360	180	120	60	30	12
	Might occur once in your career	*0.4*	*Possible*	*4*	2000	400	240	120	80	40	20	8
	Event does occur somewhere from time to time	*0.2*	*Unlikely*	*2*	1000	200	120	60	40	20	10	4
	Heard of something like this happening elsewhere	*0.1*	*Rare*	*1*	500	100	60	30	20	10	5	2
	Theoretically possible but not expected to occur	*0.01*	*Almost incredible*	*0.1*	50	10	6	3	2	1	0.5	0.2

Severity of Adverse health consequences								
Loss of health	500	100	60	30	20	10	5	2
Safety consequence	Catastrophe: multiple deaths occur, would have negative impacts to next generation	Death: People loss of life	Serious harm: Notifiable injurles. Permanent and serious harm	Serious harm: Notifiable injurles. Incident results in serious harm	Moderate incident: Incident occurs and exposure to moderate harm	Minor harm: Incident occurs and minor harm	Minor harm: Incident without harm or with tiny impacts.	No harm: Hazard could be sufficiently managed by the current preventions.
Health consequence	Health disaster: Multiple people affected with high WHODAS scores	WHODAS up to 100 Death or paralysis.	WHODAS up to 60 Half-body, or leg paralysis	WHODAS up to 30 Amputation.	WHODAS Up to 20 Long term impact e.g. hearing loss	WHODAS up to 10 Permanent but not debilitating musculoskeletal injury	WHODAS Up to 5 Temporary effects to human body, healed naturally.	WHODAS up to 2 No harm to human body.
Chemical Toxicity per GHS 7	GHS Category 1: Extremely toxic. Fatal if swallowed, contacted, or inhaled. One drop is fatal	GHS Category 2: Highly toxic. Fatal if swallowed, contacted, or inhaled. One teaspoon is fatal	GHS Category 3: Moderately toxic. Toxic if swallowed, contacted, or inhaled. One tablespoon is fatal	GHS Category 4: Slightly toxic. Harmful if swallowed, contacted, or inhaled. Two cups are fatal	GHS Category 5: Practically non-toxic. May be harmful if swallowed, contacted, or inhaled. One litre is fatal	Level 6: Relatively harmless. Permanent but not debilitating injury. May cause death for sensitive individuals.	Level 7: Temporary harm with no permanent loss of ability or quality of life. Exposure is below LOAEL.	Level 8: No adverse effects expected. Exposure is below NOAEL.

Risk appetite response scale	
1000+	Cessation, immediate intervention required
120+ to 1000	Unacceptable risk. Cease operations until risk is under control
60+ to 120	Urgent treatment required. Operations proceed with caution and ongoing monitoring of risk.
10+ to 60	Implement treatment in a reasonable time period. Continue operations with caution and monitoring
0-10	No special treatment required. Continue the operations with on-going monitoring of the efficacy of existing preventions.

This Chemical Toxicity scale assumes an existing safety system comprising substances that are properly labelled, securely stored, only accessible by people with sufficient training, and appropriate personal protective equipment is provided. If the safety system is not strong, then consider increasing the likelihood of the consequences arising.

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A.3 Risk register

		Risks of system in its CURRENT STATE, with its existing controls.						Risks of system in its FUTURE STATE after these treatments. Insert any new threats caused by the treatments.			
Architecture level: Work-stream, project phase, hardware category, workstation	Specific hazard	Consequence (C), as per 'Severity of harm' scale 1. Hazard occurred 2. Incident with no harm 3. Incident and Minor harm 4. Incident and exposure to serious harm 5. Serious harm Occurs 8. Death 10. Catastrophe	Likelihood (L) of that harm consequence arising 6 Almost certain 5 Likely 4 Possible 3 Unlikely 2 Rare 1 Almost incredible	Risk = (C x L)	Treatment? Consider Preventative and Recovery mechanisms. 30 or higher Unacceptable risk. 18 or higher Urgent treatment. 8 or higher Consider treatment	Action required by who? Resources required?	Monitoring required of efficacy of treatment?	Treated Consequence (C*)	Treated Likelihood (L*)	Residual Risk (C* x L*)	Is this acceptable? What further action is required?
Gripper	Cut on sharp edges	3 Serious cut unlikely	4 Reasonably possible during handling	12	Practice care when handling; safe cover for gripper during transport	All users	Ensure reasonably impossible to cut self with cover applied	3	2	6	Risk reduced sufficiently
Gripper	Finger trapped during operation	4 Fair amount of torque; potential for	3	12	Avoid actuation of gripper during testing;	All users	Ensure gripper remains closed	4	2	8	Risk reduced sufficiently

		serious injury to finger			keep fingers well away from active arm		during testing					
Main arm	Collision with operator	3 Unlikely enough momentum to seriously injure in a blunt-force collision	4 Reasonably possible during testing	12	Keep body away from arm during testing	All users	Monitor proximity of operators to arm; classify reasons to approach during operation and mitigate with alternatives		3	2	6	Risk reduced sufficiently
Main arm	Severed wires; electrocution risk	3 Comparatively low voltage; serious injury unlikely	2 Wires well protected due to need for waterproofing	6	None immediately needed; below a risk level of 10	All users	Monitor condition of wires; replace if damaged		3	2	6	Risk reduced sufficiently
Electronics	Electrocution risk due to underwater operation	3 Comparatively low voltage; serious injury unlikely	3 Wires already waterproofed	9	None immediately needed; below a risk level of 10	All users	Monitor waterproofing of wires; ensure care where new electronics installed		3	3	9	Risk reduced sufficiently

Testing Environment	Drowning	8 Easily fatal	3	24	Use contained bodies of water where possible; ensure operators can swim when operating in open water; supply life jackets as appropriate	All users	Monitor when incidents do happen, and address the root cause to avoid repeats		8	1	8	Risk reduced sufficiently

Appendix C: Python Code for Non-linear Curve-fit

```python
import numpy as np
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt

def motorResponse(X, a, b):
    t, v = X
    return np.where(t>=0, (a*v/b) * (1-np.exp(-b*t)), 0.0)

def readFile(fName):
    with open(fName) as file:
        trials = file.read().split('\n\n')
    data = [[] for i in range(3)]
    for trial in trials:
        trial = trial.split('\n')
        v = int(trial[0].split('=')[1])
        t0 = int(trial[1].split(', ')[0])
        for datapoint in trial[1:]:
            t, w = [float(i) for i in datapoint.split(', ')]
            t = (t-t0)/1000
            w = w / 21.3 #account for internal gear ratio
            [data[i].append(j) for i,j in enumerate([t,w,v])]
    return [np.array(i) for i in data]

def plot(a, b, v, data):
    ts, ws, vs = [i[data[2]==v] for i in data] #numpy boolean masking!
    graph = plt.axes()
    graph.scatter(ts, ws, marker='X', label="Measured")
    graph.plot(ts, motorResponse((ts, vs), a, b), label="Curve Fit")
    graph.set_xlabel("Time [s]")
    graph.set_ylabel("w [rad/s]")
    graph.set_title(f"Motor Response to {v} V Step Input")
    graph.grid(True, linestyle=':')
    graph.legend()
    plt.show()

ts, ws, vs = readFile(r'c:\Users\Timothy\Downloads\MotorSpeedData.txt')

popt, pcov = curve_fit(motorResponse, (ts, vs), ws, p0=(10.0, 10.0)) #seed
a,b = popt

print(f"Parameter a: {a:.4f}")
print(f"Parameter b: {b:.4f}")

plot(a,b,12,[ts,ws,vs])
```

Appendix D: Analytic Calculation Of Drag Constant

Under standard conditions, water has a density (ρ) of 1000 kg/m^3 and a kinematic viscosity (ν) of 1.0 × 10^{-6} m^2/s. The arm-simulator has a diameter (d) of 0.008 m and length (L) of 0.06.

During a trial at 12 V while submerged in water, the arm-simulator was measured at a steady-state angular velocity (ω) of 29.48 rad/s. The velocity at the tip of the arm was thus $\omega d = 29.48 \times 0.06 = 1.77\, m/s$, which, in turn, corresponds to a Reynold's number of $\frac{(\omega d)d}{\nu} = \frac{1.77 \times 0.008}{1.0 \times 10^{-6}} = 14.2 \times 10^3$. This is well within the turbulent flow regime; consulting any existing empirical data of smooth cylinders in this Reynold's number range suggests an underlying dimensionless drag coefficient (C_D) of *around* 1.

Consider now a thin section of the arm-simulator of length dr, at distance r, from the axis of rotation around the shaft. Thus considering a frontal area of $d \times dr$, drag torque on the section can be expressed as $\tau(r) = r\frac{1}{2}\rho(\omega r)^2 C_D(d \times dr) = \frac{1}{2}dr^3\rho\omega^2 C_D dr$. The total drag over the arm-simulator is then calculated as $\tau = \int \tau\,(r)dr = \frac{1}{8}\rho C_D d L^4 \omega^2$.

The lumped "drag constant" (C_d) is defined by $C_d = \frac{\tau}{\omega^2}$, and thus $C_d = \frac{1}{8}\rho C_D d L^4$. Substituting known parameters, it can be shown that $C_d = \frac{1}{8}(1000)(1)(0.008)(0.06)^4 = 1.30 \times 10^{-5}$.

Appendix E: C++ PID Controller Code

```
#include <Encoder.h>

#define encA 2
#define encB 3
#define pi 3.1415926

#define motorIn 10
#define motorDirA 9
#define motorDirB 8

#define maxVoltage 12

#define kP 5
#define kI 0.7
#define kD 0.9

Encoder myEncoder(encA, encB);

#define ENCODER_PPR 900 //(40.0 * 23.1 ish)

unsigned long prevTime = 0;
const long interval = 20; // Update speed every 20ms
long prevTicks = 0;
float target = 0;
float prevError = 0;

#define bufferSize 20
float errorBuffer[bufferSize] = {0};
int windex = 0;

float sum(float* array, int size) {
  float total = 0;
  for (int i = 0; i < size; i++) {
    total += array[i];
  }
  return total;
}

int sign(float val) {
  return val != 0? 1 - 2 * (val < 0) : 0;
}

float clamp(float value, float low, float high) {
  return (value < low) ? low : (high < value) ? high : value;
}
```

```
float minError(float target, float current) {
  return atan2(sin(target - current), cos(target - current)); //???
}

void killMotor() {
  digitalWrite(motorDirA, LOW);
  digitalWrite(motorDirB, LOW);
}

void runMotor(float voltage) {
  int command = (255 * voltage) / maxVoltage;
  if (command > 0) {
    digitalWrite(motorDirA, HIGH);
    digitalWrite(motorDirB, LOW);
  } else {
    digitalWrite(motorDirA, LOW);
    digitalWrite(motorDirB, HIGH);
  }
  analogWrite(motorIn, clamp(abs(command), 0, 255));
}

void setup() {
  Serial.begin(115200);
  Serial.println("Starting!");

  pinMode(motorIn, OUTPUT);
  pinMode(motorDirA, OUTPUT);
  pinMode(motorDirB, OUTPUT);
}

void loop() {
  unsigned long currTime = millis();

  // Non-blocking calculation loop
  if (currTime - prevTime >= interval) {
    long currTicks = myEncoder.read();

    long dTicks = currTicks - prevTicks;
    long dTime = currTime - prevTime;
    prevTicks = currTicks;


    // Calculate speed:
    float angVel = ((1000 * dTicks * 2 * pi) / (ENCODER_PPR * dTime));

    // Calculate angle:
    float theta = (currTicks * 2 * pi) / ENCODER_PPR;
    float error = minError(target, theta);
```

```
    float dError = 1000 * (error - prevError) / dTime;
    float iError = sum(errorBuffer, bufferSize) * dTime;

    prevError = error;
    if (windex >= bufferSize) {
      windex = 0;
    }
    errorBuffer[windex] = error;

    float command = kP * error + kD * dError + kI * iError;


    if (abs(command) > 2 || abs(angVel) > 0) {
      runMotor(command);
    } else {
      killMotor(); //it's not gonna run anyway. save it from burning out
    }

    prevTime = currTime;
  }

  if (Serial.available()) {
    target = (2 * pi * Serial.parseInt()) / 360;

    //discard anything still in the buffer
    while (Serial.available() > 0) {
       Serial.read();
    }
  }
}
```

Appendix F: C++ Feedforward Controller Code

```
#include <Encoder.h>

#define encA 2
#define encB 3
#define pi 3.1415926

#define motorIn 10
#define motorDirA 9
#define motorDirB 8

#define maxVoltage 12

#define Ra 7.3
#define Kt 0.37
#define D 0.00115
#define Jt 0.000452
#define Cd 0.000423

#define kP 35.0
#define kI 5.0
#define kD 1.2

Encoder myEncoder(encA, encB);

#define ENCODER_PPR 900 //(40.0 * 23.1 ish)

unsigned long prevTime = 0;
const long interval = 10; // Update speed and control every 10ms
long prevTicks = 0;
float target = 0;
float prevError = 0;
float thetaStart = 0;
float theta = 0;
float startTime = 0;

#define bufferSize 50
float errorBuffer[bufferSize] = {0};
int windex = 0;

float sum(float* array, int size) {
  float total = 0;
  for (int i = 0; i < size; i++) {
    total += array[i];
  }
  return total;
}
```

```
int sign(float val) {
  return val != 0? 1 - 2 * (val < 0) : 0;
}

float clamp(float value, float low, float high) {
  return (value < low) ? low : (high < value) ? high : value;
}

float minError(float target, float current) {
  return atan2(sin(target - current), cos(target - current)); //???
}

void killMotor() {
  digitalWrite(motorDirA, LOW);
  digitalWrite(motorDirB, LOW);
}

void runMotor(float voltage) {
  int command = (255 * voltage) / maxVoltage;
  if (command > 0) {
    digitalWrite(motorDirA, HIGH);
    digitalWrite(motorDirB, LOW);
  } else {
    digitalWrite(motorDirA, LOW);
    digitalWrite(motorDirB, HIGH);
  }
  analogWrite(motorIn, clamp(abs(command), 0, 255));
}

#define k 15
#define x0 0.5

float s(float x, float goal, float start) {
  float L = goal - start;
  return L / (1.0 + exp(-k * (x - x0))) + start;
};

float ds(float x, float goal, float start) {
    float L = goal - start;
    float sVal = s(x, goal, start);
    return k * (sVal - start) * (1.0 - (sVal - start) / L);
};

float dds(float x, float goal, float start) {
    float L = goal - start;
    float sVal = s(x, goal, start);
```

```
    return k * k * (sVal - start) * (1.0 - (sVal - start) / L) * (1.0 -
2.0 * (sVal - start) / L);
};


void setup() {
  Serial.begin(115200);
  Serial.println("Starting!");

  pinMode(motorIn, OUTPUT);
  pinMode(motorDirA, OUTPUT);
  pinMode(motorDirB, OUTPUT);
}

void loop() {
  unsigned long currTime = millis();

  // Non-blocking calculation loop
  if (currTime - prevTime >= interval) {
    long currTicks = myEncoder.read();

    long dTicks = currTicks - prevTicks;
    long dTime = currTime - prevTime;
    prevTicks = currTicks;

    // Calculate speed:
    float angVel = ((1000 * dTicks * 2 * pi) / (ENCODER_PPR * dTime));

    // Calculate angle:
    theta = (currTicks * 2 * pi) / ENCODER_PPR;

    // Calculate s-curve commands:
    float timeSince = (millis() - startTime)/1000;
    float thetaC = s(timeSince, target, thetaStart);
    float dThetaC = ds(timeSince, target, thetaStart);
    float ddThetaC = dds(timeSince, target, thetaStart);

    // Calculate errors
    float error = minError(thetaC, theta);
    float dError = 1000 * (error - prevError) / dTime;
    float iError = sum(errorBuffer, bufferSize) * dTime;

    prevError = error;
    if (windex >= bufferSize) {
      windex = 0;
    }
    errorBuffer[windex] = error;
```

```
    float command = (Ra/Kt)*(Jt*ddThetaC + D*dThetaC + Cd*dThetaC*dThetaC)
+ Kt*dThetaC + kP*error + kD*dError + kI*iError;

    if (abs(command) > 2 || abs(angVel) > 0) {
      runMotor(command);
    } else {
      killMotor(); //it's not gonna run anyway. save it from burning out
    }
    prevTime = currTime;
  }

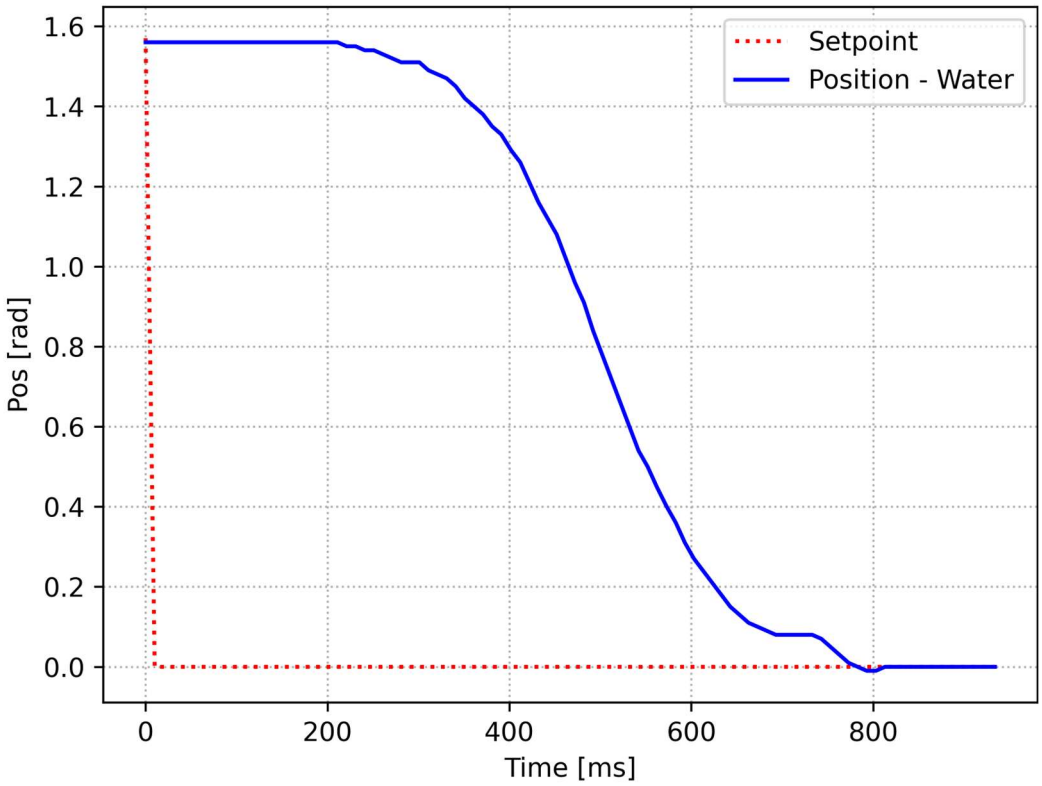
  if (Serial.available()) {
    target = (2 * pi * Serial.parseInt()) / 360;
    thetaStart = fmod(theta, 2*pi);
    target = fmod(target, 2*pi);

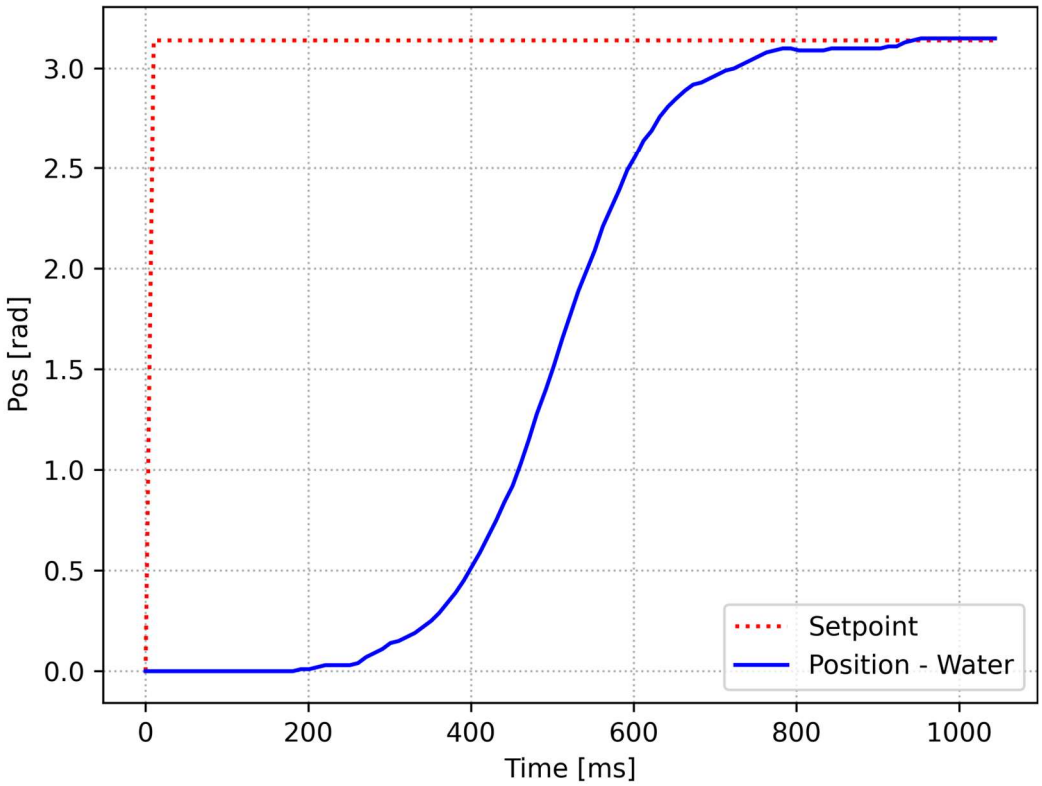
    if (abs(target - thetaStart) > pi) {
      if (target > thetaStart){
        target -= 2*pi;
      } else {
        target += 2*pi;
      }
    }

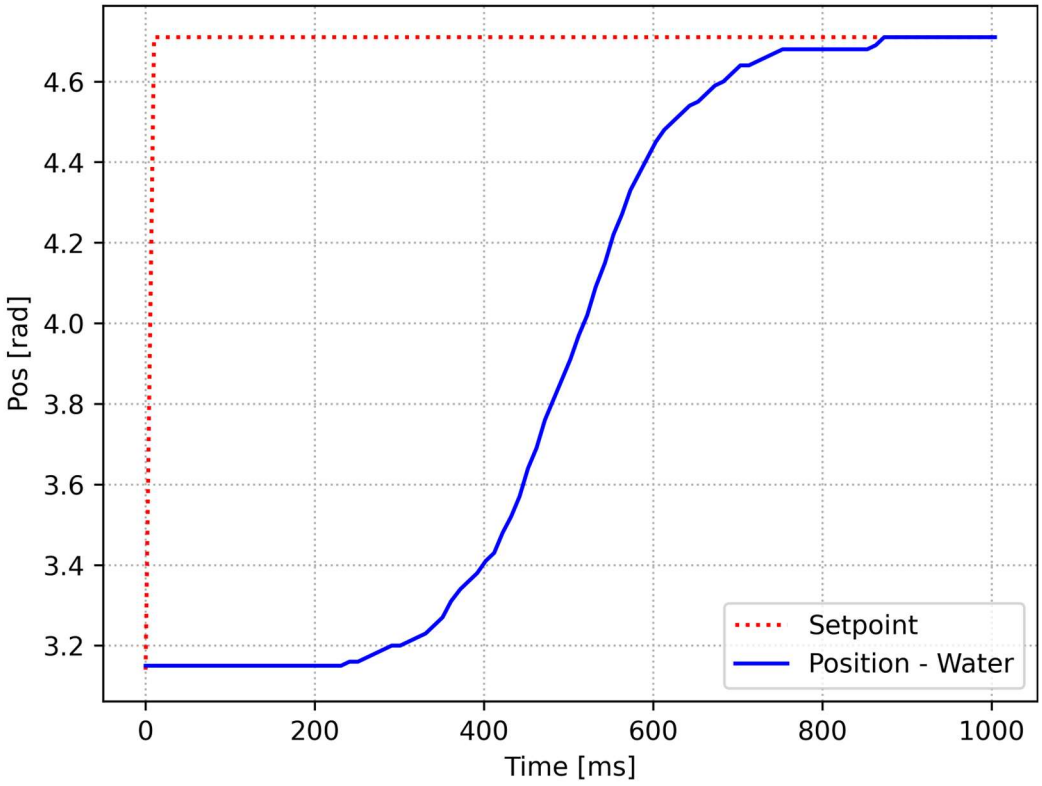
    startTime = millis();
    // Serial.println(target);

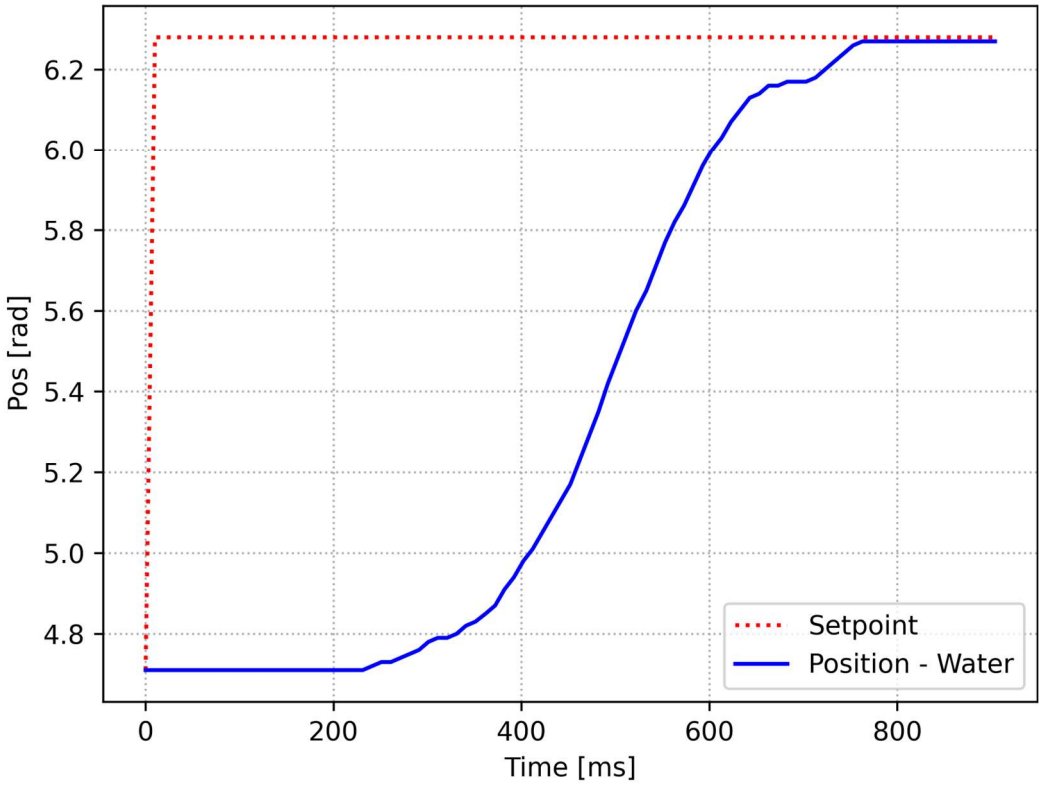
    //discard anything still in the buffer
    while (Serial.available() > 0) {
       Serial.read();
    }
  }
}
```

Appendix G: Additional Feedforward Responses

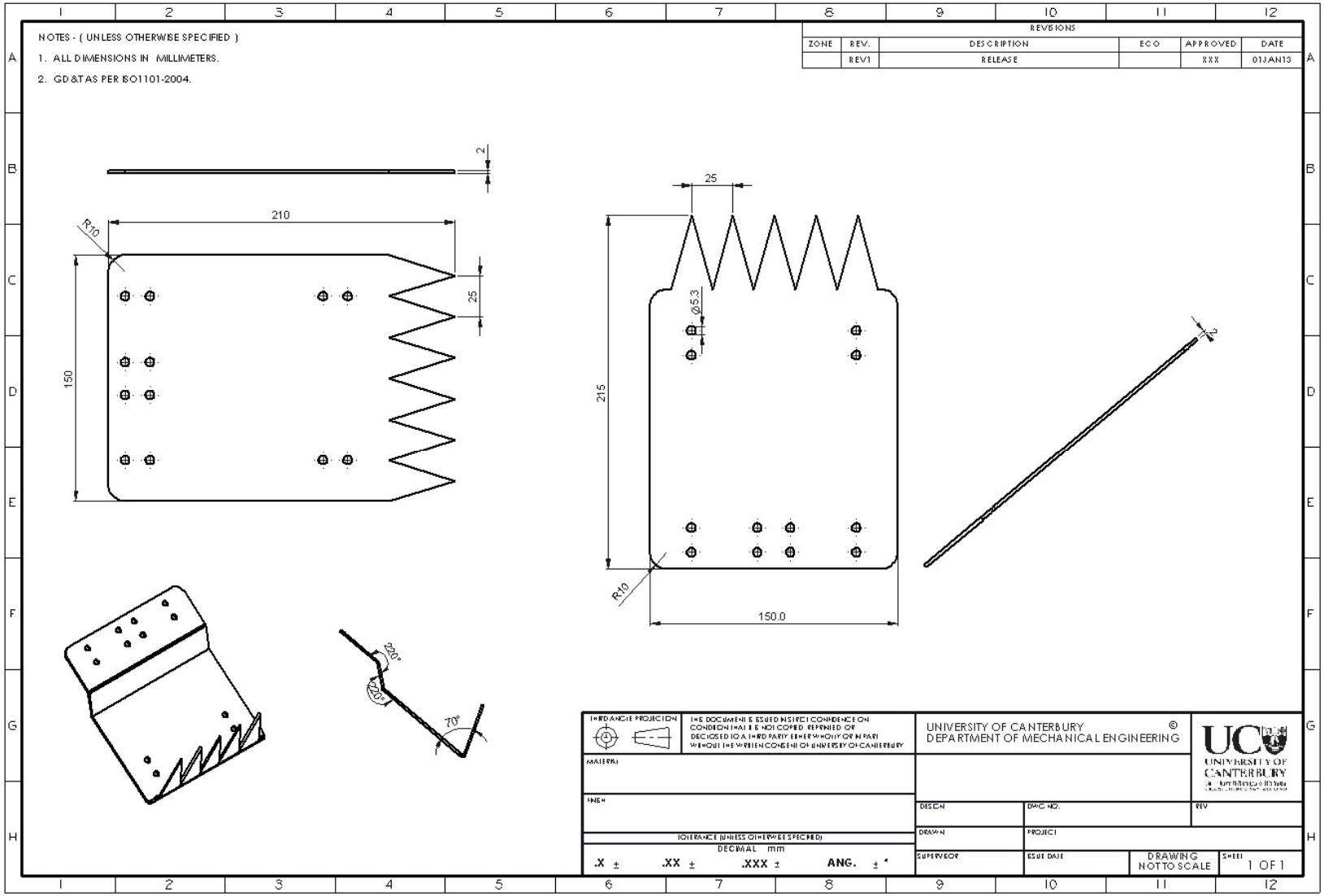





Pos [rad]
4.6
4.4
4.2
4.0
3.8
3.6
3.4
3.2
0
200
400
600
800
1000
Time [ms]
Setpoint
Position - Water


Pos [rad]
6.2
6.0
5.8
5.6
5.4
5.2
5.0
4.8
0
200
400
600
800
Time [ms]
Setpoint
Position - Water

Appendix F - Drawings



Appendix G - Costing Report

Table G.1: Predicted costs

Item	Cost
Servo Motor	$200.00
Teather tension sensor	$50.00
Electronics to process tension data	$50.00
Adjustable spring	$50.00
Teather management reel and frame components	$50.00
Sea testing	Covered by sponsor
Total	$400.00

Table G.2: Actual costs

Item	Cost
Stepper motor	$2,546.53
Drive Belt	$47.70
Control Electronics	$95.81
Stainless Fasteners	$39.53
Stainless Bearings	$39.68
Replacement ESC	$106.00
Total	$2875.25

Appendix H - List of Deliverables

H.1 – Mussel Position Prediction

- GIT repository including results and code used throughout project

H.2 – Turbidity Reduction

- GIT repository including results and code used throughout project

H.3 – Autonomous Tether Reel

- Retrofitted tether teel
- Control electronics
- Wiring diagram
- Git repository of control code
- CAD models, drawings, and DXFs and STLs (where applicable)

H.4 – Grabber System

- Physical prototype of collection system
- Full suite of CAD files (including earlier revisions and manual prototypes)
- Documentation (including wiring and code)